

DMD2050

Universal Satellite Modem

Installation and Operation Manual

TM110
Revision 1.3





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3138 E. Elwood St.

Phoenix, Arizona 85034 (USA)

ATTN: Customer Support

Phone: (602) 437-9620

Fax: (602) 437-4811

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Preface

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This manual provides installation and operation information for the Radyne DMD2050 Universal Satellite Modem. This is a technical document intended for use by engineers, technicians, and operators responsible for the operation and maintenance of the DMD2050.

Conventions

Whenever the information within this manual instructs the operator to press a pushbutton switch or keypad key on the Front Panel, the pushbutton or key label will be shown enclosed in "less than" (<) and "greater than" (>) brackets. For example, the Reset Alarms Pushbutton will be shown as <RESET ALARMS>, while a command that calls for the entry of a '7' followed by 'ENTER' Key will be represented as <7,ENTER>.

Cautions and Warnings



A caution icon indicates a hazardous situation that if not avoided, may result in minor or moderate injury. Caution may also be used to indicate other unsafe practices or risks of property damage.



A warning icon indicates a potentially hazardous situation that if not avoided, could result in death or serious injury.



A note icon identifies information for the proper operation of your equipment, including helpful hints, shortcuts, or important reminders.

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Record of Revisions

Revision Level	Date	Reason for Change
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1.1	12-16-04	Revised Table of Contents
1.2	10-12-05	Revised Section 3 & 4, Appendices
1.3	11-10-06	Revised to reflect Web Browser, Drop & Insert, TCP/IP Internet sections

Comments or Suggestions Concerning this Manual

Comments or suggestions regarding the content and design of this manual are appreciated. To submit comments, please contact the Radyne Inc. Customer Service Department.

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Introduction

1

This chapter provides an overview of the DMD2050 Universal Satellite Modem. The DMD2050 may be referred to in this manual as the DMD2050 or “the unit”.

1.0 Overview

Radyne 's new DMD2050 Satellite Modem (Figure 1-1) breaks new ground in flexibility and performance. With standards based upon MIL-STD-188-165A and the ability to run commercial industry standard functions including IDR, IBS and DVB, while covering data rates up to 52 Mbps, this duplex modem covers virtually all Satellite IP, Telecom, Video and Internet applications, while allowing switching between spur-free 70/140 MHz operation to L-Band without any configuration changes.

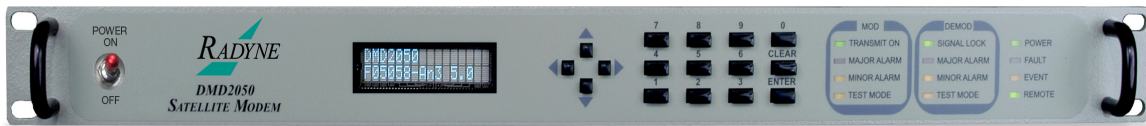


Figure 1-1. DMD2050 Universal Satellite Modem Front Panel

The extensive list of software options allows for the deployment of a modem with today's needs while keeping an eye toward tomorrow. Additional options and configuration (such as Monitor and Control (M&C) Functions) can be activated in seconds via the front panel. Additional hardware options like Turbo, Interface Expansion, and DC operation complete the modem's dynamic feature coverage.

The DMD2050's impressive remote accessibility surpasses all others in the field. Remote control via Radyne 's trusted RLLP (Radyne Link Level Protocol), Ethernet 10 Base-T SNMP and Web Browser includes control of all the modem's features plus software maintenance. Additionally, the Vacuum Fluorescent Display (VFD) can be supplemented with terminal software running on a PC or laptop.

The modem now presents its entire monitor and control functions on the big screen. Supported by Radyne 's extensive line of redundancy switches, converters, encoders and decoders, the DMD2050 can be built into any satellite requirement. Compatibility with current modems, such as Radyne 's DMD50, DMD2401, DMD15, and DISA certified MIL-188-165 compliant DMD15L are maintained for seamless substitution and addition to existing systems.

A full range of Industry Standard Interfaces is available for the DMD2050. Interface types are selectable from MIL-188-114A, ITU G.703, HSSI, ASI, DVB/M2P and Ethernet Bridge.

1.1 DMD2050 Configurations

The DMD2050 can be configured in the following different ways:

- features and options that are installed when the unit is ordered
- feature upgrades
- hardware options that the user can install at their own location
- options that are installed to a unit that is sent to a Radyne facility

1.1.1 Features/Options Installed at Time of Order

Features installed at the time of ordering are the options pre-installed/initialized in the factory prior to shipment. These can be reviewed from the front panel. Refer to Section 4, User Interfaces for information on how to view these features.

Factory installed options are chassis and board configurations that are introduced during manufacture.

1.1.2 Feature Upgrades

Feature Upgrades are a simple and quick way of changing the feature set of an installed modem. Feature upgrades are how most DMD2050 options are implemented. Features may be purchased at any time by contacting a Radyne Inc. salesperson. Refer to Section 4 and Appendix D, for information on how upgrade features.

1.1.3 Hardware Options

Hardware options (refer to Appendix A) are purchased parts that can be installed into the unit at the customer's site. A screwdriver is normally the only tool required. Please contact the Radyne Inc. Customer Service Department for information not limited to availability and to shipping costs.



Only authorized service personnel should handle and install optional hardware options.

1.1.4 Radyne Installed Options

Units may also be sent to the Radyne Inc. facility for hardware option installation. Please contact the Radyne Inc. Customer Service Department for information not limited to availability and to shipping costs.

1.2 Function Accessibility

All functions can be accessed with a terminal or personal computer via a serial link for complete remote monitoring and control capability.



Installation

2

This section provides instructions on unpacking and installation, as well as storage of the unit.

2.0 Installation Requirements

The DMD2050 Modem is designed to be installed within any standard 19-inch (48.26 cm) wide equipment cabinet or rack. It requires 1 rack unit (RU) of mounting space (1.75 inches/4.45 cm) vertically and 19.25 inches (48.89 cm) of depth. Including cabling, a minimum of 19.5 inches (49.53 cm) of rack depth is required. The rear panel of the modem is has power entering from the left and IF Cabling entering from the right (as viewed from the rear of the unit). Data and Control Cabling can enter from either side. The modem can be placed on a table or suitable surface if required.



WARNING!!

There are no user-serviceable parts or configuration settings located inside the DMD2050 Chassis. There is a potential shock hazard internally at the power supply module. DO NOT open the DMD2050 Chassis under any circumstances.



CAUTION!!

Before initially applying power to the unit, it is a good idea to disconnect the transmit output from the operating ground station equipment. This is especially true if the current DMD2050 configuration settings are unknown, where incorrect settings could disrupt existing communications traffic.



*The DMD2050 contains a Lithium Battery. **DANGER OF EXPLOSION** exists if the battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the manufacturer. Dispose of used batteries in accordance with local and national regulations.*

2.1 Unpacking

The DMD2050 Universal Satellite Modem was carefully packaged to avoid damage and should arrive complete with the following items for proper installation:

- DMD2050 Modem Unit
- Power Cord, with applicable AC Connector
- Installation and Operation Manual

2.2 Removal and Assembly

The DMD2050 Modem Unit is shipped fully assembled. It does not require removal of the covers for any purpose in installation.



Always ensure that power is removed from the DMD2050 before removing or installing any optional modules. Failure to do so may cause damage to the equipment.

Carefully unpack the unit and ensure that all of the above items are in the carton. If the available AC mains power at the installation site requires a different cordset from the one included in the package, then a suitable and approved cordset (for the country where the equipment is to be installed) will be required before proceeding with the installation.

Should the Power Cable/AC Connector be of the wrong type for the installation, either the cable or the power connector end should be replaced. The power supply itself is designed for universal AC application. See specifications for appropriate voltages and currents.

2.3 Mounting Considerations

When mounted in an equipment rack, adequate ventilation must be provided. The recommended ambient temperature in the rack should preferably be between 10° and 35°C, and held constant for best equipment operation. The air available to the rack should be clean and relatively dry. The modems may be stacked one on top of the other to a maximum of 10 consecutive units before providing one (1) RU of space for airflow. Modems should not be placed immediately above a high-heat or EMF Generator to ensure the output signal integrity and proper receive operation.

Do not mount the DMD2050 in an unprotected outdoor location where there is direct contact with rain, snow, wind or sun. The only tools required for rack mounting the DMD2050 are four (4) customer supplied rack-mounting screws and the appropriate screwdriver. Rack mounting brackets are an integral part of the front bezel of the unit and are not removable.

2.4 DMD2050 Initial Configuration Check

The DMD2050 is shipped from the factory with preset factory defaults. Upon initial power-up, a user check should be performed to verify the shipped modem configuration. Refer to Section 4, User Interfaces to locate and verify that the following configuration settings are correct:



The DMD2050 Interface Type (MIL-188-114A, G.703, etc.) MUST be selected from the Front Panel BEFORE the mating connectors are installed. Failure to do so may cause damage to the Universal Interface Module. Power up the DMD2050, select the appropriate interface type, and then install the mating connectors.



Transmit (Tx) and Receive (Rx) Interface types are dependent upon the customer's order.

**NOTE**

Implementing Strap Code 26 can set the following modem configuration. Refer to Table 4-4 for an explanation and tabular listing of available Strap Codes. The Frequency and Modulator Output Power are set independently of the strap code.

Standard DMD2050 Factory Configuration Settings**Modulator:**

Data Rate:	2.048 Mbps
Mode:	Closed Network
Satellite Framing:	None
Scrambler:	V.35 (IESS)
Drop and Insert:	Disabled
Inner FEC:	1/2 Rate Viterbi
Outer FEC:	Disabled
Modulation:	QPSK
Frequency:	70.000000 MHz
Modulator Output Power:	-20 dBm

Demodulator:

Data Rate:	2.048 Mbps
Mode:	Closed Network
Satellite Framing:	None
Scrambler:	V.35 (IESS)
Drop and Insert:	Disabled
Inner FEC:	1/2 Rate Viterbi
Outer FEC:	Disabled
Modulation:	QPSK
Frequency:	70.000000 MHz

To lock up the modem, enter 'IF Loopback Enable' under the Test Menu, or connect a Loopback Cable from J1 to J2 on the rear panel of the modem.

2.5 Modulator Checkout

The following descriptions assume that the DMD2050 is installed in a suitable location with prime AC power and supporting equipment available.

2.5.1 Initial Power-Up



Before initial power up of the DMD2050, it is a good idea to disconnect the transmit output from the operating ground station equipment. This is especially true if the current Modulator Configuration Settings are unknown, where incorrect settings could disrupt the existing communications traffic. New units from the factory are normally shipped in a default configuration which includes setting the transmit carrier off.

Turn on the unit by placing the Rear Panel Switch (located above the power entry connector) to the On Position. Upon initial and subsequent power-ups, the DMD2050 Microprocessor will test itself and several of its components before beginning its Main Monitor/Control Program. These power-up diagnostics show no results if successful. If a failure is detected, the Fault LED will illuminate.

The initial field checkout of the modem can be accomplished from the Front Panel or in the Terminal Mode. The Terminal Mode has the advantage of providing full screen access to all of the modem's parameters, but requires a separate terminal or computer running a Terminal Program. The Terminal Mode is enabled from the front panel in the System M&C Submenus.

2.5.2 Factory Terminal Setup

The factory terminal setup is as follows:

Emulation Type:	VT-100 (can be changed)
Baud Rate:	19.2 K (Can be changed via Front Panel)
Data Bits:	8
Parity:	No Parity (Fixed)
Stop Bits:	1 Stop Bit

2.6 Storage

It is recommended that the unit be stored in its original sealed packing. The unit should be stored in a dry location where the temperature is stable, away from direct contact with rain, snow, wind, sun, or anything that may cause damage.



Theory of Operation

3

3.0 DMD2050 Hardware

The DMD2050 is based on a two printed circuit card (minimum configuration) design with additional optioned printed circuit cards available for additional features. The minimum configuration consists of an L-Band/IF Assembly and a Digital Baseband Assembly. The optional printed circuit cards include a Turbo Codec printed circuit card and one of several types of Interface printed circuit card (refer to Appendix A). A block diagram of the DMD2050 is shown in Figure 3-1.

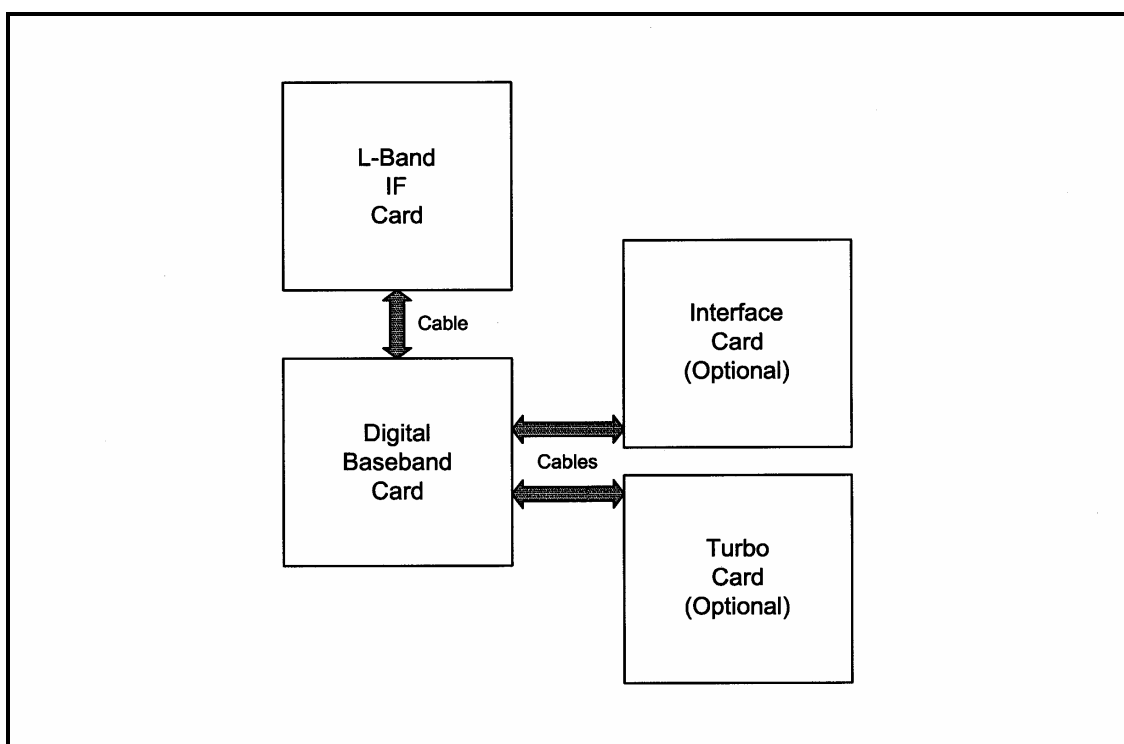


Figure 3-1. DMD2050 Block Diagram

3.0.1 DMD2050 L-Band/IF Printed Circuit Card

The L-Band/IF Printed Circuit Card consists of an analog modulation function, an analog complex downconversion, and two wide-band digital synthesizers. The block diagram of the L-Band/IF Assembly is shown in Figure 3-2.

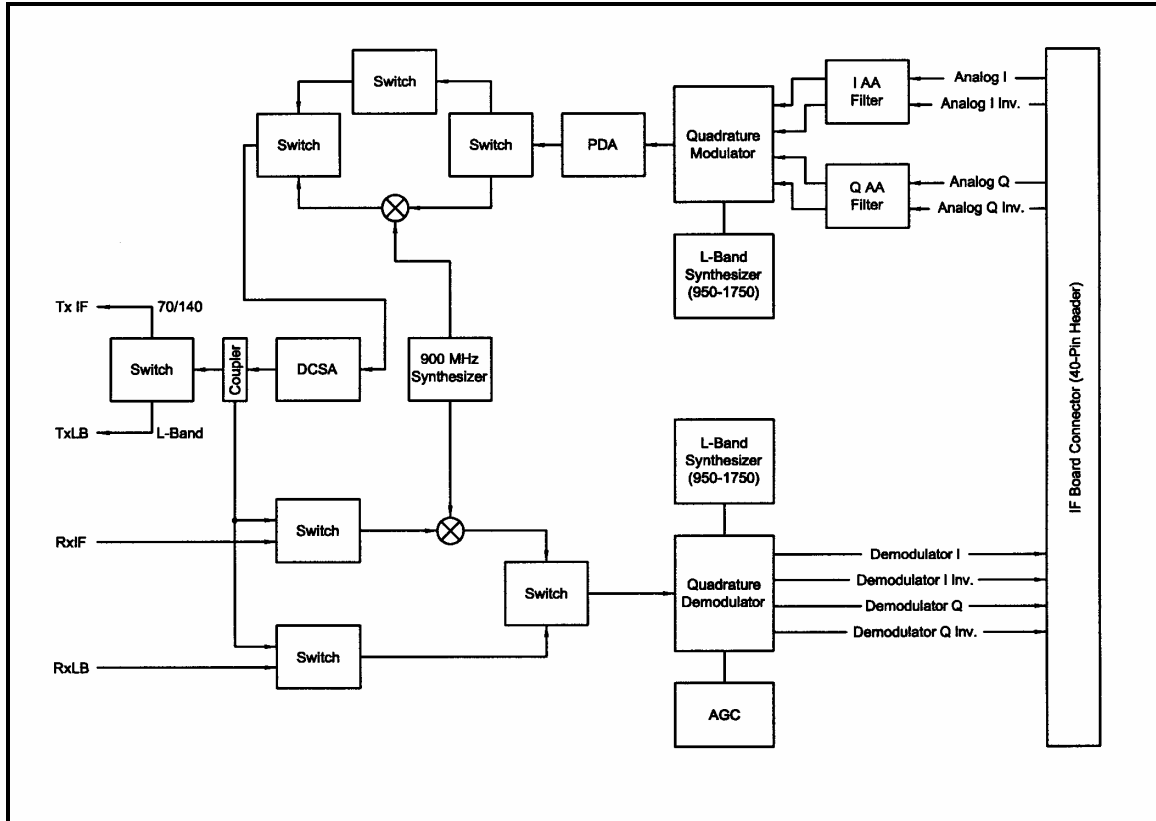


Figure 3-2. IF Card Block Diagram

In the modulator, analog in-phase (I) and quadrature (Q) signals are generated on the Digital Baseband Printed Circuit Card, routed to the L-Band/IF Printed Circuit Card, and modulated at the desired frequency. The L-Band or 70/140 modulated signal is then passed through a microprocessor controlled variable attenuator providing gain control of the output signal.

In the complex downconverter, the signal for demodulation is amplified and sent through a variable wideband attenuator for AGC. The gain-controlled signal is then passed through a complex downconverter to a low IF.

3.0.2 DMD2050 Baseband Processing Printed Circuit Card

The advent of million-plus gate count FPGAs, advanced logic synthesis tools, and DSPs providing hundreds of MIPs enabled the design of a software configurable modem. Large, fast FPGAs now provide designers with what is essentially an on the fly programmable ASIC. High speed, complex digital logic functions that previously could only be implemented in dedicated integrated circuits are now downloaded from a micro-controller through a serial or peripheral interface. When a new digital logic function is needed, a new configuration file is loaded into the FPGA. There is no limit to the number of digital logic configurations available to the FPGA, aside from the amount of Flash memory available to the system microprocessor for storage of configuration files.

The DMD2050 Baseband Processing Printed Circuit Card provides a flexible architecture that allows many different modes of terrestrial and satellite framing, various FEC options, digital voice processing, and several different modulation/demodulation formats. Also included on the Baseband Printed Circuit Card are three synchronous interfaces, an EIA-530 Interface supporting MIL-188-114A and RS-422. All three interfaces are provided on the same DB-25 Connector, and are selectable from the front panel.

A block diagram of the Baseband Processing Card is shown in Figure 3-3.

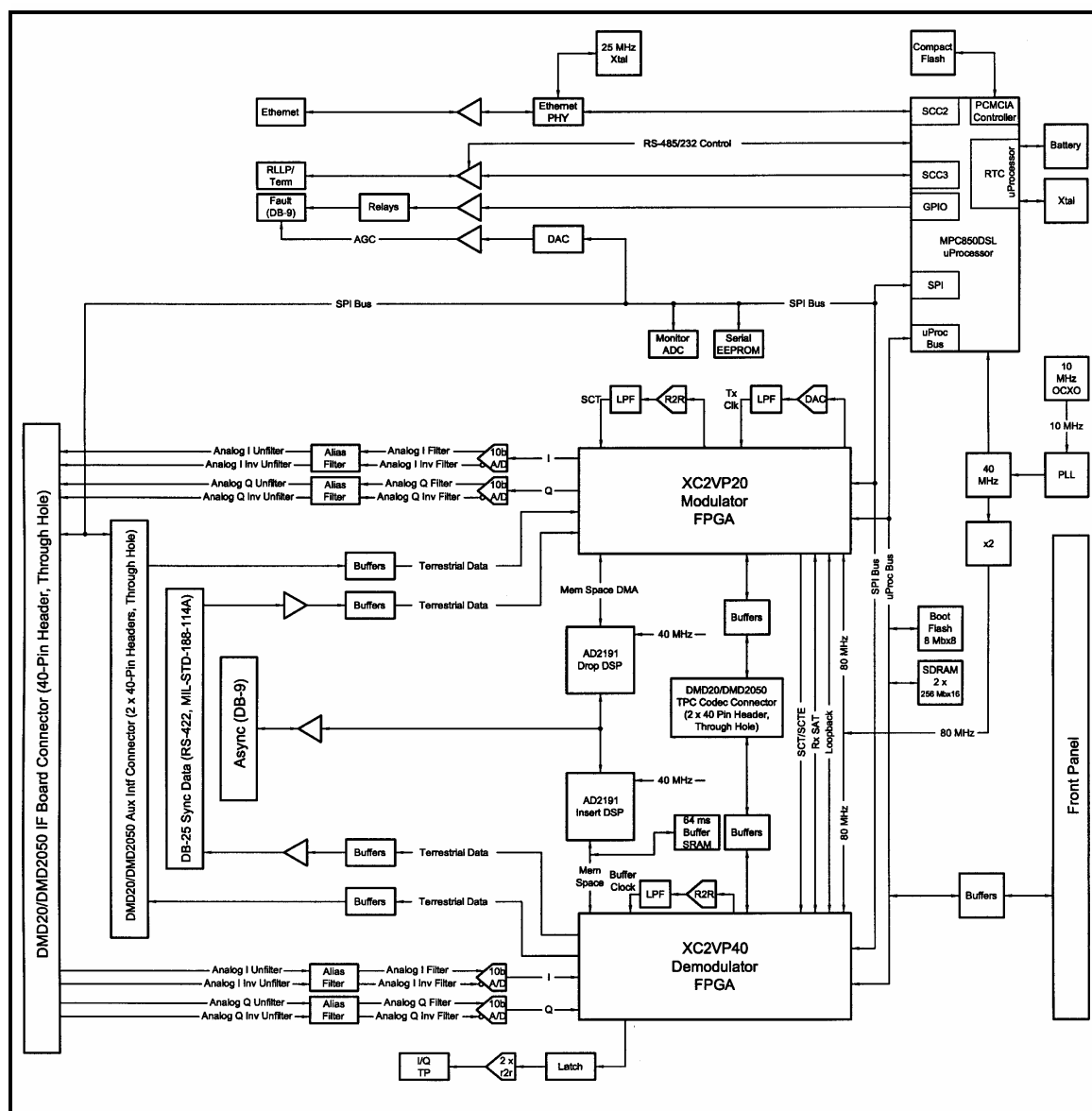


Figure 3-3. DMD2050 Baseband Processing Card Block Diagram

The Baseband Printed Circuit Card also contains the Monitor and Control (M&C) Circuitry responsible for:

- Programmable part setup and initialization
- Continuous control and adjustment of some functions
- Calibration
- Monitoring fault status
- Calculating and displaying measurements
- Calculations
- User monitor and control interface including front panel and remote
- Unit's configuration and feature set

The M&C System is based on a powerful microprocessor with a large amount of Flash memory. Several bus architectures are used to interconnect the M&C to all components of the DMD2050. Communication to the outside world is done via connections to the remote port, terminal port, Ethernet port, and alarm ports. The M&C runs off software programmed into its Flash memory. The memory can be reprogrammed via the Ethernet port to facilitate changes in software.

3.0.3 Enhanced Interface Printed Circuit Card

The normal terrestrial data for the Baseband Processing Card can be re-routed to the enhanced interface card. The enhanced interface card adds a variety of connections to the modem for additional applications

3.1 DMD2050 Functional Block Diagram

Figure 3-4 represents the DMD2050 Functional Blocks. The modem is shown in a typical application with customer data, Tx/Rx RF equipment and an antenna.

3.1.1 Front Panel

The Front Panel includes a 2 x 16 vacuum fluorescent display, Indicator LEDs, and a Numeric Keypad (refer to Section 4.1).

3.1.2 Baseband Processing

The Baseband Processor performs all of the functions required for an IBS/IDR Framing Unit, a Reed-Solomon Codec, and an E1/T1 Drop and Insert System. In addition, the Baseband Processing Section provides for transmit clock selection and rate adaptation as well as a rate adapter and Plesiochronous/Doppler (PD) Buffer in the receive direction. A multiplexer is also provided for the SCT Clock Source for Loop Timing Applications. The transmit and receive paths may be configured independently under processor control.

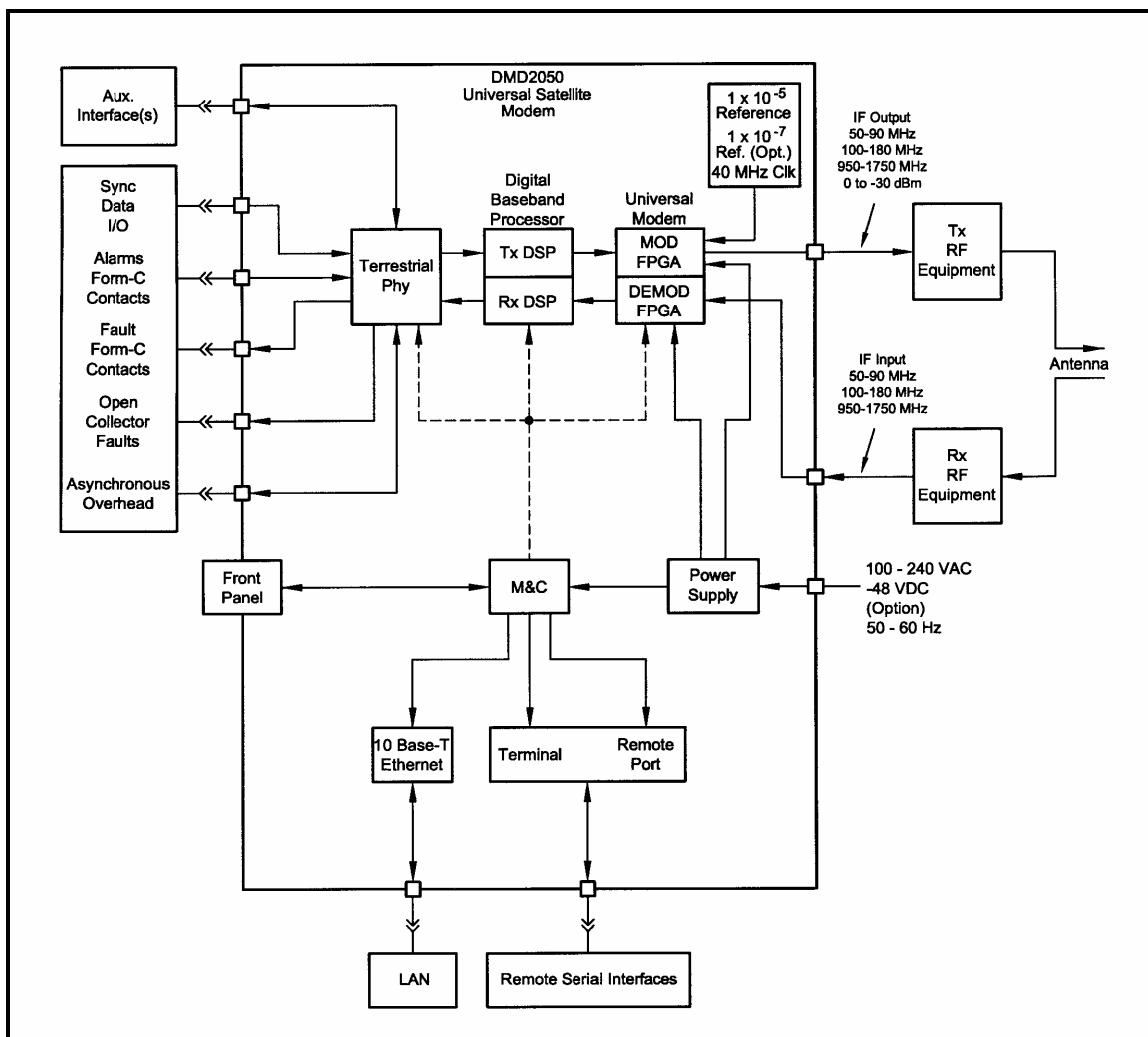


Figure 3-4. DMD2050 Universal Satellite Modem Functional Block Diagram

3.1.3 Tx Baseband Processing

The Tx Data and Clock enters the Baseband Processor, passes through a Rate Adapting FIFO and enters the Framer/Drop Processor. In Closed-Net Mode, the data passes through the framer unaltered. In IDR, IBS, and D&I Modes, the framer adds the appropriate framing and ESC as defined in IESS-308 and 309. In D&I Mode, the framer acquires the terrestrial framing structure, E1 or T1, and synchronizes the Drop Processor. The Drop Processor extracts the desired time slots from the terrestrial data stream and feeds these channels back to the framer. The framer then places the 'dropped' terrestrial time slots into the desired satellite channel slots. The data is then sent to the Reed-Solomon Encoder.

The Reed-Solomon Encoder, encodes the data into Reed-Solomon Blocks. The blocks are then interleaved and synchronized to the frame pattern as defined by the selected specification (IESS-308, IESS-309, DVB, etc.). After Reed-Solomon Encoding, the composite data and clock are applied to the BB Loopback Circuit.

3.1.4 Rx Baseband Processing

The Receive Processor performs the inverse function of the Tx Processor. Data received from the satellite passes through the BB Loopback Circuit to the Reed-Solomon Decoder to the Deframer. The Deframer acquires the IBS/IDR/DVB frame, synchronizes the Reed-Solomon Decoder and extracts the received data and overhead from the frame structure, placing the data into the PD Buffer, sending the overhead data to the UIM. In Closed-Net Mode, the data is extracted from the buffer and is sent to the UIM. Backward Alarm indications are sent to the M&C Subsystem. In Drop and Insert Mode, the Insert Processor synchronizes to the incoming terrestrial T1/E1 Data Stream, extracts satellite channels from the PD Buffer, and then inserts them into the desired terrestrial time slots in the T1/E1 Data Stream.

3.1.5 Monitor & Control (M&C) Subsystem

Also contained on the BB Card is the M&C Subsystem. The M&C contains a high-performance microprocessor and is responsible for overall command and control of modem functions. The M&C is constantly monitoring all subsystems of the modem by performing a periodic poll routine and configures the modem by responding to commands input to the system. During each poll cycle, the status of each of the subsystems is collected and reported to each of the external ports and Front Panel. Performance statistics such as Eb/No, buffer fill %, etc. are compiled. If faults are detected, the M&C will take appropriate actions to minimize the effect of such faults on the system (refer to the Fault Matrices in Section 6).

The following sections describe the M&C Subsystem features:

3.1.5.1 Remote/Terminal Port

This port functions as the Modem Remote Port or Terminal Port. This port supports an asynchronous control protocol as described in Section 4. It may be configured to support either RS-232 or RS-485 signal levels. This port is intended for use in computer-based remote M&C. All functions of the modem may be monitored and controlled from this port via a common terminal connected to the Terminal Port. This function is front panel selectable.

3.1.5.2 Async Port

This port is dedicated for ES-ES Communications. The port may be configured for a number of communications protocols. Overhead data to/from the UIM is routed to/from the framer/deframer. This port may be configured to support either RS-232 or RS-485 signal levels. This port is also used by SCC Framing for the in-band data.

3.1.5.3 Ethernet Port

This port is dedicated for Ethernet Communications. The port is configured for 10 Base-T communications protocols.

3.1.5.4 Clock

The time and date is kept in order to 'time-tag' system events.

3.2 Earth Station to Earth Station Communications Port

The modem contains a selectable RS-232, or RS-485 Asynchronous Communications Port for Earth Station-to-Earth Station Communications. The baud rate and protocol can be selected from the Front Panel.

3.3 Terrestrial Loopback

The modem provides for terrestrial loopback. For Tx Terr Loopback the Tx Data port is looped back to the Rx Data port after the interface driver/receiver. For RX Terr Loopback, the Receive Data from the satellite is looped back for retransmission to the satellite, providing a far end loopback. Tx/Rx Loopback provides both loopbacks simultaneously. Refer to Figures 3-6 through 3-8 for loopback functional block diagrams.

3.4 Modem Status

The modems M&C system is connected to most of the circuitry on any board contained in the chassis. These connections provide status on the working condition of the circuitry as well as providing the data required for the various measurements the modem provides. The M&C processes this information and generates status indications as well as alarms when necessary. Detailed status information is available via the modems various user interfaces (front panel, remote and terminal). A summary of this information can be connected to external equipment, switches or alarms via the open collector and/or form-C fault connections.

Form-C Contacts:

The UIM provides three Form-C Relays under processor control that appear at J11.

Mod Fault:	De-energized when any transmit side fault is detected.
Demod Fault:	De-energized when any receive side fault is detected.
Common Fault:	De-energized when any fault that is not explicitly a Tx or Rx Fault such as an M&C or Power Supply Fault.

Open Collector Faults:

The UIM provides two Open Collector Faults that appear at Pins 28 & 10 on J8.

Mod Fault:	Will sink up to 20 ma (maximum) until a transmit or common fault is detected. Will not sink current if a fault is detected.
Demod Fault:	Will sink up to 20 ma (maximum) until a receive or common fault is detected. Will not sink current if a fault is detected.

The open collector faults are intended for use in redundancy switch applications in order to provide quick status indications.

3.5 Baseband Processor Card

The Baseband Processor Card (BB Card) contains two major subsystems – the Baseband Processing System and the Monitor and Control Subsystem.

3.5.1 Baseband Processing

The Baseband Processor performs all of the functions required for an IBS/IDR Framing Unit, a Reed-Solomon Codec, an E1/T1 Drop and Insert System, a Turbo Codec, and Sequential/Viterbi. In addition, the Baseband Processing Section provides for Transmit clock selection and rate adaptation as well as a rate adapter and Plesiochronous/Doppler (PD) Buffer in the receive direction. A multiplexer is also provided for the SCT Clock Source for Loop Timing Applications. The transmit and receive paths may be configured independently under processor control.

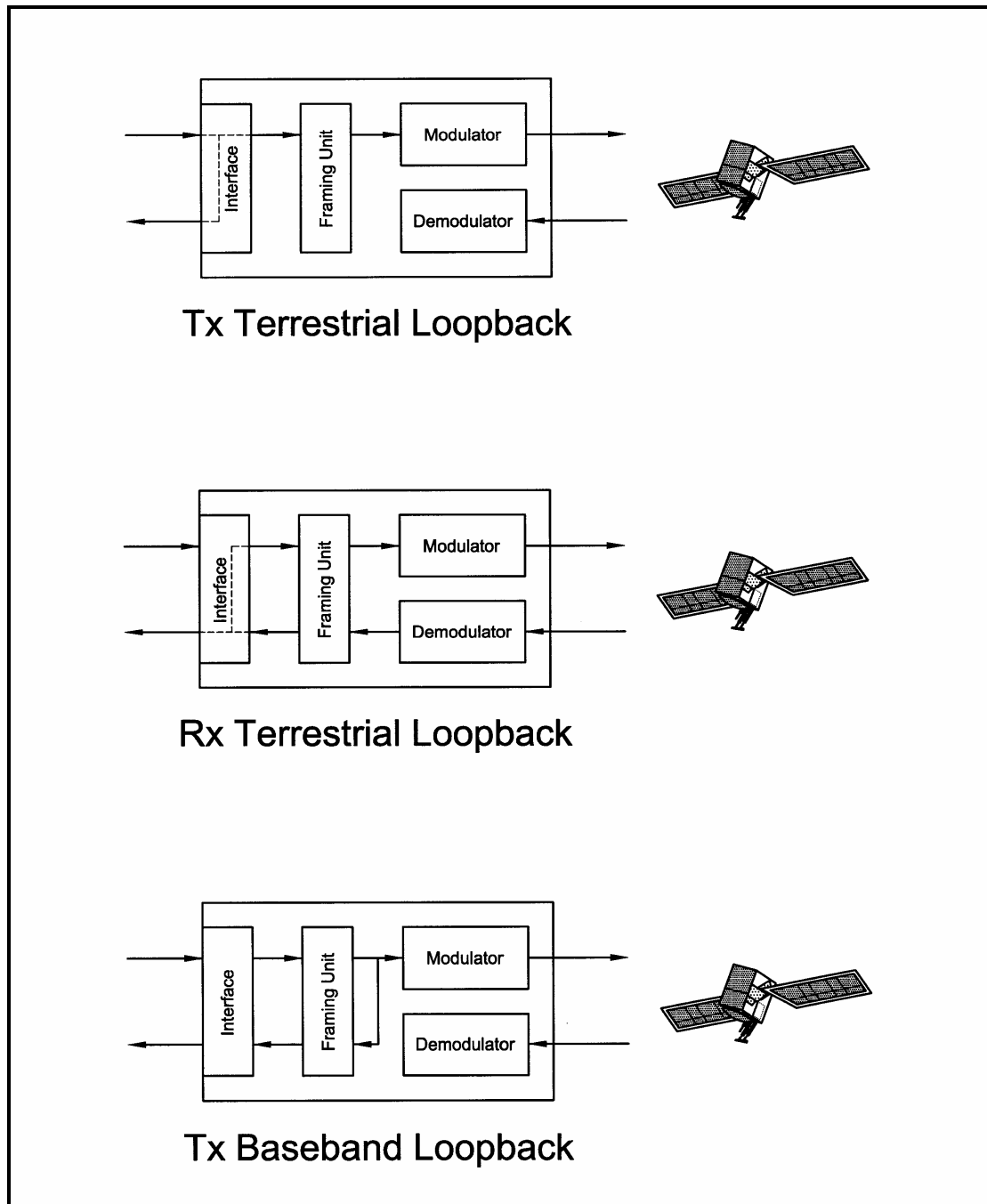


Figure 3-6. Loopback Functional Block Diagram

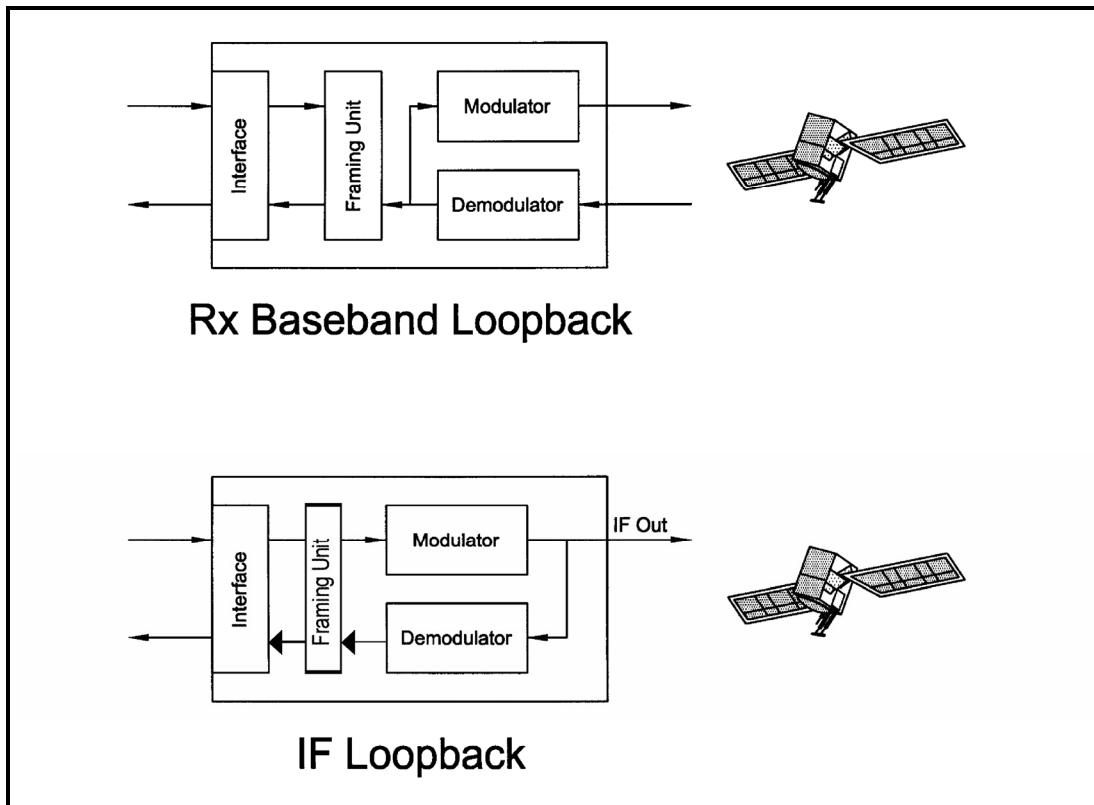


Figure 3-7. Loopback Functional Block Diagram

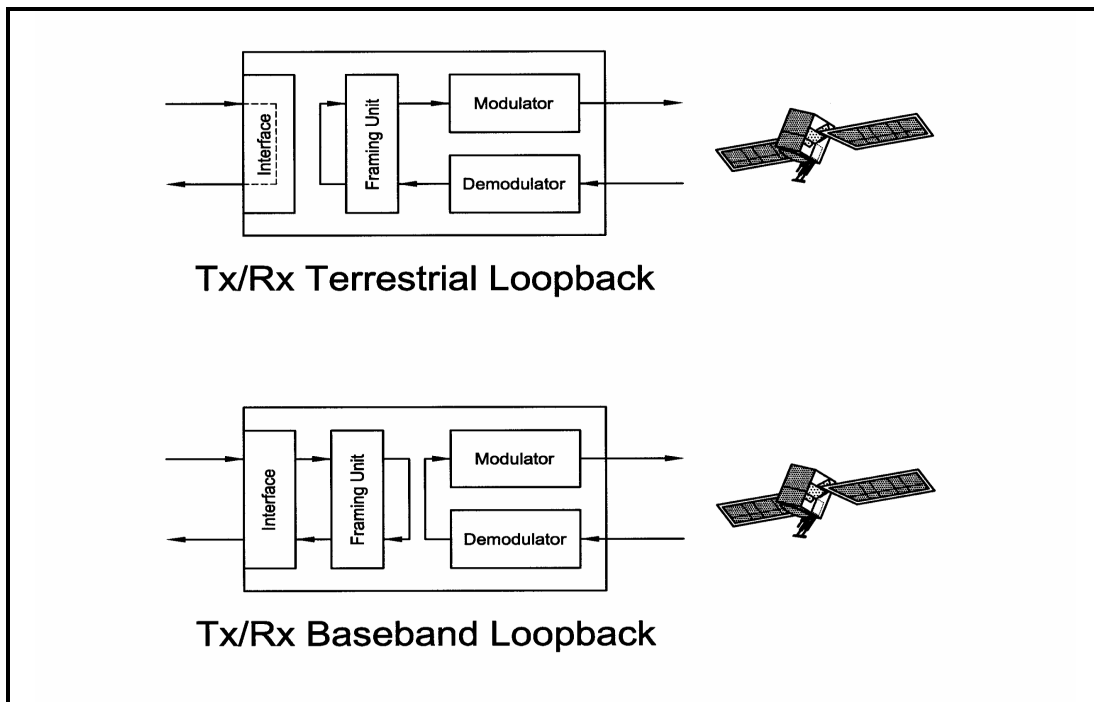


Figure 3-8. Loopback Functional Block Diagram

3.6 DMD2050 Clocking Options

The following paragraphs define the types of clocking options available to the user at the Front Panel of the DMD2050. Refer to Figure 3.9 for clocking and polarity.

3.6.1 Clock Selection

As shown in Figure 3-9, both the Tx Clock and the Buffer Clock source may be independently locked. Clock sources may be:

- SCT (Internal Oscillator)
- SCTE (External Tx Terrestrial Clock)
- EXT BNC (External Clock Source)
- Rx Satellite Clock (Loop Timing)
- EXT IDI (Drop and Insert Clock)

For loop timing applications the SCT Clock Source can be selected to be Rx Satellite Clock.

DMD2050 CLOCKING AND POLARITY

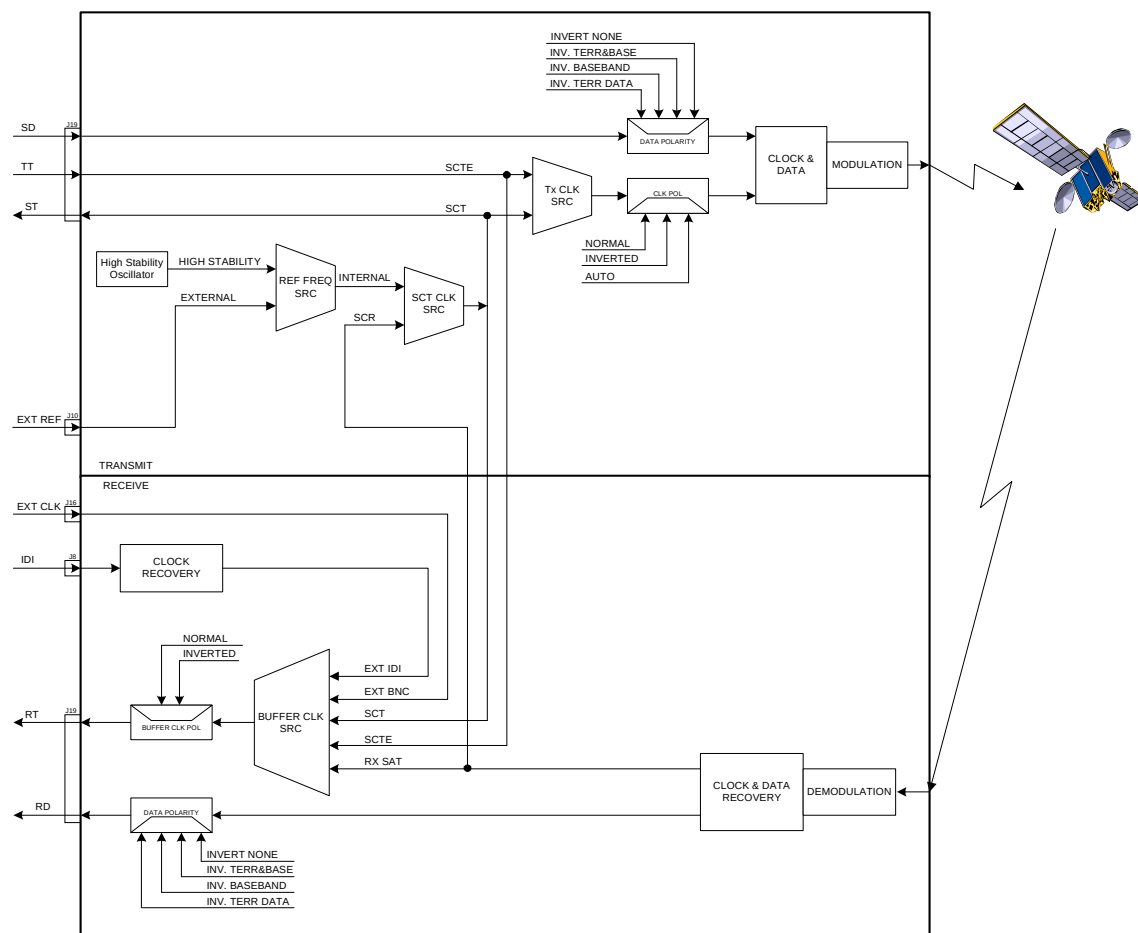


Figure 3-9. Clocking and Polarity Diagram

3.6.2 SCTE: Serial Clock Transmit External

This clock is the Transmit Terrestrial Clock associated with the interface. With the G.703 Interface selected, SCTE is the clock that is recovered from the G.703 data stream. SCTE is sometimes referred to as Tx Terrestrial Timing and for Synchronous Interfaces such as RS-422, SCTE is sometimes referred to as TT (Terminal Timing).

3.6.3 SCT: Serial Clock Transmit

This clock is an internally generated clock that is output from the modem. The clock is generally used by the Terrestrial Terminal equipment for clocking the transmit data. The frequency of the clock is set the same as the Transmit Terrestrial Clock rate if internal is selected, or is the receive

3.6.4 EXT CLK: External Clock, J16

This is an independent clock source. This clock is most often used if there is a station master clock. The EXT EXC can be selected, in the Interface/General Menu, to be None, BNC EXC, or IDI. IDI is used ONLY for D&I cases where external framing is selected. In this case, the EXT EXC must be set to IDI where the Receive Buffer Clock is derived from the external Receive T1 or E1 Trunk.

3.6.5 BNC EXC: BNC External Clock, j16

Unbalanced external clock input into a BNC connector. This clock can be used to source the EXT CLK.

Clock specification:

Frequency:	1 MHz – 20 MHz
Level:	0dBm $\pm$ 3dB

3.6.6 IDI: Insert Data In

This clock source is only used as an external frame source selected in D&I Mode. If External Frame Source is selected, then IDI **must** be selected for the buffer clock. For this case, a Receive T1/E1 Trunk is input and a buffer clock is derived.

3.6.7 SCR: Serial Clock Receive

This Receive Clock is recovered from the satellite's receive signal from the satellite. SCR is sometimes referred to as Receive Clock, Satellite Clock, or as RT (Receive Timing).

3.6.8 EXT IF REF: External IF Reference

This is not actually a clock, but does have some clocking implications. When the external reference is used, the master oscillator within the DMD2050 is locked to the external reference, and the internal accuracy and stability of the DMD2050 assumes that of the External Reference. Therefore, not only are the transmit frequencies of the DMD2050 locked to the external reference, but the modem's internal SCT Oscillator is locked to the external reference as well.

3.7 Transmit Timing

As shown in Figure 3-7, Transmit Terrestrial Data enters the modem and is clocked into a dejitter FIFO. Data is clocked out of the FIFO by the Modulator Clock. The Modulator Clock and Phase-Locked Loop (PLL), in conjunction with the Dejitter FIFO, reduces the input jitter. Jitter reduction exceeds the jitter transfer specified in CCITT G.821.

3.7.1 EXT CLK as TX Clock Source (MIL-188-114A, RS-422)

Data must be clocked into the modem by either the SCTE or SCT Source. If EXT CLK is selected as the Tx Clock Source, then SCTE must be supplied to the modem. The output of the dejitter buffer will be clocked with EXT CLK. This case should only be used if SCTE has excessive jitter and will degrade link performance.

3.7.2 SCT or SCTE

If SCT is selected, then only data that is synchronous to the SCT Clock is required to be supplied to the modem. It is intended for the terminal equipment to use the SCT as its clock source. The Autophase Circuit will automatically ensure that the data is clocked correctly into the modem. Therefore, a return clock is not necessary. The Clock Polarity should be set to Auto.

If SCTE is selected, then SCTE **must** be supplied to the modem. The Clock Polarity should be set to AUTO.

3.7.3 G.703 Interface

If the G.703 Interface is selected, then the Tx Clock Source will default to SCTE and the Clock Polarity will default to Auto.

3.7.4 Ethernet Data Interface (Optional)

If the Ethernet Data Interface is selected, then the Tx Clock Source will default to SCTE and the Clock Polarity will default to Normal.

3.8 Receive Timing

Any of the clocking selections, SCTE, SCT, EXT CLK, or RxSat (SCR) may be selected as the Buffer Clock. Data will be clocked out of the buffer at the data rate synchronous to the selected clock source.

3.9 Loop Timing

If loop timing is desired (i.e.; the modem timing is slaved to the far end master station), the modem clocks can be configured as follows:

3.9.1 Transmit (MIL-188-114A, RS-422)

Set SCT Source to 'SCR'. The Tx Terminal Equipment must clock the TX Data with the SCT Clock and return data and SCTE (Optional). If SCTE is returned to the modem from the terminal equipment, set TX CLK SRC to SCTE. If SCTE is not returned to the modem, set TX CLK SRC to SCT. The TX CLK POL should be set to AUTO.

3.9.2 G.703 Interface or Asymmetrical Data Rates

Loop timing with a G.703 Interface or Asymmetrical Data Rates requires external equipment at the remote end that is capable of using the recovered RD Clock as source timing for (SCTE) SD. The modem will not manipulate the clock frequency. Therefore, the transmit and receive clock rates must be equal in order for the modem to perform loop timing.

3.9.3 Receive

Select the Buffer clock to RxSAT (SCR).

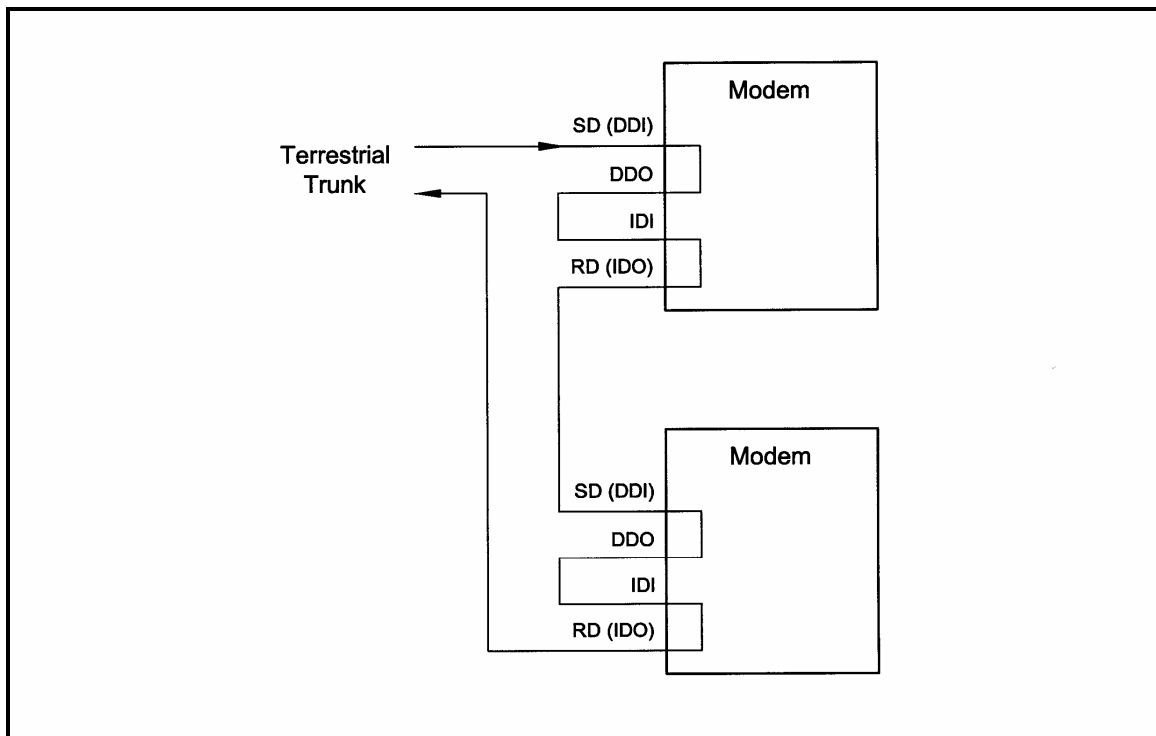
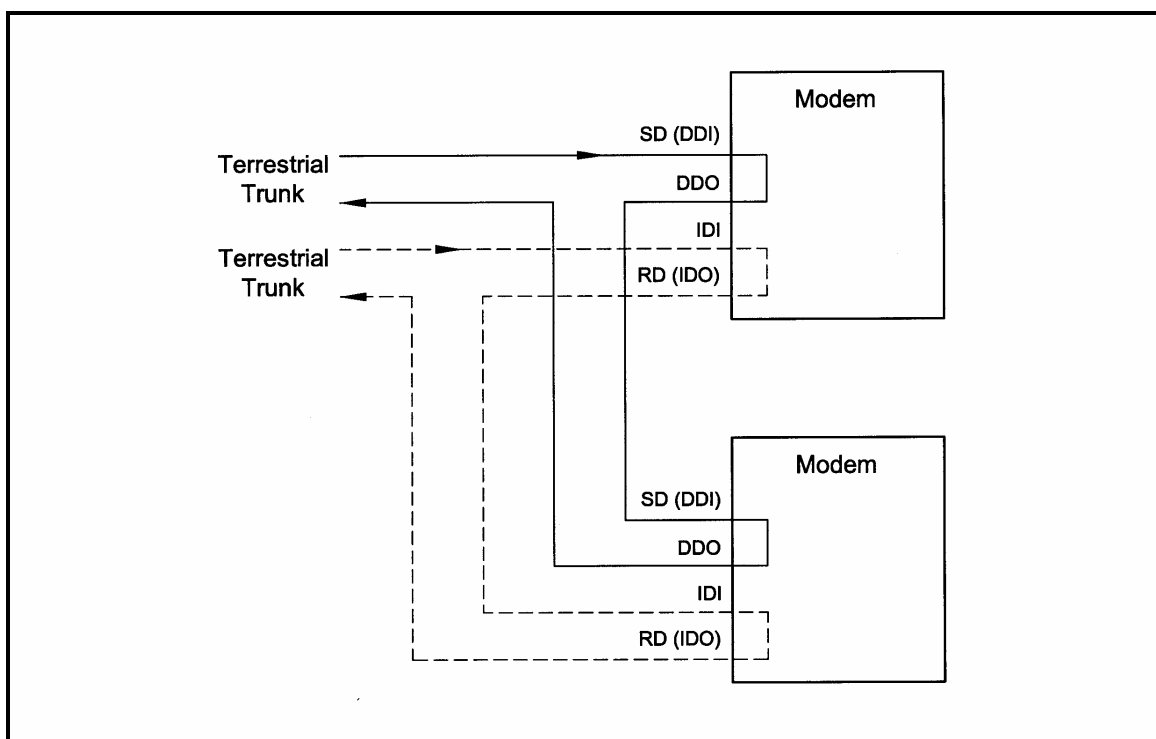
3.10 Drop and Insert (D&I)

The Radyne DMD2050 Drop and Insert (D&I) function provides an interface between a full T1 or E1 Trunk whose framing is specified in CCITT G.704 and a fractional Nx64 Kbps Satellite Channel that conforms to the IBS and small IDR Framing Structures. For information pertaining to Radyne's proprietary Drop & Insert, refer to Appendix E.

The Drop function allows the user to select the terrestrial T1 or E1 timeslots that are to be dropped off for transmission over the link in the specified satellite channels. The Insert function allows the user to select the T1 or E1 timeslots into which the received satellite channels are to be inserted. The two functions are completely independent allowing maximum flexibility in choosing configurations. The four-port G.703 Interface allows one or more modems to be looped together using the same T1 or E1 trunk.

The Transmit Data Trunk is brought into the modem via the Send Data In (SDI) Port. From there, the TX Baseband Processor extracts the selected timeslots from the G.704 Frame and prepares them for transmission. The original trunk data is sent out of the modem unaltered via the Send Data Out (SDO) Port. The Receive Data Trunk is brought into the modem via the Insert Data In (IDI) Port. The data is buffered inside the modem and the RX Baseband Processor inserts satellite data into the selected timeslots in the G.704 Frame. The modified terrestrial trunk is then output via the Receive Data Out (RDO) Port.

Figure 3-10 shows two modems looped together. This configuration could be simplified to just use one modem, or extended to use more than two modems. Figure 3-11 shows an alternative method of looping where all of the drop (transmit) data is processed prior to performing any insert (receive) processing. In both configurations, the terrestrial trunk is providing the timing for the satellite transmission and for the terrestrial receive.

**Figure 3-10. Looped Modems****Figure 3-11. Looped Modems with Separate D&I Trunks**

3.10.1 Drop Only

When Drop is enabled and Insert is disabled, the DMD2050 performs a drop-only function. Framed E1 or T1 Data is input via the Send Data In Port, the selected timeslots are dropped into the IBS frame structure, and the unaltered terrestrial data is output via the Send Data Out Port (refer to Figure 3-12).

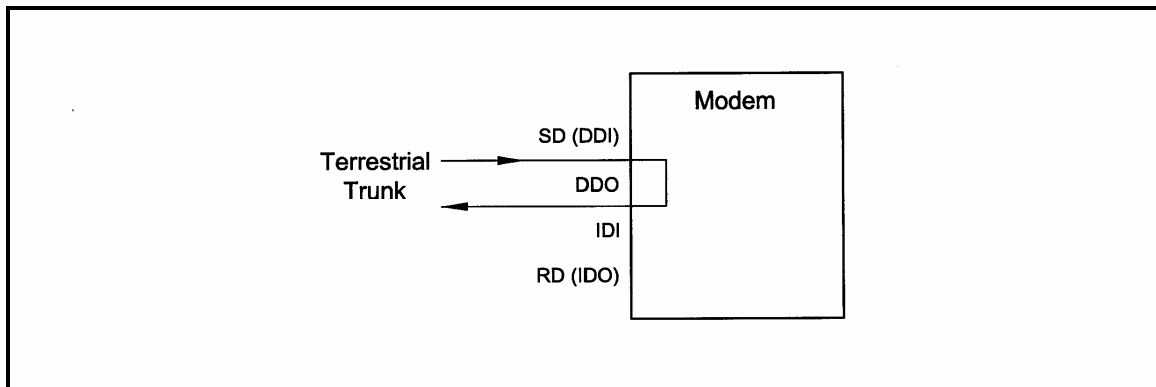


Figure 3-12. Drop Only

3.10.2 Insert Only

When Insert is enabled and Drop is disabled, the DMD2050 performs an insert-only function. If framed terrestrial E1 or T1 Data is available, it should be input via the Insert Data In Port. The Terrestrial Data is buffered inside the Modem. The RX Baseband Processor inserts satellite data into the selected timeslots in the G.704 Frame and the modified terrestrial data is then output via the Receive Data Out Port (refer to Figure 3-13).

If framed terrestrial data is not available, selection of the Internal T1/E1 frame source will cause the modem to generate the required G.704 Frame. The Satellite Data will be inserted into the selected timeslots, and the resulting terrestrial data will be output via the Receive Data Out Port. Any non-inserted timeslots in the G.704 Frame will be filled with the appropriate Idle Code (refer to Figure 3-14).

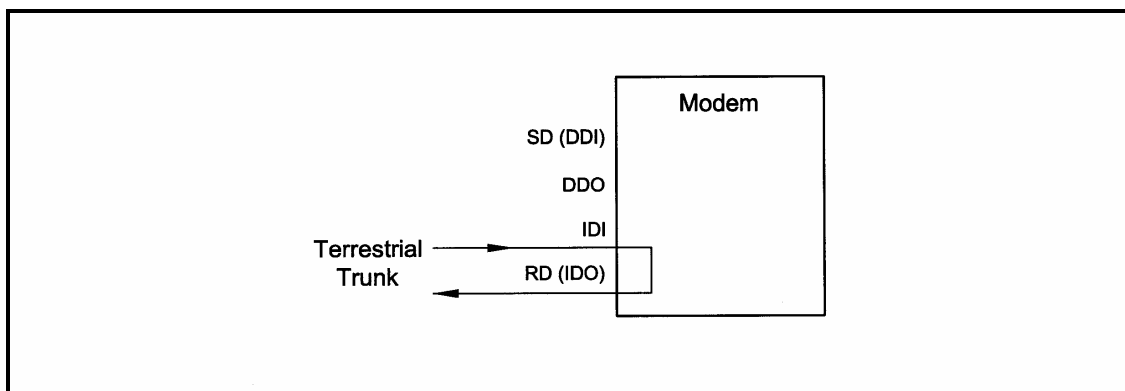


Figure 3-13. Insert Only with External Frame Source

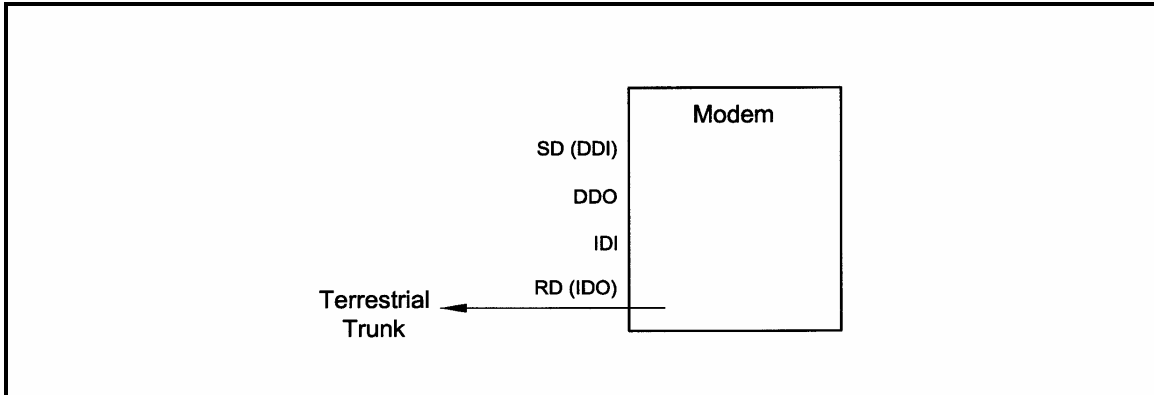


Figure 3-14. Insert Only with Internal Frame Source

3.11 Mode Selection

The DMD2050 D&I can be easily configured to support several commonly used terrestrial data formats. For E1 Data, the user can choose between PCM-30, PCM-30C, PCM-31 and PCM-31C. For T1 Data, the user can choose between T1-D4, T1-ESF, and SLC-96. The following paragraphs provide more information on the various mode selection capabilities of the DMD2050.

3.11.1 PCM-30

The PCM-30 Mode of Operation supports an E1 Interface with Multiframe Alignment (MFAS) and Channel Associated Signaling (CAS). The user may independently program n timeslots to drop and n timeslots to insert where $n = 1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 16, 20, 24, \text{ or } 30$. In addition to the selected drop timeslots, the Transmit Function also extracts the appropriate ABCD signaling bits from terrestrial timeslot 16 for transmission in IBS Frame as required. Conversely, the Receive Function extracts received ABCD signaling bits from the IBS Frame and inserts them in timeslot 16 of the appropriate terrestrial frame. This transmission and reception of ABCD signaling based upon the drop and insert timeslots is performed automatically and is transparent to the user. In PCM-30 mode, the user may *not* select timeslot 16 as a Drop or Insert Timeslot.

3.11.2 PCM-30C

The PCM-30C Mode of Operation supports an E1 Interface with Multiframe Alignment (MFAS) and Channel Associated Signaling (CAS). In addition, the Drop function verifies the received terrestrial CRC checksum and the Insert function calculates the required CRC checksum. The user may independently program n timeslots to drop and n timeslots to insert where $n = 1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 16, 20, 24, \text{ or } 30$. In addition to the selected Drop timeslots, the Transmit Function also extracts the appropriate ABCD signaling bits from terrestrial timeslot 16 for transmission in IBS Frame as required. Conversely, the Receive Function extracts received ABCD signaling bits from the IBS frame and inserts them in timeslot 16 of the appropriate terrestrial frame. This transmission and reception of ABCD signaling based upon the Drop and Insert timeslots is performed automatically and is transparent to the user. *In PCM-30C Mode, the user may not select timeslot 16 as a Drop or Insert Timeslot.*

3.11.3 PCM-31

The PCM-31 Mode of Operation supports an E1 Interface with no Multiframe Alignment (MFAS) or Channel Associated Signaling (CAS). The user may independently program n timeslots to drop and n timeslots to insert where $n = 1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 16, 20, 24, \text{ or } 30$. Because there is no implied ABCD signaling, the user is free to select timeslot 16 as a Drop or Insert Timeslot.

3.11.4 PCM-31C

The PCM-31C Mode of Operation supports an E1 Interface with no Multiframe Alignment (MFAS) or Channel Associated Signaling (CAS). In addition, the Drop Function verifies the received terrestrial CRC checksum and the Insert Function calculates the required CRC checksum. The user may independently program ' n ' timeslots to drop and ' n ' timeslots to insert where ' n ' = 1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 16, 20, 24, or 30. Because there is no implied ABCD signaling, the user is free to select timeslot 16 as a Drop or Insert Timeslot.

3.11.5 T1-D4/T1-D4-S

The T1-D4 Mode of Operation supports a T1 Interface with 12 frames per multiframe. The user may independently program n timeslots to drop and n timeslots to insert where $n = 1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 16, 20, 24, \text{ or } 30$. In the DMD2050, Robbed Bit Signaling (RBS) is handled without any need for operator intervention and is transparent to the user.

3.11.6 T1-ESF/T1-ESF-S

The T1-ESF Mode of Operation supports a T1 Interface with 24 frames per multiframe. The CRC-6 checksum is automatically checked by the Drop Function and generated by the Insert Function and placed in the appropriate F-bit positions in the terrestrial multiframe. The user may independently program n timeslots to drop, and n timeslots to insert, where $n = 1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 16, 20, 24, \text{ or } 30$. In the DMD2050, Robbed Bit Signaling (RBS) is handled without any need for operator intervention and is transparent to the user.

3.12 Multidestinational Systems

Because the Drop and Insert Functions are completely independent, the DMD2050 easily supports multidestinational communications. Figure 3-15 illustrates a Multidestinational System with one Hub site and three remote sites. At the Hub site, thirty channels are being transmitted to all three remote sites and a fractional set of channels is being received from each remote site. At the other end of the link, each remote site is transmitting a fractional E1 to the Hub site as well as receiving all 30 channels from the Hub site. It also identifies those channels intended for it, and inserts them into the terrestrial data stream.

3.13 Drop and Insert Mapping

The following displays under Interface D&I Setup (both Tx and Rx), are editing displays only:

SATCh TS
Enter to Edit

Any changes made in these displays are made on the screen, **but are not entered into the modem**. Once these menus are configured, the Mapping Menu must be used to actually enter the settings into the modem.

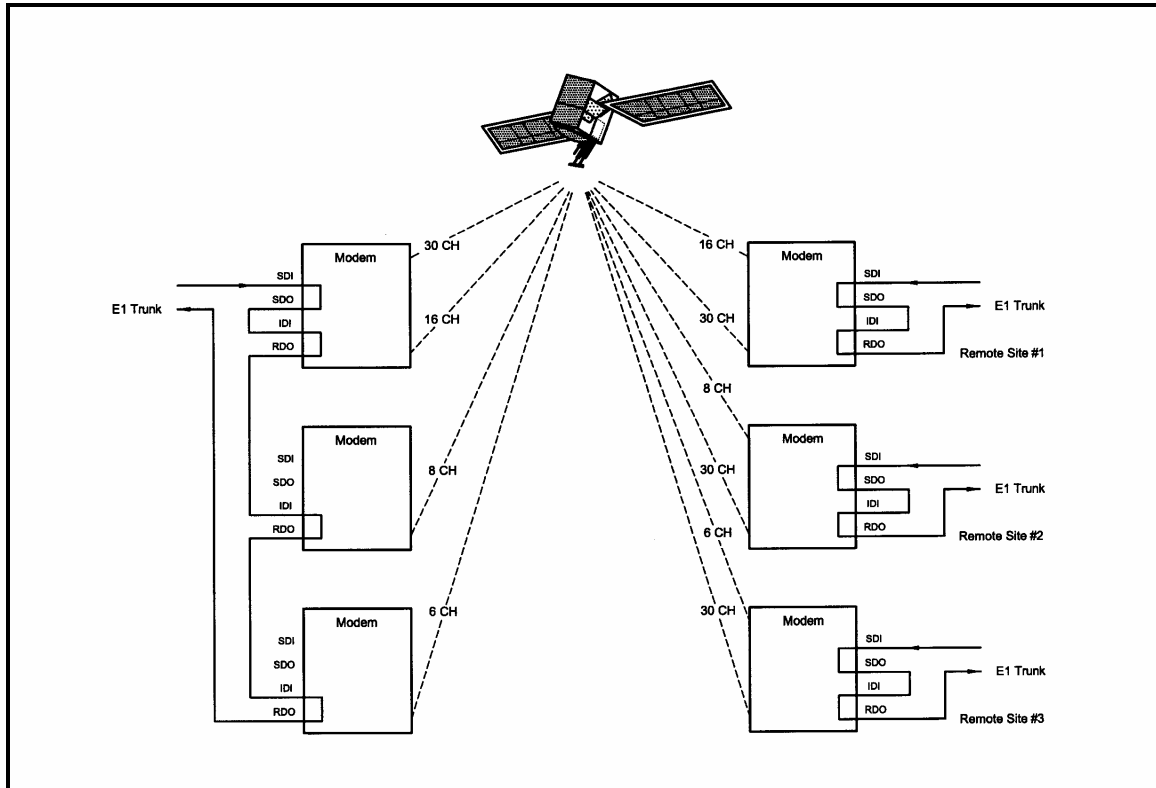


Figure 3-15. Multidestinational Communications

Example:

For a modem w/ Drop & Insert enabled at a data rate of 256 (with timeslots assigned 1 - 1, 2 - 2, etc.). At a data rate of 256, the modem will allow 4 channels to assign timeslots. Under the Tx Menu, assign the timeslots that are to be used to the 4 channels. CH1 is assigned to TS1 (Timeslot #1), CH2 to TS 2, CH3 to TS3 and CH4 to TS4, <ENTER> must be depressed after assigning each individual TS. Once the timeslots are assigned to the channels, use the Left or Right Arrow Key to scroll to the Mapping Menu. This menu will appear in the following way:

```

      Map      Copy
    *****
  
```

This is the menu where the channel assignments are actually entered into the modem. To do this, perform the following steps:

For the Transmit Side:

1. Push <ENTER> to get the flashing cursor.
2. Use the Up Arrow Key to make the left portion of the display read "TX EDIT".
3. Use the Right or Left Arrow Keys to switch the flashing cursor to the right portion of the display.
4. Use the Up or Down Arrow Key to until the right hand portion displays "TX ACTIVE".
5. The mapping display should now look like this:

```

      Map      Copy
    TX EDIT > TX ACTIVE
  
```

6. Push <ENTER> to enter this command. This tells the modem to configure to the settings that were assigned in the Channel/Timeslot display.

For the Receive Side:

1. With Rx Side Channels configured as follows: CH1 to TS1, CH2 to TS2, CH3 to TS3, and CH4 to TS4.
2. After the timeslots are assigned properly, scroll to the Mapping Menu and use the above procedure to enter the settings into the modem.
3. Set the display to read:

```

      Map      Copy
    RX EDIT > RX ACTIVE
  
```

4. Press <ENTER> to enter the settings into the modem.

To View the current Timeslot Assignment:

1. If there is a question of the channels not being entered properly, the Mapping Menu may be used to see how the channels/timeslots are configured in the modem.
2. Use <ENTER> and the Arrow Keys to make the mapping menu read (for the Tx Side):

Map Copy
TX ACTIVE > TX EDIT

3. Press <ENTER>. The modem has now copied the current Tx Settings to the Tx Channel/Timeslot Display.
4. For the Rx Side:

Map Copy
RX ACTIVE > RX EDIT

5. Press <ENTER>. The modem has now copied the current Rx Settings to the Rx Channel/Timeslot display).

**NOTE**

It is not mandatory to assign timeslots in sequential order, although the lowest timeslot must be entered in the lowest channel. For example: timeslots may be assigned 1 - 2, 2 - 5, etc. but not 1 - 5, 2 - 2.

3.14 Reed-Solomon Codec**NOTE**

Refer to Figures 3-16, 3-17, and Table 3-1.

Utilizing a Reed-Solomon (R-S) Outer Codec concatenated with a Convolutional Inner Codec is an effective way to produce very low error rates even for poor signal-to-noise ratios while requiring only a small increase in transmission bandwidth. Typically, concatenating an R-S Codec requires an increase in transmission bandwidth of only 9 – 12% while producing a greater than 2 dB improvement in E_b/N_o . R-S is a block Codec where K data bytes are fed into the encoder which adds $2t = (N - K)$ check bytes to produce an N byte R-S block. The R-S decoder can then correct up to “t” erred bytes in the block.

3.14.1 Reed-Solomon Operation in the DMD2050

When the Reed-Solomon Codec is enabled, data is fed to the R-S Encoding Section of the DMD2050 where it is scrambled, formed into blocks, R-S encoded, and interleaved. Unique words are added so that the blocks can be reformed in the Receiving Modem (Refer to Figures 3-16 and 3-17). Data is then sent to the modulator where it is convolutionally encoded, modulated and transmitted to the satellite.

When the signal is received and demodulated by the Receiving Modem, it is fed to a Viterbi Decoder for the first layer of error correction. After error correction is performed by the Viterbi Decoder, the unique words are located and the data is deinterleaved and reformed into blocks. The R-S Decoder then corrects the leftover errors in each block. The data is then descrambled and output from the R-S Section.

3.14.2 Reed-Solomon Code Rate

The R-S Code Rate is defined by (N, K) where N is the total R-S block size in bytes - data + check bytes - and K is the number of data bytes input into the R-S Encoder. The transmission rate expansion required by the R-S Codec is then defined by N/K . The DMD2050 automatically sets the correct R-S code rate for IDR/IBS open network operation in accordance with the data shown in Table 3-1. In Closed Net Mode, the DMD2050 allows the following N and K setting: (126, 112), (219, 201), (194, 178), (225, 205).

Variable Reed-Solomon rates are available on the optional AS/5167 Super Card. Refer to Appendix A for further information.

3.14.3 Interleaving

The DMD2050 allows for interleaving depths of 4 or 8 R-S Blocks. This allows burst errors to be spread over 4 or 8 R-S blocks in order to enhance the error correcting performance of the R-S Codec. For Intelsat Network Modes, the DMD2050 automatically sets the interleaving depth to 4 for QPSK or BPSK, or 8 for 8PSK. In Closed Network Mode, the interleaver depth can be manually set to 4 or 8, and in DVB Network Mode, the DMD2050 automatically sets the interleaver depth to 12.

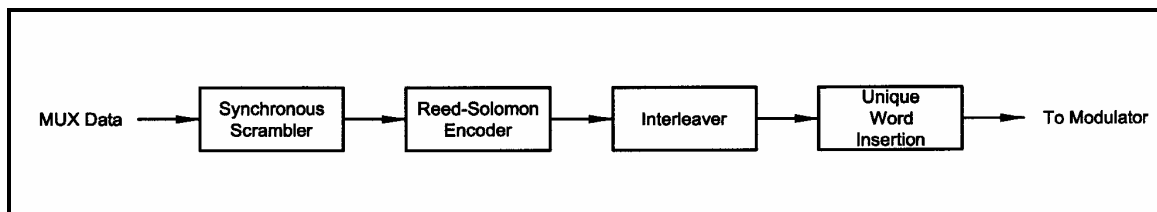


Figure 3-16. Reed-Solomon Encoder Functional Block Diagram

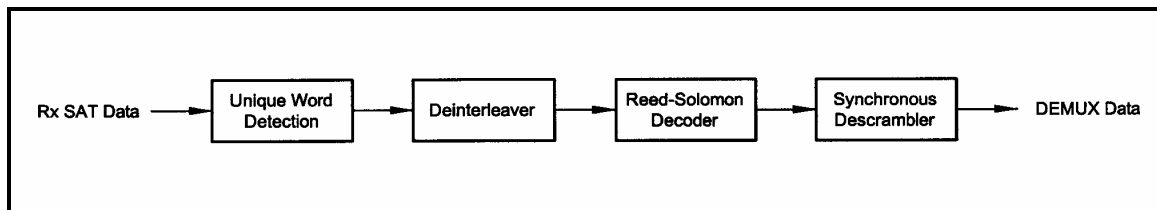


Figure 3-17. Reed-Solomon Decoder Functional Block Diagram

Table 3-1. Reed-Solomon Codes					
Type of Service	Data Rate (Kbps)	R-S Code (n, k, t) ¹	Bandwidth Expansion [(n/k) -1]	Interleaving Depth	Maximum ² R-S Codec Delay (ms)
Small IDR (With 16/15 O/H)	64	(126, 112, 7)	0.125	4	115
	128	(126, 112, 7)	0.125	4	58
	256	(126, 112, 7)	0.125	4	29
	384	(126, 112, 7)	0.125	4	19
	512	(126, 112, 7)	0.125	4	15
	768	(126, 112, 7)	0.125	4	10
	1024	(126, 112, 7)	0.125	4	8
	1536	(126, 112, 7)	0.125	4	5
IDR (With 96 Kbps O/H)	1544	(225, 205, 10)	0.0976	4	9
	2048	(219, 201, 9)	0.0896	4	7
	6312	(194, 178, 8)	0.0899	4	2
	8448	(194, 178, 8)	0.0899	4	<2
8PSK	1544	(219, 201, 9)	0.0896	8	18
	2048	(219, 201, 9)	0.0896	8	13
	6312	(219, 201, 9)	0.0896	8	4
	8448	(219, 201, 9)	0.0896	8	3
DVB	All	(204, 188, 8)	0.0851	12	-
1. <i>n</i> = code length, <i>k</i> = information symbols and <i>t</i> = symbol error correcting capability. 2. Design objective.					

3.15 DMD2050 Automatic Uplink Power Control (AUPC Operation)

The DMD2050 Modem has an optional built-in provision for Automatic Uplink Power Control (AUPC). AUPC attempts to maintain a constant E_b/N_0 at the receive end of an SCPC link. This is especially useful when operating over a satellite at Ku-Band Frequencies in locations with high rainfall periods.



NOTE

Implementing Strap Code 26 can set the following modem configuration. Refer to Table 4-4 for an explanation and tabular listing of available Strap Codes. The Frequency and Modulator Output Power are set independently of the strap code.

The IBS Async Framed Data Mode provides a service channel between the two sites of a link permitting the modem processors to send messages and get responses over this channel. AUPC can be set to operate on either or both directions of a link but always requires a bi-directional channel. Therefore, both the Modulator and Demodulator interface mode must be set to IBS Async for the AUPC Menus to be visible and for the AUPC function to operate properly. The AUPC Functions and their descriptions are shown on Table 3-2.

Table 3-2. AUPC Functions	
Function	Description
AUPC ENABLE/DISABLE	Enables/Disables the AUPC to function locally
AUPC Eb/No	Desired E_b/N_0 of remote modem
AUPC MIN LVL	Sets minimum output power to be used
AUPC MAX LVL	Sets maximum output power to be used
AUPC DEF LVL	Sets default output power to be used
<i>The AUPC Menus are located under the Modulator Menu as shown in Section 4. The EFAUPC Menu displays when EFAUPC Framing is enabled.</i>	

The basic AUPC Operation is described as follows:

Assume that the two modems, one at each end of the link, are set to AUPC operation. Only one direction is discussed, but the same functions could be occurring in both directions simultaneously. Modem "A" is transmitting to modem "B" under normal conditions and modem "B" has a receive E_b/N_0 of 7.5 dB. Modem "A" has been set to an AUPC E_b/N_0 on the Front Panel of 7.5 dB, and is currently outputting -15 dBm. Next, it begins raining at location "B", and the E_b/N_0 drops to -7.0 then -6.8 dB. Modem "B" is constantly sending update messages to "A" and reports the current E_b/N_0 . When "A" sees the drop in E_b/N_0 , it slowly begins to raise the output power, and raises it again when it sees further drops. As the rain increases in intensity, and the E_b/N_0 decreases again, "A" continues to increase its power level to compensate, and when the rain diminishes and quits, it lowers its power level to compensate. The operation is therefore a feedback control loop with the added complication of a significant time delay.

There are safeguards built into the AUPC System. First, the Modulator has two additional parameters, which allow control of the Maximum and Minimum Power Output Levels. Second, a default power level is specified which takes precedence over the output power level during signal loss or loss of AUPC Channel Communication. The default power level should normally be set to a high enough level to reestablish communication regardless of rain fade. The other controls are built into the operating control software to limit response times and detect adverse operating conditions.

3.16 Asynchronous Overhead Operation (Framing/Multiplexer Capability)

The Asynchronous Framing/Multiplexer is capable of multiplexing a relatively low-speed overhead channel onto the terrestrial data stream resulting in a slightly higher combined or aggregate data rate through the modem. The overhead channel is recovered at the far end. This added channel is termed variously "An Overhead Channel", "Service Channel", "Async Channel" or in IESS terminology an "ES to ES Data Channel." The basic frame structure used by the multiplexer is that specified in the IESS-309 Standard, resulting in a 16/15 Aggregate to through-Data Ratio.

For Regular Async: (Standard IBS), the Baud Rate is approximately 1/2000 of the Data Rate listed in Table 3-3.

For Enhanced Async: (IBS Async.), the Baud Rate is selectable, but Data Rate is limited.

The maximum Baud Rate is 19,200 bps for IBS Async. Two software-controlled modes are designed into the card to best utilize the available bits; "Standard IBS" and "IBS (Async)". The characteristics of the Channel Interface is also determined by the standard or Async mode.

The Async Channel can be set under software-control to either RS-232 or RS-485 mode. The pin assignments for both modes are shown in Table 3-3. The "RS-485" Setting controls the output into tri-state when the modem is not transmitting data, allowing multiple modem outputs to be connected together.

Table 3-3.			
Kbps	Baud Rate Example for Standard IBS	Kbps	Baud Rate Example for Enhanced Mode
128	64	9.6	300
256	128	19.2	600
384	192	32	600
512	256	64	1200
640	320	128	2400
768	384	192	4800
896	448	256	4800
1024	512	320	9600
1152	576	384	9600
1280	640	448	9600
1408	704	512	9600
1536	768	576	9600
1664	832	640	19200
1792	896	704	19200
1920	960	768	19200
1920	960	768	19200

2048	1024	832	19200
		896	19200
		960	19200
		1024	19200
		1088	19200
		1152	19200
		1216	19200
		1280	19200
		1344	19200
		1408	19200
		1472	19200
		1536	19200
		1600	19200
		1664	19200
		1728	19200
		1792	19200
		1856	19200
		1920	19200
		1984	19200
		2048	19200

3.17 Standard IBS Mode

In the first or "Normal" mode, all bit assignments are per the IBS standard. The bits of Overhead Housekeeping byte 32 are implemented as shown in Table 3-4 below:

Table 3-4.		
Bit 1	ES to ES Data Channel	This bit is routed directly to the ES to ES Data Channel. Its data rate is 1/512 th of the aggregate rate (or 1/480 th of the through terrestrial data rate), and is normally used to super-sample an asynchronous data channel.
Bit 2	Frame Alignment	Part of the Frame Alignment word.
Bit 3	Backward Alarm	Transmit and Receive with main processor to activate Main Alarm/LED.
Bit 4	Multiframe Message	As per IBS.
Bits 5 and 6	Spare	Not currently utilized.
Bits 7 and 8	Encryption Utilization	Not currently utilized.

The ratio of the Through Terrestrial Data Channel Rate to the aggregate rate is 15/16. The standard transmit and receive channels of the ES to ES Data Channel in Standard IBS Mode are raw channels operating at the specific bit rate as controlled by the data channel rate, without buffering. In addition, no clocks are provided with this channel. Since it would be rare that the data rate provided was exactly that required for a standard rate device, the only method of communicating using this channel is to allow it to super-sample the user data.

3.18 Asynchronous Multiplexer Mode

Since many of the frame bits in the standard IBS mode are not used, an “Enhanced” Multiplexer Mode has been implemented that can be engaged under software control. Since this mode changes the use of many of the framed non-data bits, this mode is only usable when the DMD2050 is at both ends of a link. In this mode, the overhead signaling bytes 16 and 48 can be used to implement a significantly higher speed ES to ES Data Channel under software control. When implemented, this rate is 16 times that of the normal IBS standard, or $1/30^{\text{th}}$ of the terrestrial data rate ($1/32^{\text{nd}}$ of the aggregate rate).



NOTE

The IBS Async mode MUST be selected for true Asynchronous channel operation to be available.

3.19 ESC Backward Alarms

When running in IDR Mode and if the modem has the ESC Option, there will be four Backward Alarms available for use by the earth stations at each end of the link (both ends must have the ESC option). These alarms are accessed via the ESC ALARMS Port. The four alarms are controlled by four relays, each having a normally open, normally closed, and a common connection. The common connections of these relays (referred to as Backward Alarm Inputs) can be connected to whichever system on the earth station that the user wishes to trigger the backward alarm. When ground is applied to the Common (Input) Connection of one of these relays, that relay and associated backward alarm will then be in a “no fault” state. When the ground is removed, the relay and the associated Tx Backward Alarm will toggle to the faulted state. When in the faulted state, the receive end of the link will receive that backward alarm that is initiated at the transmit end of the link.

The user can connect whichever systems on the earth stations that they desire to these Backward Alarms Relays as long as they will supply ground to the Backward Alarm Relay Input in the “no fault” condition and the ground will be removed in the “faulted” condition.

For example: the user could connect the Demod Summary Fault of the modem to the Backward Alarm 1 Input, so that if the demod went into Major Alarm (such as a Carrier Loss), Backward Alarm 1 would be transmitted to the receive end of the link. At the receive end, it would show up as Rx Backward 1 (Receive Backward Alarm 1).

3.19.1 To Disable the ESC Backward Alarms

If the ESC ALARMS Port will not be used and the Backward Alarm Indications are to be disabled, you will need to connect pins 19, 11, 22 and 23 to pin 1 (Gnd).

3.20 Satellite Control Channel (SCC)

The SCC format uses a variable overhead rate to transmit an asynchronous data channel in addition to the normal data channel. The SCC asynchronous mode implemented on the DMD2050 is "PassThru" Mode.

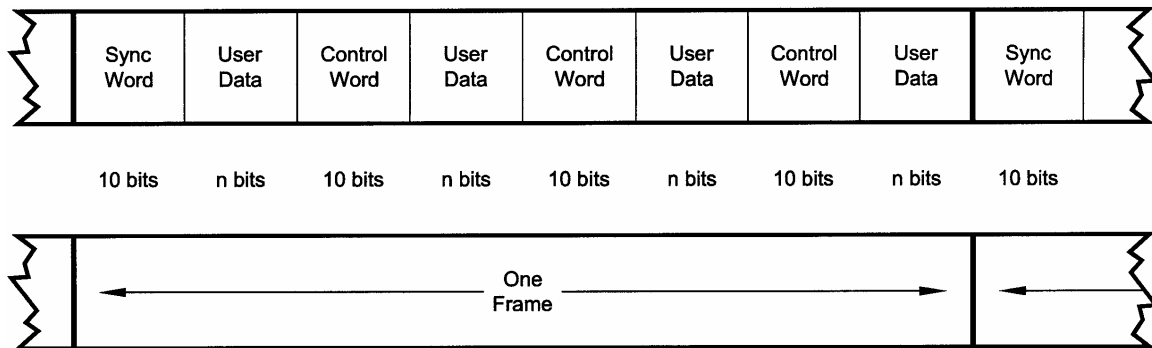
3.20.1 SCC Framing Structure

Each SCC frame consists of the following:

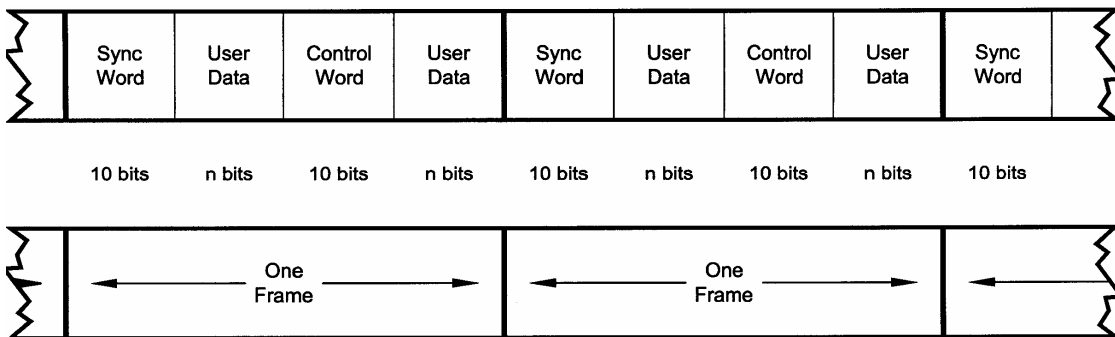
- A 10-bit synchronization pattern called the Synchronizing Word.
- Multiple variable length slots filled with user data.
- Multiple 10-bit control words that contains eight bits of in-band data (the extra two bits are for the async start/stop).

The number of user data slots and control words per frame is selected by the SCC Control Ratio Parameter. This can be any value from 1 to 1 through 1 to 7. A higher ratio allows a lower overhead rate but since there are less Sync Words, there is a higher acquisition time.

The following examples show a control ratio of 1 to 3 and 1 to 1. Example 1 shows three Control Words for every Synchronizing Word, and Example 2 shows one Control Word for every Synchronizing Word.



1 to 3 Control Ratio



1 to 1 Control Ratio

The Control Ratio of the receiving units must match the Control Ratio of the transmitting unit.

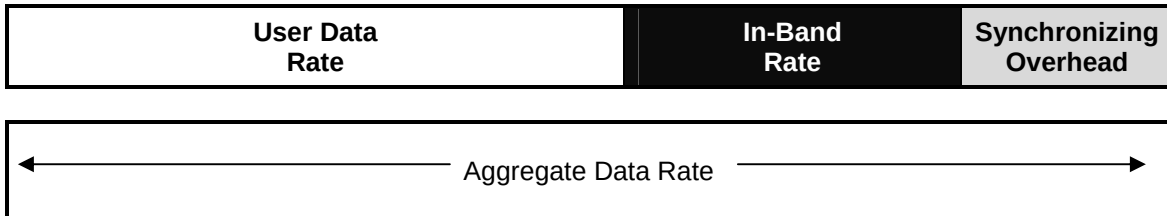
3.20.2 Aggregate Data Rate

The aggregate data rate equals the following:

$$\text{User Data Rate} + \text{In-Band Rate} + \text{Synchronizing Overhead Rate}$$

Because SCC must adjust the overhead so that there are an equal number of user data bits in each slot, the synchronizing overhead cannot be easily calculated. However, dividing the In-Band Rate by the Control Ratio can approximate it. The following equation shows the basic calculation of this rate:

$$\text{Aggregate Data Rate} = \text{User Data Rate} + \text{In-Band Rate} + (\text{In-Band Rate}/\text{Control Ratio})$$



As an example, given the following parameters:

User Data Rate: 1,024,000 bps
 In-Band Rate: 19,200 bps
 Control Ratio: 1 to 7

Aggregate data rate = 1,024,000 + 19,200 + (19,200/7) or approximately 1,045,942 (actually 1045974).

This gives an overhead ratio of 1,045,974/1,024,000 = 1.021

In addition, another constraint changes the actual Aggregate Data Rate. The user data slot size is limited to 2,500 bits. Because of this, the modem increases the in-band rate to reduce the user data slot size. This only happens at higher user data rates.

3.20.3 Overhead Rate Comparison

The SCC Overhead Ratio varies depending on the User Data Rate, the In-Band Rate, and the Control Ratio. This gives SCC the advantage of lower overhead rates when compared to IBS, which has a fixed overhead ratio of 16/15 or 1.067. The following table gives some example overhead rates for different user data and control ratios.

User Data Rate	In-Band Rate	Control Ratio	Aggregate Data Rate	Overhead Ratio
512,000	19,200	1/7	533,974	1.043
1,024,000	19,200	1/7	1,045,974	1.021
2,048,000	19,200	1/7	2,069,951	1.011
3,072,000	19,200	1/7	3,093,943	1.007
4,096,000	19,200	1/7	4,117,951	1.005
6,312,000	19,200	1/7	6,337,248	1.004
6,312,000	19,200	1/3	6,337,606	1.004
6,312,000	19,200	1/1	6,350,418	1.006

3.20.4 Actual Overhead Rate Calculation

The following is the actual calculation the modem does to calculate the overhead ratio:

1. The modem calculates the minimum in-band rate to limit the size of the user data slots to 2,500 bits (the result is truncated to an integer).

$$\text{Minimum In-Band} = (\text{User Data Rate} * \text{Control Ratio}) / ((\text{Control Ratio} + 1) * 250)$$

2. Using the bigger of Minimum In-Band or the selected In-Band, the modem calculates the number of bits for each user data slot (result is truncated to an integer).

$$\text{Slot Bits} = (\text{User Data Rate} * (\text{Control Ratio} * 10)) / (\text{In-band Rate} * (\text{Control Ratio} + 1))$$

The actual ratio the modem uses is:

$$\text{Actual Ratio} = (\text{Slot Bits} + 10) / \text{Slot Bits}$$

Example 1:

User Data Rate: 1,024,000 bps
 In-Band Rate: 19,200 bps
 Control Ratio: 1 to 7

$$\text{Minimum In-Band} = (1,024,000 * 7) / ((7 + 1) * 250) = 3,584 \text{ (less than In-Band Rate)}$$

$$\text{Slot Bits} = (1,024,000 * (7 * 10)) / (19,200 * (7 + 1)) = 466$$

$$\text{Actual Ratio} = (466 + 10) / 466 = 1.021$$

Example 2:

User Data Rate: 6,312,000 bps
 In-Band Rate: 19,200 bps
 Control Ratio: 1 to 7

$$\text{Minimum In-Band} = (6,312,000 * 7) / ((7 + 1) * 250) = 22,092 \text{ (more than In-Band Rate)}$$

$$\text{Slot Bits} = (6,312,000 * (7 * 10)) / (22,092 * (7 + 1)) = 2,500$$

$$\text{Actual Ratio} = (2,500 + 10) / 2,500 = 1.004$$

3.21 SCC OVERHEAD CHANNEL SETUP

1. Set the Framing Mode (located under Mod and Demod Data Menus) to SCC. After doing this, two new menus will appear to the right of the Framing Menu, for both the Mod and Demod. The new menus will be:

SCC CTL RATIO
SCC INBAND RATE

2. Set the desired SCC control ratio:

SCC CTL RATIO {1/1, 1/2, 1/3, 1/4, 1/5, 1/6, 1/7}

This allows the user to simulate the framing used by the Satellite Control Channel Option (Pass-Thru Mode only). The SCC CTL RATIO is the ratio of overhead in-band data to synchronizing words.

3. Set the desired SCC in-band rate:

SCC INBAND RATE {300 to 200000}

This allows the user to request the rate of in-band data for the overhead channel. This sets the overhead amount only. The actual amount of data that can be passed through the overhead channel will be set under "ES Baud Rate" (see Step 6 below).

4. Under the Interface > General menus, locate the TX ASYNC MODE (menu).

5. Under the TX ASYNC MODE Menu, set the desired ES Interface type:

ES INTERFACE {RS-232, RS-485}

This allows the user to select the interface type.

6. Under TX ASYNC MODE Menu, set the desired baud rate for the ASYNC Port (J17). This will be the baud rate that will pass through the overhead channel:

ES BAUD RATE {150 - 19200}

This allows the user to select the baud rate of the ASYNC port (J17) in SCC Mode.

7. Under TX ASYNC MODE Menu, set the desired ES BITS/CHAR:

ES BITS/CHAR {7,8}

This allows the user to choose between 7 or 8 bits of data.

8. Repeat Steps 4 through 7 under the RX ASYNC MODE (menu)

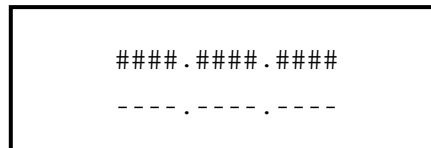
9. The physical connection to the DMD2050 for the overhead channel will be the DB-9 Female Port labeled ASYNC (J17).

3.22 Locating the DMD2050 ID Code Operational Procedure

The modem has unique ID codes that allow the user to add feature upgrades to the modem without the unit having to be returned to the factory. Users are required to identify these ID codes when they want additional features added to their unit. Radyne will supply a new ID code that is required to be entered in the ID code field. Once the new ID code is entered, the modem will activate the new features.

Locating the unique ID Code on the modem:

1. Verify that the unit is powered up.
2. Use the Arrow Keys on the Front Panel to navigate to the System Menu. This is the top-level menu.
3. Press the Down Arrow one time, then the left arrow one time to navigate to the HW/FW Config Screen.
4. Press the Down Arrow one time, then the Left Arrow one time to navigate to the Features Screen.
5. Press the Down Arrow one time. A two-row screen will display . The top row is the unique ID number for that unit (which is actually the ID number of the Main Board). It is a 12-digit code (3 sets of 4 digits). The second row is three groups of dashes (----.----.----). The dashes are the location where the given code is to be entered. Refer to the following display:



```
####.####.####
----.----.----
```

The “#” on the screen are actually the numbers of the ID code. Write down this unique ID code and present it to a Radyne Sales Representative. You will then be given a 12-digit code that is to be entered in place of the ----.----.----.

6. Enter the code that was just given you and press <ENTER>. If the code is accepted (within three attempts), the screen will display “Upgrade Mgr: Entry Accepted” and the new features will be unlocked, otherwise the display will show “Further Attempts Require a Reboot”, and power must be cycled for further attempts. When the new features are unlocked, the bottom row will return to displaying ----.----.----. The keyed in code will no longer appear on the screen.



User Interfaces

4

4.0 User Interfaces

There are three user interfaces available for the DMD2050. These are:

- Front Panel Interface
- Remote Port Interface
- Terminal Interface.

4.1 Front Panel User Interface

The Front Panel of the DMD2050 allows for complete control and monitor of all DMD2050 parameters and functions via a keypad, VFD (Vacuum Flourescent Display) and status LEDs.

The front panel layout is shown in Figure 4-1, showing the location and labeling of the front panel. The front panel is divided into four functional areas: the VFD Front Panel Display, the Cursor Control Arrow Keys, the Numeric Keypad, and the Front Panel LED Indicators, each described below in Table 4-1.

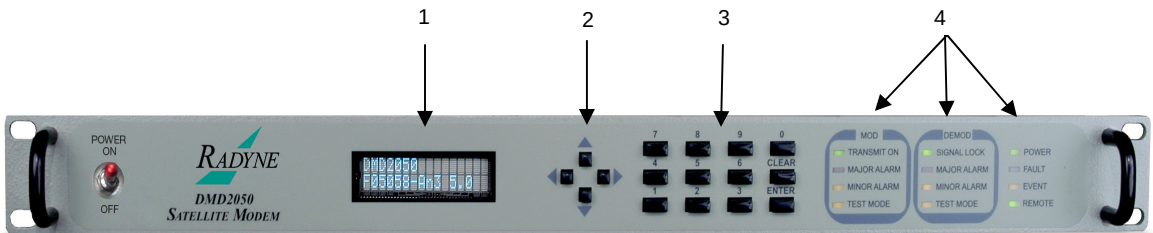


Figure 4-1. DMD2050 Front Panel

Table 4-1.		
Item Number	Description	Function
1	VFD Front Panel Display	Displays DMD2050 operating parameters and Configuration data
2	Cursor Control Arrow Keys	Controls the up, down, right and left motion of the cursor in the VFD window
3	Numeric Keypad	Allows entry of numeric data and Clear and Enter function keys
4	Front Panel LED Indicators	See Paragraph 4.1.4 below for an itemized description of these LEDs

4.1.1 VFD Front Panel Display

The front panel display is a 2 line by 16-character vacuum fluorescent display. The display is lighted and the brightness can be set to increase when the front panel is currently in use. The VFD automatically dims after a period of inactivity. The display has two distinct areas showing current information. The upper area shows the current parameter being monitored, such as 'Frequency' or 'Data Rate'. The lower line shows the current value of that parameter. The VFD is a single entry window into the large matrix of parameters that can be monitored and set from the Front Panel.

4.1.2 Cursor Control Arrow Keys

A set of 'Arrow' or 'Cursor' keys ($\uparrow$), ($\downarrow$), ($\rightarrow$), ($\leftarrow$), is used to navigate the parameter currently being monitored or controlled. Table 4-2 describes the functions available at the Front Panel.

4.1.3 Numeric Keypad

A 10 Key Numeric Keypad with 2 additional keys for the 'Enter' and 'Clear' function allows the entry of data into the system. Table 4-2 describes the functions available at the Front Panel.

Table 4-2. Edit Mode Key Functions (Front Panel Only)							
Parameter Type	0 – 9	$\uparrow$	$\downarrow$	$\leftarrow$	$\rightarrow$	'Clear' & $\leftarrow$	'Clear' & $\rightarrow$
Fixed Point Decimal	Changes Digit	Toggles $\pm$ (If Signed)	Toggles $\pm$ (If Signed)	Moves Cursor 1 Position Left	Moves Cursor 1 Position Right	N/A	N/A
Unsigned Hexadecimal	Changes Digit	Increments Digit Value	Decrements Digit Value	Moves Cursor 1 Position Left	Moves Cursor 1 Position Right	N/A	N/A
Enumerated	N/A	Previous Value in List	Next Value in List	N/A	N/A	N/A	N/A
Date/ Time	Changes Digit	N/A	N/A	Moves Cursor 1 Position Left	Moves Cursor 1 Position Right	N/A	N/A
IP Address	Changes Digit	Increments Digit Value	Decrements Digit Value	Moves Cursor 1 Position Left	Moves Cursor 1 Position Right	N/A	N/A
Text Strings	Changes Character	Increments Character Value	Decrements Character Value	Moves Cursor 1 Position Left	Moves Cursor 1 Position Right	Clears to Left of Cursor Inclusive	Clears to Right of Cursor Inclusive

4.1.4 Front Panel LED Indicators

Eight LEDs on the DMD2050 Front Panel (Refer to Table 4-3) indicate the status of the DMD2050's operation. The LED colors maintain a consistent meaning. Green signifies that the indication is appropriate for normal operation, Yellow means that there is a condition not proper for normal operation, and Red indicates a fault condition that will result in lost communications.

Table 4-3.		
LED	Color	Function
Modem LED Indicators		
Power	Green	Indicates that the unit is turned on.
Fault	Red	Indicates a hardware fault for the unit.
Event	Yellow	Indicates that a condition or event has occurred that the modem has stored in memory. The events may be viewed from the Front Panel or in the Terminal Mode.
Remote	Green	Indicates that the unit is in the process of updating firmware with FTP.
Modulator LED Indicators		
Transmit On	Green	Indicates that the transmitter is on.
Major Alarm	Red	Indicates that the Transmit Direction has failed, losing traffic.
Minor Alarm	Yellow	Indicates that a Transmit Warning Condition exists.
Test Mode	Yellow	Indicates that the transmitter is involved in a current Test Mode activity.
Demodulator LED Indicators		
Signal Lock	Green	Indicates that the receiver locked to an incoming carrier and data, including FEC Sync.
Major Alarm	Red	Indicates that the Receive Direction has failed, losing traffic.
Minor Alarm	Yellow	Indicates that a Receive Warning Condition exists.
Test Mode	Yellow	Indicates that the receiver is involved in a current Test Mode activity.

4.2 Parameter Setup

The four Cursor Control Arrow Keys are used to navigate the menu tree and select the parameter to be set. After arriving at a parameter that needs to be modified, depress <ENTER>. The first space of the modifiable parameter highlights (blinks) and is ready for a new parameter to be entered. After entering the new parameter using the keypad (Refer to Figure 4-2), depress <ENTER> to lock in the new parameter. If a change needs to be made prior to pressing <ENTER>, depress <CLEAR> and the display defaults back to the original parameter. Depress <ENTER> again and re-enter the new parameters followed by <ENTER>.

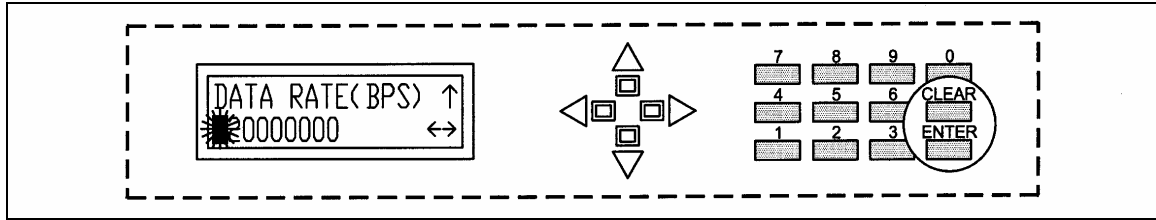


Figure 4-2. Entering New Parameters

Following a valid input, the DMD2050 will place the new setting into the nonvolatile EEPROM making it available immediately and available the next time the unit is powered-up.

4.3 Front Panel Control Screen Menus

The DMD2050 Front Panel Control Screens are broken down into sections under several Main Menus.

4.3.1 Main Menus

MODULATOR

DEMODULATOR

INTERFACE

MONITOR

ALARMS

SYSTEM

TEST

4.3.2 Modulator Menu Options and Parameters

NETWORK SPEC

{MIL 188-165A, IDR, IBS, DROP & INSERT, CLOSED NET, DVB}

Used with IDR, or IBS Interface Only.

The Network Spec Command sets a number of parameters within the modem to meet a set specification. The purpose is to eliminate keystrokes and potential compatibility problems.

Additionally, data rates not covered by the given network specification of operation will not be allowed. If the mode of operation is selected after the data rate has been entered, then the data rate must be compatible with the desired mode of operation or the mode will not be allowed. The following parameters are set for the given mode of operation and cannot be changed while the unit is in the given mode of operation:

MIL-STD 188-165A:

(MILSTD-188-165A)

For Data Rates:	9.6 K – 51.848 Mbps
Framing:	None
Scrambler:	OM73
Spectrum	MIL 188 165A

IDR:

(IESS-308)

For Data rates	1.544, 2.048, 6.312, 8.448 Mbps
Framing Type:	96 Kbps (IDR)
Scrambler Type:	V.35
Spectrum Mask:	Intelsat

For Data Rates < 1.544

Framing Type:	1/15 (IBS)
Scrambler Type:	IESS-309
Spectrum Mask:	Intelsat

IBS:

(IESS-309)

For Data Rates $\leq$ 2048

Framing Type:	1/15 (IBS)
Scrambler Type:	IESS-309
Spectrum Mask:	Intelsat

Drop & Insert:

Data Rates:	n x 64 n = 1, 2, 3, 4, 5, 6, 8, 10,12, 15, 16, 20, 24, 30
Framing Type:	1/15 (IBS)
Scrambler Type:	IESS-309
Spectrum Mask:	Intelsat

DVB:

EN301-201 and EN300-421

Data Rates: All Rates

Framing Type: DVB

Scrambler Type: DVB

Spectrum Mask: DVB 0.25, 0.35

Closed Net:

All possible combinations allowed, however, DVB settings requires the DVB network spec.

STRAP CODE**{Refer to Strap Code Guide, Table 4-4}**

The Strap Code is a quick set key that sets many modem parameters. Consult the strap code guide for available strap codes. Parameters set by strap code:

Data Rate

Inner Code Rate

Satellite Framing

Scrambler

Drop and Insert

Outer Code Rate (Reed-Solomon)

Modulation

Network Spec

IF (menu)**FREQUENCY (MHz)****{50 – 90 MHz, 100 – 180 MHz, or 950 - 2050 MHz}**

Allows the user to enter the Modulator IF Output Frequency of the modem in 1 Hz increments.

POWER (dBm)**{0 to -25 dBm}**

Allows the user to enter the Transmitter Power Level.

CARRIER**{ON, OFF, AUTO, VSAT, RTS}**

Allows the user to select the carrier type. Refer to Appendix E for further information.

SPECTRUM	{NORMAL, INVERTED} Allows the user to invert the direction of rotation for PSK Modulation. Normal meets the IESS Specification.
MODULATION	{QPSK, BPSK, OQPSK, 8PSK, 16QAM} Allows the user to select the modulation type.
SPECTRAL MASK	{MIL-188-165A, INTELSAT 0.35, DVB SAT 0.35, DVB SAT 0.25} Allows the user to set the spectral shape of Tx Data Filter.
COMPENSATION	{0 to 1 dB} Allows the user to set the offset on the transmit output power will will allows for some specific cable loss up to 1dB.
DATA (menu)	
DATA RATE (bps)	{Refer to Technical Specs for Data Rates} Allows the user to set the Data Rate in bps steps via the Front Panel Arrows or Keypad.
SYMB RATE (sps)	Allows the user to view the Symbol Rate.
INNER FEC	Viterbi {1/2, 3/4, 7/8, None} Optional FEC Rates: Sequential {1/2, 3/4, 7/8} Trellis {8PSK} Turbo ≤ 20Mbps {.793, .495, 3/4, 7/8} Turbo >20Mbps {.750, .875} CSC {3/4} DVB VIT {2/3, 5/6} DVB Trellis {3/4, 5/6, 7/8, 8/9} Allows the user to select the Tx Code Rate and Type.
TPC INTERLEAVER	{DISABLE, ENABLE} Allows the user to disable or enable the TPC Interleaver. Valid only on Radyne Turbo Codes TPC .495 and TPC .793.
DIFF CODING	{ENABLED, DISABLE} Allows the user to enable or disable the Differential Encoder. Having the encoder enabled ensures proper phase lock. May not be adjustable in some modes.
SCRAMBLER SEL	{NONE, V.35-IESS, V.35 CITT, V.35 EF, IBS w/Optional Framing and optional Reed-Solomon, Reed-Solomon Scrambler w/Optional Framing, CCITT, V.35FC, OM-73, V.35EF_RS, TPC SCRAMBLER (Turbo Codec), DVB, EDMAC} Allows the user to select the descrambler type.
SCRAMBLER CTRL	{ENABLED, DISABLE} Allows the user to enable or disable scrambler operation.

SAT FRAMING

{1/15 (IBS), 1/15 (Async), 96 Kbps (IDR), DVB, EDMAC, EFAUPC, SCC, NONE, EFFICIENT D&I}
Used with IDR, IBS, or Asynchronous Interface Only.
 Allows the user to select the framing type.

**NOTE**

When SCC Framing is selected above, the following two screens are available.

SCC CTL RATIO

{1/1, 1/2, 1/3, 1/4, 1/5, 1/6, 1/7}
 Allows the user to simulate the framing used by the Satellite Control Channel Option (Pass Thru Mode only). The SCC CTL RATIO is the ratio of overhead in-band data to synchronizing words.

SCC INBAND RATE

{300 to 200000}
 Allows the user to request the rate of in-band data for the overhead channel.

IN-BAND RATE

{150, 300, 600, 1200, 2400, 4800, 9600 19200}
ED&I Only
 Available when enhanced async is enabled, this field allows the operator to set the Earth Station to Earth station in-band rate. In most cases, this should be set to the same rate or higher than the ES port baud rate. When this rate is set lower than the ES port baud rate, the user must insure that the actual transmission rate does not exceed the in-band rate, otherwise, characters will be dropped.`

TERR FRAMING

{NONE, 188, 204} DVB Only

DATA POLARITY

{INV. TERR & BASE, INV. BASEBAND, INV.TERR DATA, NONE}
 Allows the user to invert the Tx Data polarity.

SYMBOL PAIR

{NONE, SWAPPED}
 Allows the user to swap the I & Q Channels.
(BPSK Mode Only)

ESC OVERHEAD

{VOICE X2, DATA 64KBPS}
 IDR ESC Channel used for Voice or 64 K data channel.

AUPC (menu)

Displays when IBS Framing is selected for the Modulator and Demodulator.

AUPC MODE

{DISABLED, RADYNE}

	Displays the AUPC Mode.
AUPC Eb/No	{4 to 20dB} Allows the user to enter the target Eb/No value.
AUPC MIN LVL	{0 to -25dB} Allows the user to set the minimum Transmit Power. The minimum Transmit Power is the lowest power setting that will be used when the remote modem commands a decrease of the Transmit Power.
AUPC MAX LVL	{0 to -25 dB} Allows the user to set the maximum Transmit Power. The maximum Transmit Power is the highest power setting that will be used when the remote modem commands an increase of the Transmit Power.
AUPC DEF LVL	{0 to -25dB } Allows the user to set the default Transmit Power. The nominal Transmit Power is the setting that will be used when the remote modem indicates that its receiver has lost lock.
EF AUPC (menu)	
LOCAL AUPC (menu)	The 'LOCAL AUPC CONFIGURATION' Menu contains the local configuration parameters for the AUPC Function.
AUPC ENABLE	{DISABLED, EFDATA} Allows the user to enable or disable the Local AUPC Function of the local modem. The Local AUPC Function is the response to the commands for an increase or decrease of the Transmit Power in 0.5 dB steps. The command to change to the setting is indicated in the 'REMOTE CL ACTION' Menu upon receiver loss of lock of the remote modem.
NOMINAL TX POWER	{0 to -25dB} Allows the user to set the nominal Transmit Power. The nominal Transmit Power is the setting that will be used when the remote modem indicates that its receiver has lost lock and commands a change to the setting indicated in the 'REMOTE CL ACTION' Menu. That change will only be implemented if the 'REMOTE CL ACTION' Menu is set to "NOMINAL".
MINIMUM TX POWER	{0 to -25dB } Allows the user to set the minimum Transmit Power. The minimum Transmit Power is the lowest power setting that will be used when the remote modem commands a decrease of the Transmit Power.
MAXIMUM TX POWER	{0 to -25dB} Allows the user to set the maximum Transmit Power. The maximum Transmit Power is the highest power

setting that will be used when the remote modem commands an increase of the Transmit Power. It is the setting that will be used when the remote modem indicates that its receiver has lost lock and commands a change to the setting indicated in the 'REMOTE CL ACTION', Menu. That change will only be implemented if the 'REMOTE CL ACTION' Menu is set to "MAXIMUM".

TARGET Eb/No	{4.0 to 20dB} Allows the user to set the desired E_b/N_o for the local receiver. This setting is compared against the receive E_b/N_o and commands the remote modem to increase or decrease transmit power accordingly are sent.
TRACKING RATE	{0.5 to 6.0} Allows the user to set the rate at which the commands to increase or decrease Transmit Power are sent. Each command will result in a 0.5 dB increase or decrease in Transmit Power from the remote transmitter. The tracking rate is adjustable from 0.5 dB per minute to 6.0 dB per minute in 0.5 dB steps. The resulting 'command rate' is 1 command every minute to 1 command every five seconds.
LOCAL CL ACTION	{HOLD, NOMINAL, MAXIMUM} Allows the user to set the Transmit Power Setting to be used when the local modem receiver loses lock. The setting can be 'HOLD' (no action taken), 'NOMINAL' (the nominal Transmit Power Setting is used), and 'MAXIMUM' (the maximum Transmit Power Setting is used).
REMOTE CL ACTION	{HOLD, NOMINAL, MAXIMUM} Allows the user to set the Transmit Power Setting to be used when the remote modem receiver loses lock. The setting can be 'HOLD' (no action taken), 'NOMINAL' (the nominal Transmit Power Setting is used), and 'MAXIMUM' (the maximum Transmit Power Setting is used).
REMOTE AUPC (menu)	The 'REMOTE AUPC CONFIGURATION' Menu contains the remote configuration parameters for the AUPC Function.
AUPC ENABLE	{OFF, ON} Allows the user to enable or disable the AUPC Function of the remote modem. The remote AUPC Function is the response of the local modem to commands for an increase or decrease of the Transmit Power in 0.5 dB steps and the command to change to the setting indicated in the 'REMOTE CL ACTION' Menu of the remote modem upon receiver loss of lock.
LOOPBACK	Allows the user to enable or disable the Baseband Loopback Test Mode of the remote modem.

TX 2047 PATTERN	Allows the user to enable or disable the Transmit 2047 Pattern Test Mode of the remote modem.
REMOTE AUPC MONITOR	The 'REMOTE AUPC MONITOR' Menu contains the remote monitor status for the AUPC Function.
REMOTE 2047 BER:	Reports the BER measurement of the receiver 2047 Pattern Test Mode of the remote modem. BER is reported from the 1×10^{-5} to 1×10^{-7} in tenth decade steps. If the pattern does not synchronize or is out of range, 'NO DATA' will be displayed.
REED-SOLOMON (menu)	These selections are visible only when the Reed-Solomon Option is installed.
ENABLE/DISABLE	{ENABLED, DISABLE} Allows the user to Enable/Disable the Reed-Solomon Encoder.
RS RATE	{Refer to Table 3-1 for valid n/k values} Displays the currently used n, k Reed-Solomon Codes. In Closed Net Mode, the user may select custom R-S Codes.
INTERLVR DEPTH	{4, 8, 12} Allows the user to select the Reed-Solomon interleaver depth. In Closed Net Mode, a depth of 4 or 8 may be selected.

4.3.3 Demodulator Menu Options and Parameters

NETWORK SPEC	{MIL-STD 188-165A, IDR, IBS, DROP & INSERT, CLOSED NET, DVB} <i>Used with IDR, or IBS Interface Only.</i> The Mode Command sets a number of parameters within the modem to meet a set specification. The purpose is to eliminate keystrokes and potential compatibility problems. Additionally, data rates not covered by the given mode of operation will not be allowed. If the mode of operation is selected after the data rate has been entered, then the data rate must be compatible with the desired mode of operation or the mode will not be allowed. The following parameters are set for the given mode of operation and cannot be changed while the unit is in the given mode of operation: MIL188-165A: (MILSTD-188-165A) <table> <tr> <td>For Data Rates:</td><td>9.6 K – 51.848 Mbps</td></tr> <tr> <td>Framing:</td><td>None</td></tr> <tr> <td>Scrambler:</td><td>OM73</td></tr> <tr> <td>Spectrum</td><td>MIL 188 165A</td></tr> </table>	For Data Rates:	9.6 K – 51.848 Mbps	Framing:	None	Scrambler:	OM73	Spectrum	MIL 188 165A
For Data Rates:	9.6 K – 51.848 Mbps								
Framing:	None								
Scrambler:	OM73								
Spectrum	MIL 188 165A								

IDR:

(IESS-308)

For Data rates 1.544, 2.048, 6.312, 8.448 Mbps

Framing Type: 96 Kbps (IDR)

Descrambler type: V.35

Spectrum Mask: Intelsat

For Data Rates < 1.544 Mbps

Framing Type: 1/15 (IBS)

Descrambler Type: IESS-309

Spectrum Mask: Intelsat

IBS:

(IESS-309)

For Data Rates ≤ 2.048 Mbps

Framing Type: 1/15 (IBS)

Descrambler Type: IESS-309

Spectrum Mask: Intelsat

Drop & Insert:Data Rates: $n \times 64$, $n = 1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 16, 20, 24, 30$

Framing Type: 1/15 (IBS)

Descrambler Type: IESS-309

Spectrum Mask: Intelsat

DVB:

(Per EN301 201 and 300 421)

Data Rates: All Rates

Framing Type: DVB

Scrambler Type: DVB

Spectrum Mask: DVB 0.25, 0.35

Closed Net:

All possible combinations allowed, however, DVB settings requires the DVB network spec.

STRAP CODE**{Refer to Strap Code Guide, Section 4.3, Table 4-4}**

The Strap Code is a quick set key that sets many modem parameters. Consult the strap code guide for available strap codes. Parameters set by strap code:

Data Rate
 Inner Code Rate
 Satellite Framing
 Scrambler
 Drop and Insert
 Outer Code Rate (Reed-Solomon)
 Modulation
 Network Spec

IF (menu)

FREQUENCY (MHz)	{50 – 90 MHz, 100 – 180 MHz, or 950 - 2050 MHz} Allows the user to enter the Modulator IF Frequency in 1 Hz increments.
SPECTRUM	{NORMAL INVERTED} Allows the user to invert the direction of rotation for PSK Modulation. Normal meets the IESS Specification.
MODULATION	{QPSK, BPSK, OQPSK, 8PSK, 16QAM} Allows the user to select the demodulation type.
SPECTRAL MASK	{MIL-188-165A, INTELSAT 0.35, DVB 0.35, DVB 0.25} Allows the user to set the spectral shape of Tx Data Filter.
SWEEP RANGE (kHz)	{±0 to 255 kHz} Allows the user to set the acquisition range for the demodulator
SWEEP DELAY (Sec)	{0.0 – 900.0 sec} Allows the user to set the reacquisition delay time in 1/10 th second increments.
REACQ RANGE (Hz)	{0 – 32000 Hz} Allows the user to set the reacquisition sweep in 1 Hz increments.
ADJ CARRIER PWR	{Normal, Suppressed} Allows the user to indicate adjacent carrier as Normal or Suppressed (High Power). Unit will increase or decrease post discrimination gain appropriately.
FAST ACQUISITION	{DISABLE, ENABLED} Allows the user to disable or enable the RX Fast Acquisition for low symbol rates.
INPUT THRESHOLD (dBm)	{-30 to -90} Sets a minimum carrier receive level when a carrier drops below this level. The modem will sweep and try to acquire a new signal. This prevents extremely small carriers from falsely locking the modem.
EB/NO ALARM	{0.0 – 9.9 dB} Allows user to select an EB/NO level that will trigger an alarm when the received EB/NO is worse.

DATA (menu)

DATA RATE (bps)	{Refer to Technical Specs for Data Rates} Allows the user to set the Data Rate in bps steps via the Front Panel Arrows or Keypad.
SYMB RATE (sps)	Allows the user to view the Symbol Rate.

INNER FEC	Viterbi {1/2, 3/4, 7/8, None} Optional FEC Rates: Sequential {1/2, 3/4, 7/8} Trellis {8PSK} Turbo ≤ 20Mbps {.793, .495, 3/4, 7/8} Turbo >20Mbps {.750, .875} CSC {3/4} DVB VIT {2/3, 5/6} DVB Trellis {3/4, 5/6, 7/8, 8/9} Allows the user to select the Rx Code Rate and Type.
TPC INTERLEAVER	{DISABLE, ENABLE} Allows the user to disable or enable the TPC Interleaver. Valid only on Radyne Turbo Codes TPC .495 and TPC .793.
DIFF CODING	{ENABLED, DISABLE} Allows the user to enable or disable the Differential Decoder. Having the decoder enabled ensures proper phase lock. May not be adjustable in some modes.
SCRAMBLER SEL	{NONE, V.35-IESS, V.35 CITT, V.35 EF, IBS w/Optional Framing and optional Reed-Solomon, Reed-Solomon Scrambler w/Optional Framing, CCITT, V.35FC, OM-73, V.35EF_RS, TPC SCRAMBLER (Turbo Codec), DVB, EDMAC} Allows the user to select the descrambler type.
SCRAMBLER CTRL	{ON, OFF} Allows the user to enable or disable the descrambler operation.
SAT FRAMING	{1/15 (IBS), 1/15 (Async), 96 Kbps (IDR), EDMAC, EFAUPC, SCC, None} <i>Used with IDR, IBS, or Asynchronous Interface Only.</i> Allows the user to select the Framing Type.



NOTE

When SCC Framing is selected above, the following two screens are available.

SCC CTL RATIO	{1/1, 1/2, 1/3, 1/4, 1/5, 1/6, 1/7} Only available when using SCC framing. Allows the user to simulate the framing used by the Satellite Control Channel Option (Pass Thru Mode only). The SCC CTL RATIO is the ratio of overhead in-band data to synchronizing words.
SCC INBAND RATE	{300 to 200000} Only available when using SCC framing. Allows the user to request the rate of in-band data for the overhead channel.
TERR FRAMING	{NONE, 188, 204} DVB only
DATA POLARITY	{INV. TERR & BASE, INV. BASEBAND, INV.TERR DATA, NONE} Allows the user to invert the Rx Data polarity.
SYMBOL PAIR	{NONE, SWAPPED} Allows the user to swap the I & Q Channels. (BPSK Mode Only)
ESC OVERHEAD	{VOICE X2, DATA 64KBPS} IDR ESC Channel used for Voice or 64 K data channel.
REED-SOLOMON (menu)	These selections are visible only when the Reed-Solomon Option is installed.
ENABLE/DISABLE	{ENABLED, DISABLED} Allows the user to Enable/Disable the Reed-Solomon Encoder.
RS RATE	{Refer to Table 3-1 for valid n/k values} Displays the currently used n, k Reed-Solomon Codes. In Closed Net Mode, the user may select custom R-S Codes.
INTERLVR DEPTH	{4, 8, 12} Allows the user to select the Reed-Solomon interleaver depth. In Closed Net Mode, a depth of 4 or 8 may be selected.

4.3.4 Interface Menu Options and Parameters

TX SETUP (menu)

CIRCUIT ID	Allows the user entry of a Tx Circuit Identifier. Circuits can be given up to an 11 Character alphanumeric identity such as LINK1.
TERR INTERFACE	{RS423 SERIAL, M2P PARALLEL, DVB PARALLEL, G.703 E2, G.703 T2 BAL, G.703 T2 UNBAL, G.703 E1 BAL, G.703 E1 UNBAL, G.703 T1 AMI, G.703 T1 B8ZS, V.35, HSSI, ASI, ETHERNET, MIL-188-114A} Allows the user to select the Transmit Type.
ETH FLOW CONTROL	{Disabled, Enabled} Allows the user to disable or enable flow control. <i>Only visible when Ethernet is selected as the interface type.</i>
ETH DAISY CHAIN	{Disabled, Port 4} Allows the user to disable the Ethernet Port Daisy Chaining or enable it on Port 4. <i>Only visible when Ethernet is selected as the interface type.</i>
TX CLK SRC	{SCTE, SCT} Allows the user to select the Transmit Clock Source.
TX CLK POL	{AUTO, NORMAL, INVERTED} Allows the user to select the Clock Polarity for the Tx Terrestrial Clock relative to the Tx Data. "Auto" detects wrong polarity and automatically corrects. If G.703 Interface is selected, this selection cannot be changed.
SCT CLK SRC	{INTERNAL (SCT), SCR (Rx SAT CLK)} Allows the user to select SCT Source to be either the SCT Oscillator or RX Satellite Clock. Rx SAT CLK is used for loop timing.

DROP & INSERT (menu)

DROP MODE	{NONE, T1-D4, T1-ESF, PCM-30, PCM-30C, PCM-31, PCM-31C, T1-D4-S, T1-ESF-S.} Allows the user to select any of the above.
MAP COPY	{Tx ACT, Rx ACT, Tx EDIT, Rx EDIT, USER1.....USER8, ROM1.....ROM83} Allows the user to copy drop and insert maps. Tx ACT map is the drop map correctly being used by the modem.

SAT CH TERRCH

Allows the user to edit the Tx edit map to specify the terrestrial slots that will be dropped into assigned satellite channels. The satellite channels are fixed and the number of channels are determined by the data rate. The terrestrial time slots available are determined by the drop mode. When the user has finished editing the Tx edit map, it will be copied to the Tx ACT map before it can be used by the modem.

RX SETUP (menu)**CIRCUIT ID**

Provides entry of Rx Circuit Identifier. Circuits can be as DLINK1

TERR INTERFACE

{RS423 SERIAL, M2P PARALLEL, DVB PARALLEL, G.703 E2, G.703 T2 BAL, G.703 T2 UNBAL, G.703 E1 BAL, G.703 E1 UNBAL, G.703 T1 AMI, G.703 T1 B8ZS, V.35, HSSI, ASI, ETHERNET, MIL-188-114A}
Allows the user to select the Receive Type.

BUFF SIZE (msec)

{0 - 64 msec}
Allows the user to set the Doppler Buffer Size in msec.

BUFFER CLK SRC

{SCTE (External), SCT (Internal), Rx Sat, EXTBNC, EXT IDI}

Priority {1-5}

SRC DEPTH {1-5}

The user must assign priorities to the clock sources. 1 being the highest priority and 5 being the last resort. The menu has three fields; the first field is the name of the clock source, the second field is the priority entry area, and the last field is the depth of the list. In the priority field, the up/down arrow keys will scroll through the list displaying the names and the current priority. When the desired clock name is displayed, the number keys may be used to assign a priority value. Pressing <Enter> will re-sort the list. Do this until the clock sources are prioritized in the order desired. Use the left/right arrow keys to move the cursor to the depth field. This field assigns the number of entries to use. The number keypad or the up/down arrows can be used to change the value.

<u>Clock Source</u>	<u>Priority</u>		<u>SRC DEPTH</u>		
RX SAT	1	of	3	}	Only these will be used
SCTE	2	of	3		
SCT	3	of	3		
EXT BNG	4	of	3	}	Will not be used since 4>3 and 5>3
EXT IDI	5	of	3		

BUFFER CLOCK POL**{NORMAL, INVERTED}**

Allows the user to select the Buffer Clock Polarity for the Tx Terrestrial Clock relative to the Tx Data. If G.703 Interface is selected, this selection cannot be changed.

DROP & INSERT (menu)**INSERT MODE****{NONE, T1-D4, T1-ESF, PCM-30, PCM-30C, PCM-31, PCM-31C, T1-D4-S, T1-ESF-S.}**

Allows the user to select any of the above.

T1/E1 FRAME SRC**{INTERNAL, EXTERNAL, IDI/DDO LOOPBACK}**

Selects the frame source for T1 or E1 framing.

Note: IDI/DDO loop is an internal loopback of the terrestrial clocks not to be used simultaneously with external loopback cable.

MAP COPY**{TxACT, RxACT, TxEDIT, RxEDIT, USER1....USER8, ROM1....ROM8}**

Allows the user to copy drop and insert maps. RxACT map is the insert map currently being used by the modem.

SAT CH TERRCH

Allows the user to edit the Tx edit map to specify the terrestrial slots that will be dropped into assigned satellite channels. The satellite channels are fixed and the number of channels are determined by the data rate. The terrestrial time slots available are determined by the drop mode. When the user has finished editing the Tx edit map, it will be copied to the Tx ACT map before it can be used by the modem.

TERR STREAMING**{BYTE OUTPUT, PACKET OUTPUT} ASI only**

Byte Output = Continuous Mode

Packet Output = Burst Mode

GENERAL (menu)**EXT FREQ (MHz)****{Variable Through Data Rate}**

Allows the user to select the external clock frequency in MHz.

REF FREQ SRC**{EXTERNAL, HIGH STABILITY}**

Allows the user to select the Frequency Reference Source.

REF FREQ (MHz)

Allows the user to select the reference clock frequency in MHz.

TX ASYNC MODE (menu)**TX ASYNC MODE****{ES-ES, ESC ENHANCED}**

ES-ES is the normal IBS Async Channel. ESC Enhanced can be selected in Closed Net and uses the

	Overhead Signaling bytes in the IBS Overhead to pass asynchronous data.
ES INTERFACE	{RS-232, RS-485} Allows the user to select the interface type.
ES BITS/CHAR	{7,8} Allows the user to choose between 7 or 8 bit data.
RX ES ENHANCED (menu)	
RX ASYNC MODE	{ES-ES, ESC ENHANCED} ES-ES is the normal IBS Async Channel. ESC Enhanced can be selected in Closed Net and uses the Overhead Signaling bytes in the IBS Overhead to pass asynchronous data.
ES INTERFACE	{RS-232, RS-485} Allows the user to select the interface type.
ES BAUD RATE	{150 - 19200} Allows the user to select the baud rate in Enhanced Async Mode. Available rates are listed in Table 3-3.
ES BITS/CHAR	{7,8} Allows the user to choose between 7 or 8 bit data.

4.3.5 AUPC Menu Options and Parameters



NOTE

AUPC Menus are only available when the Modulator is in 'Closed Net' Network Spec and Framing is 'EFAUPC 1/15'.

4.3.6 Monitor Menu Options and Parameters

EVENTS	Displays a history of events recorded in the event buffer. A maximum of 100 events may be stored in the buffer. Upon receipt of the 101 st event, the first received event is automatically deleted, and so on, maintaining the maximum 100 events.
ERASE EVENTS.. PRESS CLEAR	Allows the user to clear the contents of the Event Buffer by pressing <CLEAR> on the keypad..
INPUT LVL (dBm)	Displays the estimated receive signal level as seen by the Demodulator.

EBNO (dB)	Displays the estimated E_b/N_0 as seen by the demodulator.
RAW BER	Displays the estimated channel error rate (before decoding) measured by the modem.
CORRECTED BER	The CBER display shows an estimated corrected bit error rate of the modem. Depending on the symbol rate the modem is running, the high-end performance scale of this display will vary ($10 E^{-9}$, 10^{-10} or 10^{-11}). At some symbol rates, a better than scale reading will appear as 0.0×10^{-00} . At other symbol rates, it will appear as E^{**} . In either case, they both mean performance is better than the scale upper limit.
BIT ERRORS	Displays the current error count from the Viterbi Decoder.
FREQ OFFSET (Hz)	Displays the received carrier frequency offset as measured by the modem.
TOTAL PACKETS	Displays the total number of Ethernet packets received from the satellite <i>(Only visible when Ethernet is selected as the interface type).</i>
ERROR PACKETS	Displays the number of erred Ethernet packets received from the satellite <i>(Only visible when Ethernet is selected as the interface type).</i>
PKT ERROR RATE	Displays the satellite Packet Error Rate <i>(Only visible when Ethernet is selected as the interface type).</i>
PKT STATS RESET	Allows the user to reset the Ethernet packet statistics by pressing <Enter> <i>(Only visible when Ethernet is selected as the interface type).</i>
LINK STATUS (menu)	<i>(the following sub-menus only display when Ethernet is selected as the interface type)</i>



NOTE

The status of the following ports may be one of the following:

<i>Down:</i>	<i>The link is down.</i>
<i>Unresolved:</i>	<i>Unable to agree on connection speed.</i>
<i>10 Mbps Half:</i>	<i>Connected at 10 Base-T Half Duplex.</i>
<i>10 Mbps Full:</i>	<i>Connected at 10 Base-T Full Duplex.</i>
<i>100 Mbps Half:</i>	<i>Connected at 100 Base-T Half Duplex.</i>
<i>100 Mbps Full:</i>	<i>Connected at 100 Base-T Full Duplex.</i>

PORT 1 STATUS	{See the note above} Displays the current status of LAN Port 1.
PORT 2 STATUS	{See the note above} Displays the current status of LAN Port 2.
PORT 3 STATUS	{See the note above} Displays the current status of LAN Port 3.
PORT 4 STATUS	{See the note above} Displays the current status of LAN Port 4.
WAN STATUS	{See the note above} Displays the current status of the WAN Port.
VOLTAGES (menu)	
TEMPERATURE	Displays internal temperature.
+1.5V RX SUPPLY	Displays the measured voltage of the 1.5 Volt Rx power bus located inside the modem.
+1.5V TX SUPPLY	Displays the measured voltage of the 1.5 Volt Tx power bus located inside the modem.
+3.3V SUPPLY	Displays the measured voltage of the +3.3 Volt power bus located inside the modem.
+5V SUPPLY	Displays the measured voltage of the +5 Volt power bus located inside the modem.
+12V SUPPLY	Displays the measured voltage of the +12 Volt power bus located inside the modem.
+20V SUPPLY	Displays the measured voltage of the +20 Volt power bus located inside the modem.
-12V SUPPLY	Displays the measured voltage of the -12 Volt power bus located inside the modem.
RX BUFFER LEVEL	{0 – 100%} Displays the status of the Doppler Buffer.
RX BUFFER RESET (ENTER)	Allows the user to re-center the Doppler Buffer when <ENTER> is pressed on the keypad.

4.3.7 Alarms Menu Options and Parameters



Masking alarms may cause undesirable modem performance.

CURRENT ALARMS (menu)

TX MAJOR (menu)	<u>Status</u> <u>Edit Table</u>
FPGA CFG	{Pass/Fail, Unmasked/Masked} Indicates a transmit FPGA configuration failure.
DSP CFG	{Pass/Fail, Unmasked/Masked} Indicates a transmit DSP configuration failure.
SCT CLOCK PLL	{Pass/Fail, Unmasked/Masked} Indicates that the Tx SCT Clock PLL is not locked. This alarm will flash on during certain modem parameter changes. A solid indication points toward a configuration failure within the modem.
SYM CLOCK PLL	{Pass/Fail, Unmasked/Masked} Indicates that the Tx Symbol Clock PLL is not locked. This alarm will flash on during certain modem parameter changes. A solid indication points toward a problem with the incoming clock to the modem (SCTE).
LB SYNTH PLL	{Pass/Fail, Unmasked/Masked} Indicates that the Tx L-Band Synthesizer is not locked. This alarm will flash on during certain modem parameter changes. A solid indication points toward a configuration problem within the modem.
IF SYNTH PLL	{Pass/Fail, Unmasked/Masked} Indicates that the Tx IF Synthesizer is not locked. This alarm will flash on during certain modem parameter changes. A solid indication points toward a configuration problem within the modem.
ETHERNET WAN	{Pass/Fail/Unmasked/Masked} Indicates that the interface card is faulted and will not pass data (<i>displays only when the Ethernet Card is installed and the Ethernet Interface is selected</i>).
FORCED ALARM	{DISABLED, ENABLED} Allows the user to disable or enable forcing of the Tx summary alarm. It does not affect Tx output.
TX MINOR (menu)	
TERR CLK ACT	{Pass/Fail, Unmasked/Masked} Indicates no Terrestrial Clock activity.
TERR DATA ACT	{Pass/Fail, Unmasked/Masked} Indicates no Tx Data activity.

TX TERR AIS	{Pass/Fail, Unmasked/Masked} Indicates that AIS has been detected in the Tx Data Stream.
DnI FRAME LOCK	{Pass/Fail, Unmasked/Masked} Indicates that the TX Oversample Clock PLL is not locked. This alarm will flash on during certain modem parameter changes. A solid indication points toward a configuration problem within the modem.
DnI M-FRAME LOCK	{Pass/Fail, Unmasked/Masked} Indicates that Tx Composite Clock PLL is not locked. This alarm will flash on during certain modem parameter changes. A solid indication points toward a problem with the incoming clock to the modem (SCTE).
TX DVB FRM LOCK	{Pass/Fail, Unmasked/Masked} Indicates that Tx input data stream framing does not match the user selected TX TERR FRAMING.
RX MAJOR (menu)	
FPGA CFG	{Pass/Fail, Unmasked/Masked} Indicates a receive FPGA hardware failure.
DSP CFG	{Pass/Fail, Unmasked/Masked} Indicates a receive DSP failure.
SIGNAL LOCK	{Pass/Fail, Unmasked/Masked} Indicates that the demod is unable to lock to a signal.
INPUT LVL THRESH	{Pass/Fail, Marked/Unmarked} Indicates the Rx level has fallen below the input threshold level.
FRAME LOCK	{Pass/Fail, Unmasked/Masked} Indicates that the Framing Unit is unable to find the expected framing pattern.
MULTIFRAME LOCK	{Pass/Fail, Unmasked/Masked} This alarm will flash on during certain modem parameter changes. A solid indication points toward a problem with the incoming clock to the modem (SCTE).
LB SYNTH PLL	{Pass/Fail, Unmasked/Masked} Indicates that the Rx L-Band Synthesizer is not locked. This alarm will flash on during certain modem parameter changes. A solid indication points toward a configuration problem within the modem.
IF SYNTH PLL	{Pass/Fail, Unmasked/Masked} Indicates that the Rx IF Synthesizer is not locked. This alarm will flash on during certain modem parameter

changes. A solid indication points toward a configuration problem within the modem.

ETHERNET WAN Indicates that the interface card is faulted and will not pass data (***displays only when the Ethernet Card is installed and the Ethernet Interface is selected***).

FORCED ALARM **{DISABLED, ENABLED}**
Allows the user to disable or enable forcing of the Tx summary alarm.

RX MINOR (menu)

BUFF UNDERFLOW **{Pass/Fail, Unmasked/Masked}**
Indicates that a Doppler Buffer underflow has occurred.

BUFF NEAR EMPTY **{Pass/Fail, Unmasked/Masked}**
Indicates that the Doppler Buffer is about to underflow.

BUFF NEAR FULL **{Pass/Fail, Unmasked/Masked}**
Indicates that the Doppler Buffer is about to overflow.

BUFF OVERFLOW **{Pass/Fail, Unmasked/Masked}**
Indicates that a Doppler Buffer overflow has occurred.

RX DATA ACTIVITY **{Pass/Fail, Unmasked/Masked}**
Indicates that there is no Rx Data activity.

SAT AIS **{Pass/Fail, Unmasked/Masked}**
Indicates that AIS has been detected in the receive satellite data stream.

DnI FRAME LOCK **{Pass/Fail, Unmasked/Masked}**
Indicates if drop/insert data is frame locked.

DnI M-FRAME LOCK **{Pass/Fail, Unmasked/Masked}**
Indicates if drop/insert data has multiframe lock.

INSERT CRC **{Pass/Fail, Unmasked/Masked}**
Indicates if the Circular Redundancy Check is passing in PCM-30C and PCM-31C Modes.

T1/E1 SIGNALING **{Pass/Fail, Unmasked/Masked}**
TBD

IFEC LOCK **{Pass/Fail, Unmasked/Masked}**
Indicates that the Framing Unit is unable to find the expected framing pattern.

OFEC LOCK **{Pass/Fail, Unmasked/Masked}**
Indicates that the Reed-Solomon Decoder is not locked.

INTERLEAVER **{Pass/Fail, Unmasked/Masked}**
Indicates that the Reed-Solomon Interleaver is not synchronized.

RS UNCORR	{Pass/Fail, Unmasked/Masked} Indicates status of the Reed-Solomon uncoded word fault.
TPC IFC LOCK	{Pass/Fail, Unmaskd, Masked}
EBNO (dB)	{Pass/Fail, Unmasked/Masked} Indicates that the Eb/No is outside of limits.
RX AGC LEVEL	{Pass/Fail, Unmasked, Masked} Indicates Rx level is below allowable limts.
IBS BER	{Pass/Fail, Unmasked/Masked} Indicates that there are more than one in 1000 bits in error in IBS Mode.
RX DVB FRM LOCK	{Pass/Fail, Unmasked/Masked} Indicates that the Rx Satellite Data Stream Framing is not DVB.
COMMON (menu)	
TERR FPGA CFG	{Pass/Fail, Unmasked/Masked} Indicates an Interface Card FPGA configuration failure.
CODEC FPGA CFG	{Pass/Fail, Unmasked/Masked} Indicates Turbo Codec Card FPGA configuration failure.
CODEC DEV CFG	{Pass/Fail, Unmasked/Masked} Indicates Turbo Codec Card ASIC configuration failure.
VOLTAGES (menu)	
+1.5V RX SUPPLY	{Pass/Fail, Unmasked/Masked} Displays the measured voltage of the 1.5 Volt Rx power bus located inside the modem.
+1.5V TX SUPPLY	{Pass/Fail, Unmasked/Masked} Displays the measured voltage of the 1.5 Volt Tx power bus located inside the modem.
+3.3V SUPPLY	{Pass/Fail, Unmasked/Masked} Displays the measured voltage of the +3.3 Volt power bus located inside the modem.
+5V SUPPLY	{Pass/Fail, Unmasked/Masked} Displays the measured voltage of the +5 Volt power bus located inside the modem.
+12V SUPPLY	{Pass/Fail, Unmasked/Masked} Displays the measured voltage of the +12 Volt power bus located inside the modem.
-12V SUPPLY	{Pass/Fail, Unmasked/Masked} Displays the measured voltage of the -12 Volt power bus located inside the modem.

+20V SUPPLY**{Pass/Fail, Unmasked/Masked}**

Displays the measured voltage of the +20 Volt power bus located inside the modem.

EXT CLOCK ACT**{Pass/Fail, Unmasked/Masked}**

Indicates that the External Clock is not active.

EXT REF ACT**{Pass/Fail, Unmasked/Masked}**

Indicates no activity on the External Reference.

EXT REF LOCK**{Pass/Fail, Unmasked/Masked}**

Indicates that the External Reference PLL is not locked.

**NOTE**

Refer to Section 4.3.7, "CURRENT ALARMS (menu)" for an explanation of the Latched Alarms Menu Options and Parameters.

LATCHED ALARMS

The following alarms are latched in order to catch intermittent failures:

TX MAJOR (menu)**{Pass/Fail}****FPGA CFG****DSP CFG****SCT CLOCK PLL****SYM CLOCK PLL****LB SYNTH PLL****IF SYNTH PLL****ETHERNET WAN****TX MINOR (menu)****TERR CLK ACT****TERR DATA ACT****TX TERR AIS****DnI FRAME LOCK**

DnI M-FRAME LOCK

DROP CRC

TX DVB FRM LOCK

RX MAJOR (menu)

FPGA CFG

DSP CFG

SIGNAL LOCK

FRAME LOCK

MULTIFRAME LOCK

LB SYNTH PLL

IF SYNTH PLL

ETHERNET WAN

RX MINOR (menu)

BUFF UNDERFLOW

BUFF NEAR EMPTY

BUFF NEAR FULL

BUFF OVERFLOW

RX DATA ACTIVITY

SAT AIS

DnI FRAME LOCK

DnI M-FRAME LOCK

INSERT CRC

T1/E1 SIGNALING

IFEC LOCK

OFEC LOCK

INTERLEAVER

RS UNCORR. WORD

TPC IFEC LOCK

EBNO

RX LEVEL

IBS BER

RX DVB FRM LOCK

COMMON (menu)

TERR FPGA CFG

CODEC FPGA CFG

CODEC DEV CFG

VOLTAGE (menu)

+1.5V RX SUPPLY

+1.5V TX SUPPLY

+3.3V SUPPLY

+5V SUPPLY

+12V SUPPLY

-12V SUPPLY

+20V SUPPLY

EXT CLOCK ACT

EXT REF ACT

EXT REF LOCK

CLEAR LATCHED ((ENTER)) Allows the user to reset the latched alarms by pressing <ENTER> on the keypad.

BACKWARD ALARMS

Backward alarms are alarms that are fed back to or received from the other end of the satellite link. In IBS Mode (including Drop & Insert), Backward Alarm 1 is the only one used. It would be received if the distant end demod drops lock.



NOTE

For the following alarms: RCV = YES indicates a received backward alarm. FRC allows the user to force the alarm locally.

IDR BACKWARD 1	{RCV = NO/YES, FRC = NO/YES}
IDR BACKWARD 2	{RCV = NO/YES, FRC = NO/YES}
IDR BACKWARD 3	{RCV = NO/YES, FRC = NO/YES}
IDR BACKWARD 4	{RCV = NO/YES, FRC = NO/YES}
MAP SUMMARY	{NONE, BK 1; BK 2; BK 1, 2; BK 3; BK 1, 3; BK 2, 3; BK 1, 2, 3; BK 4; BK 1,4; BK 2,4; BK 1, 2,4; BK 3,4; BK 1, 3,4; BK 2, 3,4; BK 1, 2, 3,4} <i>TBD</i>



NOTE

For the following alarms: XMIT = YES indicates that the modem is currently transmitting the corresponding backward alarm.

IDR BACKWARD 1	{XMT = NO/YES}
IDR BACKWARD 2	{XMT = NO/YES}
IDR BACKWARD 3	{XMT = NO/YES}
IDR BACKWARD 4	{XMT = NO/YES}
IBS PROMPT	{Pass/Fail, FRC = No/Yes} Indicates that a prompt maintenance alarm is generated. Only valid in IBS Framing.
IBS SERVICE	{Pass/Fail, FRC = No/Yes} Indicates that a service alarm is generated. Only valid in IBS Framing.

4.3.8 System Menu Options and Parameters

DATE (MM/DD/YY)	Allows the user to enter the current date.
TIME {HH:MM:SS}	Allows the user to enter the current time.
CONFIG COPY	{CURRENT, CFG1....FG10} Allows the user to save and copy modem configurations.
FRONT PANEL (menu)	
BRIGHTNESS	{100%, 75%, 50%, 25%} Allows the user to control the VFD brightness.
BKLT TIMEOUT	{00 - 99} Allows the user to enter the length of time (in seconds) of keyboard inactivity before the backlight shuts off. 00 = no timeout.
KEY CLICK	{ON, OFF} Allows the user to enable or disable the audible beep each time a key is pressed. Illegal entries will still cause a beep to be heard.
MENU NAVIGATION	{STANDARD, FLOATING}
LED TEST	{ENTER} Allows user to test all front panel LEDs.
REMOTE CONTROL	{TERMINAL, COMPUTER} Allows the user to select between terminal RS-232 control and remote port M&C RS-232/-485 control.
TERMINAL (menu)	
TYPE	{VT-100, WYSE50, VIEWPOINT} Allows the user to select the emulation type.
BAUD RATE	{150, 300, 600, 1200, 2400, 4800, 9600, 19200, 38400, 57600} Allows the user to enter the terminal baud rate.
REMOTE PORT (menu)	
ADDRESS	{32 - 255} Allows the user to enter the Remote Port Multidrop Address.
BAUD RATE	{150, 300, 600, 1200, 2400, 4800, 9600, 19200, 38,400} Allows the user to enter the Remote Port Baud Rate.
INTERFACE	{RS-232, RS-485} Allows the user to enter the Remote Port interface type.

TCP/IP (menu)**BOOT MODE****{DEFAULT, NON-VOL, BOOTP}**

DEFAULT: During initialization (boot up), the modem will restore the web setting to the standard IP Mask and addresses supplied by the modem. The modem will be taken off the network and will not be accessible. The Default settings are:

IP Address Mask:	255.000.000.000	(FF.00.00.00 hex)
Modem IP Address:	010.000.000.001	(C0.A8.00.EE hex)
Server IP Address:	010.001.001.001	(0A.01.01.01 hex)
Router IP Address:	010.000.001.001	(0A.00.01.01 hex)

BOOTP: During initialization (boot up), the modem will get the names, masks, and IP Addresses of the modem, router, and server.

NON-VOL: Stores and uses IP Mask and addresses as provided by the user.

IP TEST: Stores and uses IP Mask and addresses to fixed settings as listed below.

Bootp Server Tag:	206	
IP Address Mask:	255.255.255.000	(FF.FF.FF.00 hex)
Modem IP Address:	192.168.0.238	(C0.A8.00.EE)
Server IP Address:	192.168.000.101	(C0.A8.00.65)
Router IP Address:	192.168.000.102	(C0.A8.00.66)

BOOTp SERVER**{128 – 257, default is 206}**

Only used if Bootp is selected in Boot Mode. Should be consistent with the tag expected by the users Bootp Server.

MODEM HOST

The Host Modem for the network.

IP ADDR MASK**{XXX.XXX.XXX.XXX}**

The IP Address Mask of the local network. The mask is expressed in a hexadecimal format, and must be a valid TCP/IP Mask. This field should be set before changes are made to the Modem or Router Address.

MODEM IP ADDR**{XXX.XXX.XXX.XXX}**

The IP Address of the modem. This address should be consistent for the mask defined. This address is expressed in hexadecimal format. Broadcast and loop back addresses will not be allowed. These are addresses with all subnet bits set to 0's or 1's.

SERVER IP ADDR**{XXX.XXX.XXX.XXX}**

The IP Address of the Boot Server and the address of the SNMP Trap Server when SNMP is active. If a server is used and there is no local router, this address must be consistent with the modem address. If a router has been

specified, the address is presumed to be reachable via the router. Broadcast and loop back addresses will not be allowed. These are addresses with all subnet bits set to 0's or 1's.

ROUTER IP ADDR**{XXX.XXX.XXX.XXX}**

The IP Address of the Local Network Router. If a router is present on the local network, this address must be consistent with the IP Mask and the subnet of the modem. If no router is present, then the address should be set to a foreign address. This address is expressed in hexadecimal format. Broadcast and loop back addresses will not be allowed. These are addresses with all subnet bits set to 0's or 1's.

MODEM EADDR**{001065010000}**

Displays the Ethernet address of the device. Set at the factory and is a unique identifier for the Ethernet physical interface.

ETHER RATE**{10 MBPS/HD}**

The data rate for the local Ethernet Interface.
10 Mbps/HD – for 10 Base-T in either half-duplex or full duplex.

SNMP (menu)

A description of OID organization is provided in the MIB portion of this manual (Appendix C).

SNMP VERSION**{V1 & V2, V3}**

This selection controls the SNMP Version that will be used in messaging between the equipment and it's host.

When V1 & V2 is used, RD COMMUNITY and RDWR COMMUNITY are used to determine the authorization of an incoming message.

When V3 is used, three contexts are supported: **public**, **mib2**, and **dev**. Context, Authentication and Privacy are a portion of each SNMPV3 message.

The **public** context will only allow the user to see the sysoid of the unit. This is the most restricted access possible and only allows the unit to be identified by a host SNMP Station.

The **mib2** context allows a user with appropriate authentication to access the mib2 OIDs and the SNMP OIDs. These are of interest primarily to network operators not controlling the satellite link.

The **dev** context allows a user with appropriate authentication to access the device control portion of the MIB. These OIDs are used to control the devices satellite link and operation.

TRAP VERSION**{V1, V2}**

This controls the type of message format used when a message trap is generated by the equipment and bound for a SNMP Host. Messages will only be sent if the unit has been authorized to do so.

AUTHORIZATION**{TRAPS OFF, TRAPS ON}**

This controls the type of message format used when a message trap is generated by the equipment and bound for a SNMP host. Messages will only be sent if the unit has been authorized to do so.

RD COMMUNITY**{16 characters of name}**

This menu is only displayed when SNMP VERSION is set to V1 & V2.

This is the community that a host must be acting within when an OID variable is requested by a V1/V2 SNMP message.

RDWR COMMUNITY**{16 characters of name}**

This menu is only displayed when SNMP VERSION is set to V1 & V2.

This is the community that a host must be acting within when an OID variable is being changed by a V1/V2 SNMP message.

FTP (menu)**USER ID**

Allows the user to enter the user identification for access to an FTP session.

PASSWORD

Allows the user to enter the password for access to an FTP session.

WEB**CONFIRMATION****{ENABLE, DISABLE}****USER 1****ACCESS GROUP****{NO GROUP, GUEST, OPER, ADMIN}**

Access rights represent the following:

No Group: Denies Access

Guest: Users are able to navigate most of the site, and view modem parameter settings.

Oper: Users can monitor and control parameter settings, and change their own authentication passwords.

Admin: At this highest access right, the users can monitor and control the modems parameters, change any user's name and authentication password, and modify IP network settings. Admin setting allows access to the entire site.

AUTH PASSWORD**{xxxxxxxx}**

User to select password. The user can modify the Uthorization Passwords. The user name can have up to 14 characters supporting alpha and numeric characters. Alpha characters can be entered using the up and down arrow keys. Numeric characters can be selected by using the number keys on the front panel. The user can clear all characters from the front panel screen.

USER RESET

Resets group and password.

USER 2**ACCESS GROUP****{NO GROUP, GUEST, OPER, ADMIN}**

Access rights represent the following:

No Group: Denies Access

Guest: Users are able to navigate most of the site, and view modem parameter settings.

Oper: Users can monitor and and control parameter settings, and change their own authentication passwords.

Admin: At this highest access right, the users can monitor and control the modems parameters, change any user's name and authentication password, and modify IP network settings. Admin setting allows access to the entire site.

AUTH PASSWORD**{xxxxxxxx}**

User to select password. The user can modify the Uthorization Passwords. The user name can have up to 14 characters supporting alpha and numeric characters. Alpha characters can be entered using the up and down arrow keys. Numeric characters can be selected by using the number keys on the front panel. The user can clear all characters from the front panel screen.

USER RESET

Resets group and password.

USER 3**ACCESS GROUP****{NO GROUP, GUEST, OPER, ADMIN}**

Access rights represent the following:

No Group: Denies Access

Guest: Users are able to navigate most of the site, and view modem parameter settings.

Oper: Users can monitor and control parameter settings, and change their own authentication passwords.

Admin: At this highest access right, the users can monitor and control the modems parameters, change any user's name and authentication password, and modify IP network settings. Admin setting allows access to the entire site.

AUTH PASSWORD

{xxxxxxxx}

User to select password. The user can modify the Uthorization Passwords. The user name can have up to 14 characters supporting alpha and numeric characters. Alpha characters can be entered using the up and down arrow keys. Numeric characters can be selected by using the number keys on the front panel. The user can clear all characters from the front panel screen.

USER RESET

Resets group and password.

HW/FW CONFIG (menu)

FIRMWARE REV

Displays the installed firmware revision.

M&C REV

Displays the installed Monitor and Control revision.

MAIN BOARD (menu)



NOTE

Only the appropriate of the VCO adjustment screens listed below will be displayed. These are protected fields, to prohibit accidental changes. To edit the field, the user must depress all four of the direction arrow keys simultaneously.

INT VCO ADJUST

{0% - 100%}

Allows the user to adjust the internal frequency reference for calibration. **Only displayed of the system reference clock is INTERNAL.**

HI STAB VCO ADJUST	{0% - 100%} Allows the user to adjust the internal frequency reference for calibration. Only displayed if the system reference clock source is HI STABILITY.
LARGEST HB GAP	Used for factory test only.
SOFT RESET	{ENTER} Resets the modem.
IF BOARD (menu)	Indicates the Radyne part number for the IF Board Assembly.
AGC/CTRL/VALUE	{0% - 100%} Allows the user to adjust the internal frequency reference for calibration. Only displayed if the system reference clock source is HI STABILITY.
I OFFSET	Used for factory test only.
Q OFFSET	Used for factory test only.
IF RX LVL OFFSET	Used for factory test only.
LB RX LVL OFFSET	Used for factory test only.
POWER SOURCE	Used for factory test only.
TERR INTFC BRD	Indicates the Radyne part number for the Terrestrial Interface Assembly.
CODEC BOARD (menu)	Indicates the Radyne part number for the Codec Board.
TPC FPGA IMAGE	Used for factory test only.
RS FPGA IMAGE	Used for factory test only.
TPC CODEC IMAGE	Used for factory test only.
FEATURES (menu)	
5012.2840.2417	{____.____.____} Allows the user to install purchased feature upgrades (see Appendix A). Whether or not the feature has been upgraded, or if the feature requires a hardware upgrade (contact the Radyne Customer Service Department).
UPGRADE LIST (menu)	The following menu screens display the following:
10 MBPS, 20 MBPS, 52 MBPS	(The highest option installed will hide the lower rates.)
RXIF	
RXLBAND	

TXIF
 TXLBAND
 ENH ASYNC
 IDR
 SEQ
 RS
 RS CUSTOM
 IBS
 D&I
 AUPC
 8PSK
 16QAM
 TURBO 52 MBPS
 DVB
 EDMAC

4.3.9 Test Menu Options and Parameters

TX TEST PATTERN	{NONE, 2047, 2¹⁵-1, 2²³-1} Allows the user to enable the tests listed above.
RX TEST PATTERN	{NONE, 2047, 2¹⁵-1, 2²³-1} Allows the user to enable the tests listed above.
PATTERN SYNC	{YES, NO} Yes indicates that the RX Test Pattern is in sync.
TST PAT ERR CNT	{NO SYNC, nnnn x 10ⁿ} Displays the number of errors detected by the test pattern checker.
TST PATT BER	{NO SYNC, nnnn x 10ⁿ} Displays the measured BER for the test pattern.
RESTART TST PAT ((ENTER))	Allows the user to restart the test by pressing <ENTER> on the keypad.
LOOPBACK	{IF, TERR TX/RX, BASEBAND TX/RX, NONE, TERR RX, BASEBAND RX, TERR TX, BASEBAND TX, IFEC TX} Terrestrial Loopback is performed at the Terrestrial Interface.

IF: IF loopback loops the IF output of the Modulator to the IF input of the Demodulator. If using 8PSK or 16QAM Modulation, the output power must be above -15 dB.

TERR TX/RX: Enables both. Baseband loopback is performed at interface between the Baseband Processor Card and the Modem Card. This ensures Framing/Deframer integrity.

BASEBAND TX/RX: Enables both Baseband Tx and Baseband Rx.

NONE: No loopback performed.

TERR RX: (Distant Loop) Sends received satellite data to the Modulator for transmission to the distant end.

BASEBAND RX: Sends Rx data from the Modem Card to the Tx data input to the Modem Card.

TERR TX: Sends Tx Terrestrial Data to Rx data out.

BASEBAND TX: Sends Tx data to the receive input to the BB Card.

Only available when

using SCC framing.

CARRIER TYPE

{NORMAL, CW, DUAL, OFFSET, POS FIR, NEG FIR}
Allows the user to set the type of carrier.

NORMAL: Causes the Modulator to output normal modulation.

CW: Causes the Modulator to output a pure carrier.

DUAL: Causes a double sideband output.

OFFSET: Causes a single sideband output.

POS FIR: For manufacturer's use only.

NEG FIR: For manufacturer's use only.

4.4 DMD2050 Strap Codes

The Strap Code is a quick set key that sets many of the modem parameters. For quick setup of the DMD2050, Strap Codes are very helpful. When a Strap Code is entered, the modem is automatically configured for the code's corresponding data rate, overhead, code rate, framing, scrambler type and modulation. An example of how to set a strap code follows:

Example: At the Front Panel <Modulator> Menu, depress '↓', then move '→' to the 'Strap Code' Submenu and enter #16. The DMD2050 will be automatically configured to the parameters shown below in the highlighted row 'Strap Code 16'.

Use the Strap Code Guide (Table 4-4) for available strap codes.

Table 4-4. DMD2050 Strap Codes										
Dis = Disable										
Strap Code (Decimal)	Data Rate (Kbps)	Overhead	Code Rate	Type	Framing Type	Scrambler Type	Drop and Insert	Reed-Solomon	Modulation	Mode
1	64	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
2	128	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
3	256	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
5	384	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
6	512	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
9	768	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
4	1536	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
10	1920	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
8	2048	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
12	2048	1*	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
16	1544	96K	3/4	VIT	IDR	V.35 (IESS)	Dis	Dis	QPSK	IDR
32	2048	96K	3/4	VIT	IDR	V.35 (IESS)	Dis	Dis	QPSK	IDR
64	6312	96K	3/4	VIT	IDR	V.35 (IESS)	Dis	Dis	QPSK	IDR
128	8448	96K	3/4	VIT	IDR	V.35 (IESS)	Dis	Dis	QPSK	IDR
24	56	1	1/2	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
33	56	1	3/4	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
34	64	1	1/2	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
36	64	1	3/4	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
40	128	1	1/2	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
48	128	1	3/4	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
65	256	1	1/2	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
66	256	1	3/4	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
68	320	1	1/2	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
72	320	1	3/4	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
80	384	1	1/2	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
96	384	1	3/4	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
129	512	1	1/2	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
130	512	1	3/4	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
132	768	1	1/2	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT

136	768	1	3/4	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
144	896	1	1/2	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
44	896	1	3/4	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
7	1344	1	1/2	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
11	1344	1	3/4	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
13	1536	1	1/2	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
14	1536	1	3/4	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
19	1544	1	1/2	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
21	1544	1	3/4	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
22	1920	1	1/2	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
25	1920	1	3/4	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
26	2048	1	1/2	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
28	2048	1	3/4	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
37	2368	1	1/2	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
38	2368	1	3/4	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
41	48	1	1/2	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
160	1544	965/1024	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
52	1920	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
69	6312	1	3/4	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
70	8448	1	3/4	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
73	3152	1	1/2	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
74	3152	1	3/4	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
76	3264	1	1/2	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
81	3264	1	3/4	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
88	512	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
97	1024	1	1/2	VIT	CNT	V.35 (IESS)	Dis	Dis	QPSK	CNT
98	1024	1	3/4	VIT	CNT	V.35 (IESS)	Dis	Dis	QPSK	CNT
112	64	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
131	128	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
133	256	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
134	192	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
137	192	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
138	320	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
140	320	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
145	384	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
100	448	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
146	448	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS

104	576	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
148	576	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
152	640	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
161	640	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
162	704	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
164	704	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
168	768	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
193	832	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
194	832	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
196	896	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
208	896	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
224	960	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
15	960	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
23	1024	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
27	1024	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
29	1536	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
30	1088	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
39	1088	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
43	1152	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
46	1152	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
51	1216	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
53	1216	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
54	1280	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
57	1280	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
58	1344	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
67	1408	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
71	1408	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
75	1472	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
77	1472	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
78	1600	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
83	1600	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
85	1664	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
86	1664	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
89	1728	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
90	1728	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
92	1792	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
99	1792	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS

101	2048	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
102	1856	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
105	1856	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
106	2048	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
120	1544	965/1024	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
135	1984	16/15	1/2	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
139	1984	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
45	3088	1	1/2	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
141	3088	1	3/4	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
176	4000	1	1/2	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
116	4000	1	3/4	VIT	NONE	V.35 (IESS)	Dis	Dis	QPSK	CNT
60	1344	16/15	3/4	VIT	IBS	IBS	Dis	Dis	QPSK	IBS
200	128	NONE	1/2	VIT	NONE	V.35(IESS)	Dis	Dis	QPSK	CNT

4.5 Sample DMD2050 Applications

The following section provides brief application notes for operating the DMD2050 and explains by example how to configure the DMD2050 for some of the most popular configurations.

The following information illustrates the allowable combinations for Mode and Data Rate for the DMD2050.

Allowable Combinations: Mode/Rate/Framing.

MIL STD 188 165A:

IDR:

8.448 Mbps	3/4, 7/8 Rate FEC
6.312 Mbps	1/2, 3/4, 7/8 Rate FEC
2.048 Mbps	1/2, 3/4, 7/8 Rate FEC
1.544 Mbps or Below	1/2, 3/4, 7/8 Rate FEC

IBS:

2.048 Mbps or below	1/2, 3/4, 7/8 Rate
---------------------	--------------------

Closed Network:

8.448:	96 Kb Framing or No Framing, 3/4, 7/8 Rate FEC
6.312:	96 Kb Framing or No Framing, 1/2, 3/4, 7/8 Rate FEC
2.048:	96 Kb Framing or 1/15 Framing or No Framing, 1/2, 3/4, 7/8 Rate FEC
1.544:	96 Kb Framing or 1/15 Framing or No Framing, 1/2, 3/4, 7/8 Rate FEC
Any Rate 2.048 & lower:	1/15 Framing or No Framing, 1/2, 3/4, 7/8 Rate FEC

DVB:

4.5.1 Operational Case Examples



NOTE

For best results always begin setup by setting the data rate to 512 Kbps. This data rate is applicable for all modes and as such provides a convenient launch point for setting up the modem. Any mode of operation can be entered from this starting point.

Case 1: IDR 8.448 Mbps, 3/4 Rate Viterbi

Starting with the Data Rate = 512 Kbps

Modulator:

Method 1 -

Set mode to IDR
Under Mod Data Menu:
Set code rate to 3/4 VIT
Set data rate for 8448000
Under Interface Menu:
Set Interface type
Set Tx clock selection
Under Mod IF Menu:
Set desired Tx frequency and power level
Turn IF ON

Method 2 -

Set Mod strap code to: 128
Under Interface Menu:
Set Interface type
Set Tx clock selection
Under Mod IF Menu:
Set desired Tx frequency and power level
Turn IF on

Demodulator:**Method 1 -**

Set mode to IDR
Under Demod IF Menu: Set desired Rx frequency
Under Demod data Menu:
 Set code rate to 3/4 VIT
 Set data rate for 8448000
Under Interface Menu:
 Set Interface type
 Set Buff clock selection
 Set Buffer Size

Method 2 -

Set Demod strap code to 128
Under Demod IF Menu, set desired Rx frequency
Under Interface Menu:
 Set Interface type
 Set Buff clock selection
 Set Buffer Size

Case 2: IBS 1.544 Mbps, 3/4 Rate Viterbi

Starting with the Data Rate – 512 Kbps

Modulator:**Method 1 -**

Set Framing to 1/15
Set mode to IBS
Under Mod Data Menu:
 Set code rate to 3/4 VIT
 Set data rate for 1544000
Under Interface Menu:
 Set Interface type
 Set Tx clock selection
Under Mod IF Menu: Set desired Tx frequency and power level
 Turn IF ON

Method 2 -

Set Mod strap code to: 120
Under Interface Menu:
 Set Interface type
 Set Tx clock selection
Under Mod IF Menu: Set desired Tx frequency and power level
 Turn IF on

Demodulator:

Method 1 - Set Framing to 1/15:
Set mode to IBS:
Under Demod IF Menu: Set desired Rx frequency
Under Demod Data Menu:
Set code rate to 3/4 VIT
Set data rate for 1544000
Under Interface Menu:
Set Interface type
Set Buff clock selection
Set Buffer Size

Method 2 - Set Demod strap code to: 120
Under Demod IF Menu: Set desired Rx frequency
Under Interface Menu:
Set Interface type
Set Buff clock selection
Set Buffer Size

Case 3: Closed Network, 3/4 Rate Viterbi, IBS Overhead

Starting with the Data Rate = 512 Kbps

Modulator:

Method 1 - Set mode to IDR:
Under Mod Data Menu:
Set code rate to 3/4 VIT
Set Framing for 1/15
Under Interface Menu:
Set Interface type
Set Tx clock selection
Under Mod IF Menu: Set desired Tx frequency and power level
Turn IF ON

Method 2 - Set Mod strap code to: 101
Under Interface Menu:
Set Interface type
Set Tx clock selection
Under Mod IF Menu: Set desired Tx frequency and power level
Turn IF on

Demodulator:**Method 1 -**

Set mode to: Closed Net
Under Demod IF Menu: Set desired Rx frequency
Under Demod data Menu:
 Set code rate to 3/4 VIT
 Set Framing for 1/15

Under Interface Menu:
 Set Interface type
 Set Buff clock selection
 Set Buffer Size

Method 2 -

Set Demod strap code to: 101
Under Demod IF Menu: Set desired Rx frequency
Under Interface Menu:
 Set Interface type
 Set Buff clock selection
 Set Buffer Size

Case 4: Loop Timing Example**Method 1 -**

Set mode to IBS
Under Interface Menu:
 Under Tx Setup Menu:
 Set INTF to RS-422
 Set SCT Source to SCR
 Set Tx Clock to SCTE

Method 2 -

Set mode to Closed Net
Under Interface Menu:
 Under Tx Setup Menu:
 Set INTF to RS-422
 Set SCT Source to SCR
 Set Tx Clock to SCTE

4.6 Configuring the DMD2050 for Drop and Insert

Several dependencies exist when configuring the modem for Drop and Insert (D&I). The following paragraphs explain these dependencies and provide the user with the information required to ensure smooth transition into D&I and to minimize the potential impact of these dependencies.

4.6.1 Data Rate

Data Rate affects the Drop and Insert function in the following ways:

- It determines the number of Satellite Channels that will be displayed in the Edit Maps.
- It contributes to the Operational Mode selection process. Trying to change the Operational Mode to D&I when a data rate is not set to a valid D&I rate will result in the error message 'INVALID DATA RATE,' and the mode change will not be allowed.
- It contributes to the Terrestrial Framing Mode selection process. Trying to select a T1-type Drop Mode such as T1-ESF with the mod data rate set to 1920000 bps (a valid E1 D&I rate but not a valid T1 rate) will result in the error message 'INVALID DROP MODE' and the selection will not be allowed. Trying to select a T1 type Insert Mode such as T1-D4 with the demod data rate set to 1920000 bps will result in the error message INVALID INSERT MODE and the selection will not be allowed.
- Once D&I Mode has been selected, trying to change the data rate to something other than another valid D&I data rate will result in the error message 'RATE OUT OF BOUNDS' and the change will not be allowed.
- Once D&I Mode has been selected with a T1 Terrestrial Framing Mode, attempting to change the data rate to 1920000 will result in the error message 'RATE OUT OF BOUNDS' and the change will not be allowed.

Therefore, the data rate should be entered as the first step in configuring the modem for D&I. The Mod Data Rate should be set according to the number of timeslots to be dropped and the Demod Data Rate should be set according to the number of timeslots to be inserted. The following table gives the allowable D&I data rates based on the number of slots (n) to be dropped or inserted.

n = 1, data rate = 64000
n = 2, data rate = 128000
n = 3, data rate = 192000
n = 4, data rate = 256000
n = 5, data rate = 320000
n = 6, data rate = 384000
n = 8, data rate = 512000
n = 10, data rate = 640000
n = 12, data rate = 768000
n = 15, data rate = 960000
n = 16, data rate = 1024000
n = 20, data rate = 1280000
n = 24, data rate = 1536000
n = 30, data rate = 1920000 (valid with E1 Interface only)

4.6.2 Operational Network Specification

The Network Specification of the Modem often determines which additional menus and displays are available for use by the operator. The D&I Mode-specific menus will not be displayed unless the Network Spec. of the modem is set to D&I. Therefore, the second step in configuring the modem should be to set the Network Spec. to D&I. At this point, the D&I specific menus in the Interface section will become available and will remain available until the Network Spec. of the modem is changed to something other than D&I. When the Network Spec. is changed to something other than D&I, the D&I specific menus will automatically disappear.

4.6.3 Terrestrial Framing - Drop Mode/Insert Mode

The Drop Mode Selection and the Insert Mode Selection identify the Terrestrial Data-Framing Format. As previously mentioned, their selection is influenced by the Modulator and Demodulator Data Rates, and trying to select a T1 Type Framing Format with a data rate of 1920000 bps will result in an error message. In turn, the selection of the terrestrial framing formats influences the satellite channel to terrestrial timeslot mappings in the following manner:

The selection of T1-D4, T1-ESF, or SLC-96 type terrestrial framing format limits the terrestrial timeslots to values from 1 - 24.

The selection of PCM-30 or PCM-30C type terrestrial framing limits the terrestrial timeslots to values from 1 - 15, 17 - 31. In these modes, terrestrial timeslot 16 is reserved for ABCD signaling and may not be dropped or inserted.

The selection of PCM-31 or PCM-31C type terrestrial framing limits the terrestrial timeslots to values from 1 - 31. Therefore, the terrestrial framing format should be identified via the Drop Mode and Insert Mode entries prior to editing the Drop or Insert satellite channel to terrestrial timeslot maps.

4.6.3.1 Insert Terrestrial Frame Source

The Insert Terrestrial Frame Source selection tells the Modem from where the Insert Terrestrial Frame is coming.

External: Indicates that the terrestrial frame is to be input via the Insert Data In Port.

Internal: Indicates that the modem needs to generate the terrestrial frame and that all non-inserted timeslots need to be filled with the appropriate idle code based upon the terrestrial framing (T1 or E1).

The selection of the Insert Terrestrial Frame Source also influences the Buffer Clock selection in the following manner:

When the Insert Terrestrial Frame Source selection is set to External, the received satellite data will be clocked out of the Doppler Buffer based upon the clock recovered from the insert data input. Therefore, the Buffer Clock selection will automatically be set to External and cannot be modified. Attempts to select a different buffer clock will result in the error message INVALID BUFFER CLOCK and the selection will not be allowed.

When the Insert Terrestrial Frame Source selection is set to Internal, the operator needs to specify how data should be clocked out of the Doppler Buffer. In this case, the operator will be able to select SCTE, SCT, RX SAT, or EXT EXC as the source for the Buffer Clock. Therefore, the Insert Terrestrial Frame Source selection should be made prior to attempting to change the Buffer Clock. In most instances, the Insert Terrestrial Frame Source selection will be set to External and the Buffer Clock will automatically be set to External.

4.6.4 D&I Sample Configurations and D&I Clock Setup Options

The following are several examples of how to configure the modem for D&I. Also, refer to Figures 3-14 through 3-17 for the D&I Clocking Setup Options Available.

Example 1: Drop 512 Kbps from a T1 trunk, 3/4 Rate Viterbi
Insert 512 Kbps into a T1 trunk, 3/4 Rate Viterbi
Drop 512 Kbps from a T1 trunk, 3/4 Rate Viterbi

Under Modulator:

Under Mod Data:

Set Data Rate = 512000

Set Conv Enc = 3/4 Rate VIT

Under Modulator:

Set Network Spec. = Drop & Insert

Under Interface:

Under TX Setup:

Set Tx Type according to your hardware configuration (example:

G703BT1B8ZS)

Set Tx Clock = SCTE

Under Tx D&I:

Set Drop Mode = T1-D4

Use SATCh TS edit capability to define desired mapping of
Satellite Channels to drop Terrestrial Slots

Use Map Copy to copy Tx Edit to Tx Active

Under Modulator:

Under Mod IF:

Set Frequency to desired value

Turn IF Output Power On

Under Demodulator:

Under Demod Data:

Set Data Rate = 512000

Set Conv Enc = 3/4 Rate VIT

Under Demodulator:

Set Network Spec. = Drop & Insert

Under Interface:

Under RX Setup:

Set Rx Type according to your hardware configuration

Set Buff Size to desired depth

Under Rx D&I:

Set Insert Mode = T1-D4

Set T1 E1 Frm Src = External

Use SATCh TS edit capability to define proper mapping of
Satellite Channels to insert Terrestrial Slots

Use Map Copy to copy Rx Edit to Rx Active

Under Demodulator:

Under Demod IF:

Set Frequency to desired value

Example 2: Multidestinational Remote Site Programming
 Drop 512 Kbps from a T1 trunk, 3/4 Rate Viterbi.
 Extract 512 Kbps from a 1536 Kbps carrier and insert into a
 T1 trunk, 3/4 Rate Viterbi.

Drop 512 Kbps from a T1 trunk, 3/4 Rate Viterbi
 Configuration setup is exactly as previously shown in Example 1.

Extract 512 Kbps from a 1536 Kbps carrier and insert into a T1 trunk, 3/4 Rate
 Viterbi

Under Demodulator:

Under Demod Data:

Set Data Rate = 1536000

Set Conv Enc = 3/4 Rate VIT

Under Demodulator:

Set Network Spec. = Drop & Insert

Under Interface:

Under RX Setup:

Set Rx Type according to your hardware configuration

Set Buff Size to desired depth

Under Rx D&I:

Set Insert Mode = T1-D4

Set T1 E1 Frm Src = External

Use SATCh TS edit capability to define proper mapping of Satellite
 Channels to insert Terrestrial Slots

For Satellite Channels that are not to be inserted, enter "NI" (No Insert)
 for the Terrestrial Slot

Use Map Copy to copy Rx Edit to Rx Active

Under Demodulator:

Under Demod IF:

Set Frequency to desired value.

Figures 4-3 through 4-6 illustrate D&I Clock Setup Options

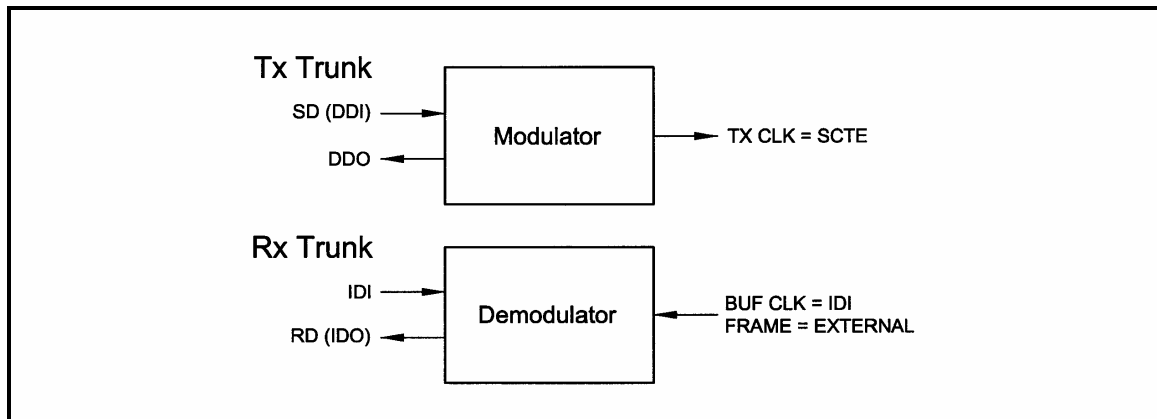


Figure 4-3. Transmit Trunk and Receive Trunk

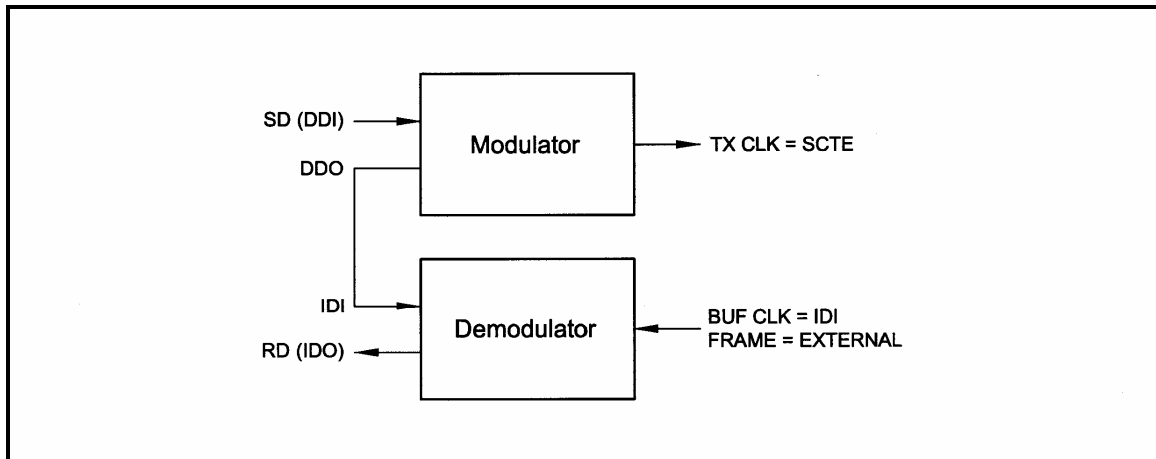


Figure 4-4. Single Trunk

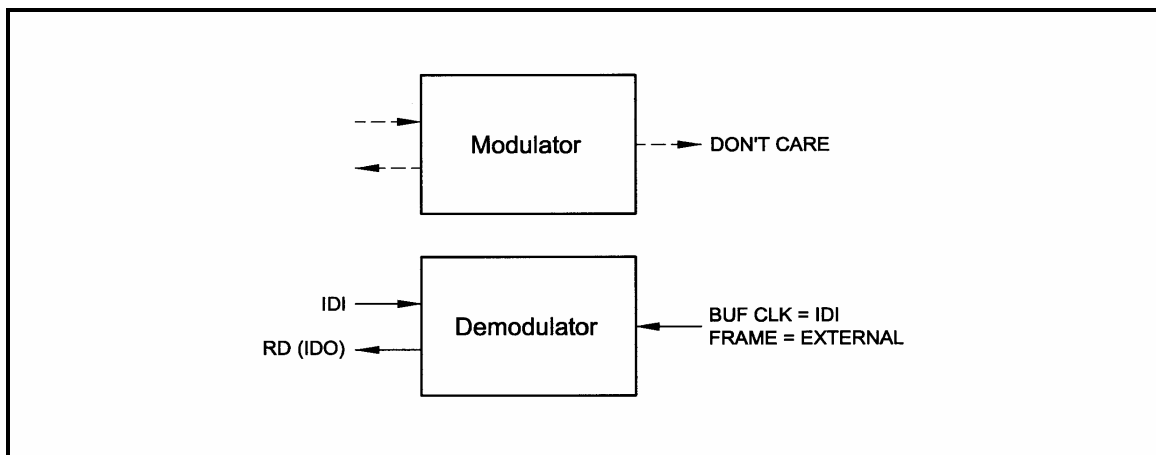


Figure 4-5. Rx Only With Trunk

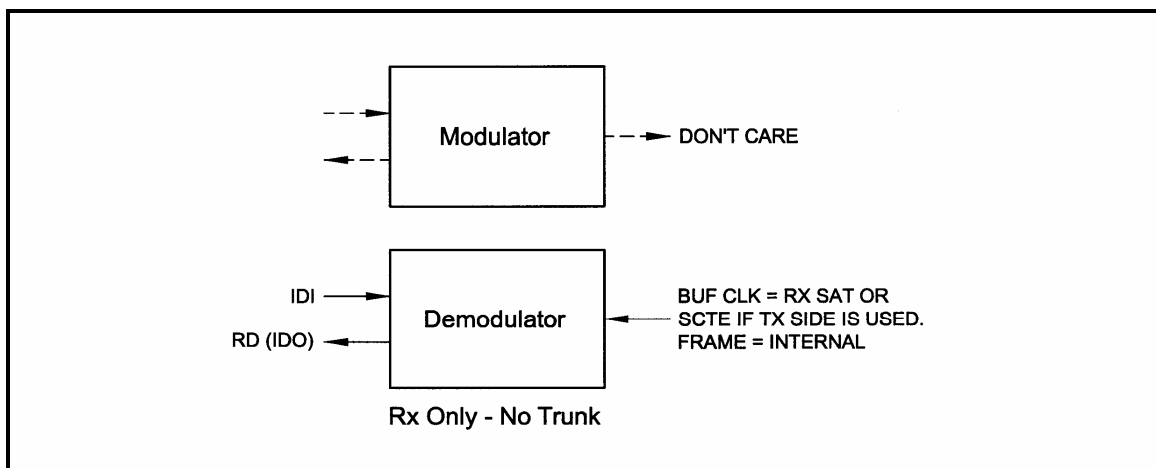


Figure 4-6. Rx Only No Trunk

4.7 D&I Maps and Map Editing

The Drop and Insert multiplexer is programmed by loading it with a transmit and receive map. Maps always contain 30 entries, although, only the first “n” entries are relevant (see Table 4-5).

The DMD2050 includes provisions to copy, change, and store the D&I transmit and receive maps directly from the Front Panel or via the remote M&C link. These maps are tables that are used to define and configure the D&I functions. Each map contains up to 30 entries, which are enough to define the channel assignments for a T1 (24 channel) or E1 (30 channel) frame structure. Maps that are created are stored in non-volatile battery backed-up memory within the modem and remain unchanged after a power-down.

Table 4-5. D&I Multiplexer Map Locations Used	
Data Rate (Kbps)	Map Locations Used (n = 1, 2,4,8,16,24,30)
64	1
128	1-2
256	1-4
384	1-6
512	1-8
768	1-12
1024	1-16
1536	1-24
1920	1-30

It is important to understand that each map contains up to 30 usable entries. In many cases a smaller number of entries will be relevant, except when the data rate is 1920 Kbps, in which case 30 entries will be used by the multiplexer. To determine the number of relevant entries, divide the data rate by 64 Kbps.

For example:

At 384 Kbps, $384/64 = 6$ entries.

Therefore, in this case only the first six entries of the map would be relevant.

The Modem is equipped with eight permanently stored default maps, which are designated ROM 1 through ROM 8. The user may also define, modify, and save an additional eight maps which are designated USER 1 through USER 8.

**NOTE**

ROM maps are read-only and may not be modified (refer to Table 4-6).

Table 4-6. D&I ROM Maps

ROM Map	T1/E1 Time Slot																																
	#	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30		
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
2	1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2	1
3	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1
4	1	2	3	4	5	6	1	2	3	4	5	6	1	2	3	4	5	6	1	2	3	4	5	6	1	2	3	4	5	6	1	2	3
5	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	1
6	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	4	5	6	7	8	9
7	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33
8	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33

Since the D&I Functions are separate and distinct, two separate maps must be configured at the start of the D&I Multiplexer Operation. These are the Tx (transmit) Active Map for Drop Mapping and the Rx (receive) Active Map for Insert Mapping. The number of entries in each map is determined by the data rates selected. Each map entry consists of an IBS Time Slot assignment and the Terrestrial (T1 or E1) Channel Number to which it is assigned. Drop Mapping and Insert Mapping are completely separate and independent.

The map that is actually used for the Drop Function is the Tx Active Map; the map that is actually used for the Insert function is the Rx Active Map. Two additional maps exist: the Tx Edit Map and the Rx Edit Map. The Edit Maps are the buffer areas that are used when creating or modifying a map through the modem's VFD; when editing is complete, the appropriate map should be copied to the Active Map.

Any map may be copied to any other map with the exception of the ROM maps. These maps may only be the source of the data used to create a User, Edit, or Active Map.

Maps can be created in the map editor and stored as "User Maps". New "Active Maps" can be downloaded during Modem Operation but this will result in a temporary disruption of service on the terrestrial line or the Satellite transmission.

The following paragraphs give examples of typical configurations that could use the ROM Maps as templates. The ROM Map used would have to be first copied to the appropriate Active Transmit (Drop) and/or Active Receive (Insert) Map(s) before it could be used. To use a modification of a ROM Map, the ROM Map must first be copied to the appropriate Edit Map, then modified, and then copied to the appropriate Active Map.



NOTE

The mapping of channels to time slots is arbitrary; it is not necessary to map CH1 to TS1, CH2 to TS2, etc. The channel to the time slot mapping may be in any order within the constraints of the number of available channels.

For example, ROM Map 1 could be used as the template for as Active Transmit (Drop) Map for a modulator configured for 64 Kbps operation. Only the first time slot of the T1 or E1 frame would be dropped into the modulator transmit path. The Drop Multiplexer would know to look only at the first entry in the Active Transmit table and would ignore the other 29 entries. If the map contained an "8" in its first entry, the eighth channel of the T1/E1 frame would be sent to the modulator.

ROM Map 2 could be used as the template for an Active Receive (Insert) Map for a demodulator configured for 128 Kbps operation. The demodulated data in the receive path would be inserted into the first two time slots of the T1 or E1 frame. The Insert Multiplexer would know to look only at the first two entries in the Active Receive table and would ignore the other 28 entries. If the first two entries were modified to contain a 27 and 28, the data would be inserted into the 27th and 28th time slots of the E1 frame.

ROM Map 3 could be used as the template for an Active Transmit (Drop) Map with a modulator and/or demodulator configured for 256 Kbps operation. The T1 or E1 Data in the transmit path or the demodulated data in the receive path would be dropped from and/or inserted into the first four time slots of the T1 or E1 frame. The Multiplexer would know to look only at the first four entries in the Active map(s) and would ignore the other 26 entries.

ROM Map 4 could be used as the template for an Active Transmit (Drop) or Active Receive (Insert) Map with a modulator and/or demodulator configured for 384 Kbps operation. The T1 or E1 Data in the transmit path or the demodulated data in the receive path would be dropped from and/or inserted into the first six time slots of the T1 or E1 frame. The Insert Multiplexer would know to look only at the first six entries in the Active map(s) and would ignore the other 24 entries. To Drop the last six channels of a T1 frame into a modulator transmit path, the first six entries of the Active Transmit map should contain 19, 20, 21, 22, 23, and 24.

ROM Map 5 could be used as the template for an Active Transmit (Drop) and/or Active Receive (Insert) Map with a modulator and/or demodulator configured for 512 Kbps operation. The T1 or E1 Data in the transmit path or the demodulated data in the receive path would be dropped from and or inserted into the first eight time slots of the T1 or E1 frame. The Multiplexer would know to look only at the first eight entries in the Active map(s) and would ignore the other 22 entries. To insert data received from a demodulator into channels 17 through 24 of an E1 frame, the first eight entries of the Active Receive map should contain 17, 18, 19, 20, 21, 22, 23, and 24.

ROM Map 6 could be used as the template for an Active Transmit (Drop) and/or Active Receive (Insert) Map with a modulator and/or demodulator configured for 768 Kbps operation. The T1 or E1 Data in the transmit path or the demodulated data in the receive path would be dropped from and or inserted into the first 12 time slots of the T1 or E1 frame. The Multiplexer would know to look only at the first 12 entries in the Active map(s) and would ignore the other 18 entries. To insert data received from a demodulator into channels 3 through 14 of an E1 frame, the first 12 entries of the Active Receive map should contain 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, and 14.

ROM Map 7 could be used as the template for an Active Transmit (Drop) and/or Active Receive (Insert) Map with a modulator and/or demodulator configured for 1920 Kbps operation. This would be used with E1 frames where time slot 16 is not used for the multiframe alignment signal and therefore channels 1 through 30 are mapped directly with time slots 1 through 30.

ROM Map 7 could also be used as the template for an Active Transmit (Drop) and/or Active Receive (Insert) Map with a modulator and/or demodulator configured for 1024 Kbps operation. This would be used with T1 or E1 frames where channels 1 through 16 are mapped into time slots 1 through 16 (in any order). Map slots 17 through 30 would be ignored.

ROM Map 7 could also be used as the template for an Active Transmit (Drop) and/or Active Receive (Insert) Map with a modulator and/or demodulator configured for 1536 Kbps operation. This would be used with T1 frames where channels 1 through 24 are mapped into time slots 1 through 24 (in any order). Map slots 25 through 30 would be ignored.

ROM Map 8 could also be used as the template for an Active Transmit (Drop) and/or Active Receive (Insert) Map with a modulator and/or demodulator configured for 1920 Kbps operation. However, this mapping would be relevant with E1 frames where time slot 16 is used for the multiframe alignment signal and therefore channels 1 through 30 are mapped to time slots 1 through 16 and 17 through 31.

4.8 Configuring the Modem to use the Ethernet Data Interface (Optional)

When the optional Ethernet Data Interface Card is installed, all of the Ethernet related menus become available and can be used to control the interface as follows:

Under the Interface Menu:

Under the Tx Setup Menu:

Set the Terrestrial Interface to Ethernet.

Set the Ethernet Flow Control as desired (refer to Section 4.8.1 for details).

Set the Ethernet Daisy Chain as desired (refer to Section 4.8.2 for details).

Set the Tx Clock to SCTE. Set the Ethernet QOS Type as desired (refer to

Section 4.8.3 for details).

Set the Ethernet QOS Queue as desired (refer to Section 4.8.4 for details).

Set the Tx Clock to SCTE.

Set the Tx Clock Polarity to Normal.

Under Drop & Insert:

Set the Drop Mode to None.

Under the Interface Menu:

Under the Rx Setup Menu:

Set the Terrestrial Interface to Ethernet.

Set the Buffer Size to Zero.

Set the Buffer Clock to Rx Sat.

Set the Buffer Clock Polarity to Normal.

Under Drop & Insert:

Set the Insert Mode to None.

4.8.1 Ethernet Flow Control

When disabled, if a packet is received for transmission and no packet buffer space is available, the incoming packet is discarded.

When enabled, flow control is used to throttle the transmission station in order to avoid overrunning the transmit buffers, which would in turn cause packets to be dropped. The throttling mechanism used depends upon the interface and whether it is half-duplex or full duplex.

4.8.1.1 Half-Duplex Flow Control

In half-duplex mode, the unit uses industry standard backpressure to support flow control as follows:

When available buffer space is almost gone, the modem will force a collision on the input port when it senses an incoming packet. This collision will cause the transmitting station to back off and retry the transmission.

The interface will stop forcing collisions as soon as free buffer space becomes available.

4.8.1.2 Full-Duplex Flow Control

In full-duplex mode, the interface implements IEEE 8802.3x flow control as follows:

When available buffer space is almost gone, the unit sends out a pause frame with the maximum pause time to stop the remote nodes from transmitting.

The interface sends out another pause frame with the pause time set to zero as soon as free buffer space becomes available.

4.8.2 Ethernet Daisy Chain

When disabled, Port 4 (JS4) on the Ethernet Data Interface operates normally. Data received on Port 4 that is not addressed to other equipment on the LAN side, is transmitted over the satellite.

When Port 4 is selected for Daisy Chain, any data received on Port 4 (JS4) is forwarded to of the other LAN side ports (Ports 1 - 3) and is not transmitted over the satellite. This is extremely useful in a point-to -multipoint configuration as illustrated in Figure 4-7.

4.8.3 Ethernet QOS Type

When Normal QOS is selected, the interface determines a packets priority based on the following:

- IEEE 803.3ac Tag when present
- IPv4 Type of Service / Differentiated Services Field
- Ipv6 Traffic Class

When Port Based QOS is selected, the interface determines the priority of a packed based upon the port on which it arrived.

- Port 1 (JS1) has the highest priority
- Port 2 (JS2) has the second highest priority
- Port 3 (JS3) has the second lowest priority
- Port 4 (JS4) has the lowest priority

4.8.4 Ethernet QOS Queue

When Fair Weighted queueing is selected, the interface transmits packets at a rate of 8, 4, 2, and 1 from the highest priority queue to the lowest respectively. With fair weighted queueing, all queues with data in them are guaranteed to receive some bandwidth.

When Strict Priority is selected, the interface transmits packets from the highest priority queue until it is empty. It then begins transmitting data from the next highest priority queue. If higher priority data arrives, the interface finishes the current packet and then goes back to transmitting packets from the higher priority queue until it is again empty. Care must be taken when selecting Strict Priority, as it is entirely possible for the lower priority queues to be stalled indefinitely.

4.8.5 Setting Up The DMD20/DMD20 LBST Ethernet Bridge To Operate Like A FIFO

In certain circumstances, it may be desirable to have the Ethernet interface operate in a FIFO like manner with no reordering of packets. This can be established by using a single port on the Ethernet interface and setting the Ethernet QOS Type to Port Based and the Ethernet QOS Queue to Strict Priority. When Setup and used in this manner, the packets will be transmitted in the exact order in which they are received.

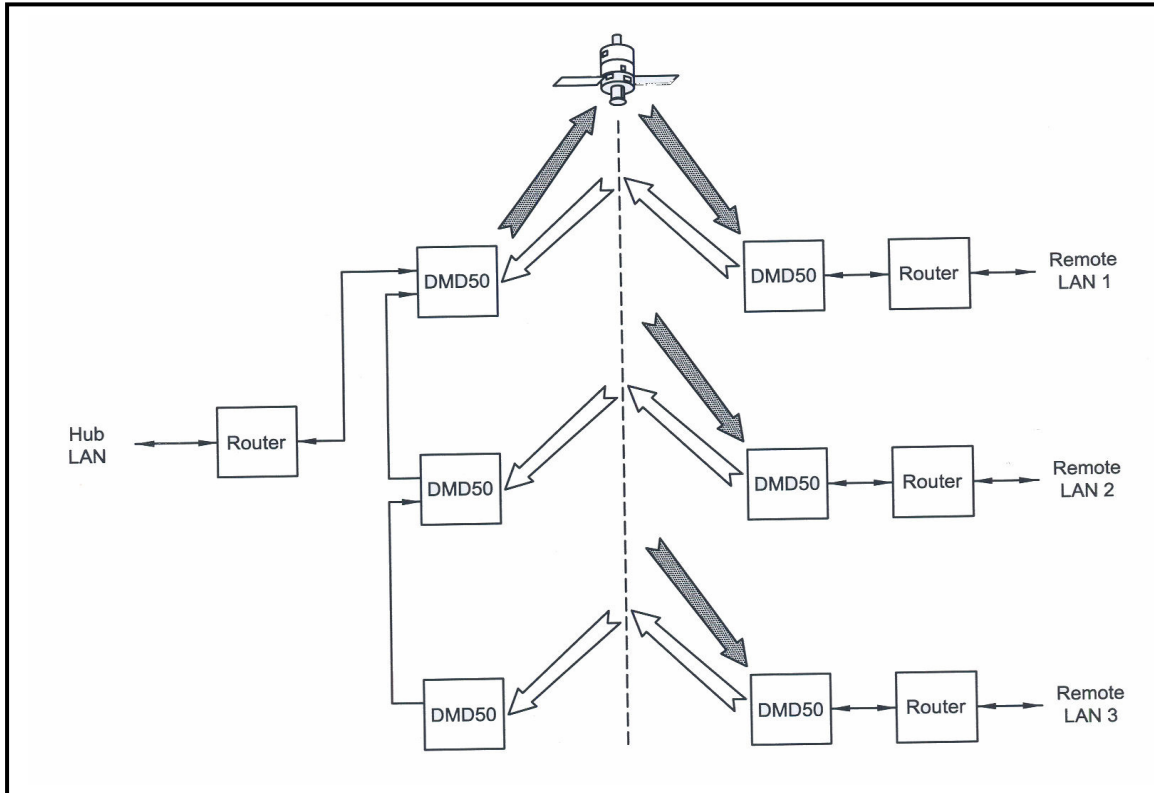


Figure 4-7. Point-to-Multipoint with Daisy Chaining

4.8.6 Packet Statistics

The following statistics are available under the Monitor Menu when the Ethernet Data Interface is selected.

Total Packets: This Counter displays the total number of Ethernet packets received from the satellite.

Error Packets: This counter displays the total number of Ethernet packets received from the satellite that had errors.

Packet Error Rate: This displays the Ethernet Packet Error Rate (PER) from the satellite.

Packet Statistics Reset: Allows the user to reset the Ethernet Total Packets and Ethernet Error Count by pressing <Enter>.

Link Status: The following status is available under the Monitor Menu/Link Status Sub-Menu when the Ethernet Data Interface is selected:

Port 1 Status: Displays the current status of LAN Port 1.

Port 2 Status: Displays the current status of LAN Port 2.

Port 3 Status: Displays the current status of LAN Port 3.

WAN Status: Displays the current status of the WAN Port.

For each of the above-listed ports, the status may take on one of the following values/meanings.

Down: The link is down.

Unresolved: Unable to agree on connection speed.

10 Mbps Half: Connected at 10 Base-T Half Duplex.

10 Mbps Full: Connected at 10 Base-T Full Duplex.

100 Mbps Half: Connected at 100 Base-T Half Duplex.

100 Mbps Full: Connected at 100 Base-T Full Duplex.

If all four LAN Ports are down, a Tx Data Activity Minor Alarm will be generated.

If the WAN Port is down, a Tx and Rx Ethernet WAN Major Alarm will be generated.

4.9 Terminal Mode Control

The DMD2050 Terminal Mode Control allows the use of an external terminal or computer to monitor and control the modem from a full screen interactive presentation operated by the modem itself. No external software is required other than VT-100 Terminal Emulation Software (e.g. "Procomm" for a computer when used as a terminal. The Control Port is normally used as an RS-232 Connection to the terminal device. The RS-232 operating parameters can be set using the modem Front Panel and stored in EEPROM for future use (refer to Section 4.11 for setup and terminal screens).



NOTE

Refer to the DMD20 Remote Protocol Manual (TM117) for the terminal screens.

4.9.1 Modem Terminal Mode Control

The modem can be interactively monitored and controlled in the Terminal Mode, with a full screen presentation of current settings and status. Programming is accomplished by selecting the item to be modified and pressing the terminal key of the option number. For example, to change the transmit data rate, enter '33' at the terminal. The modem will respond by presenting the options available and requesting input. Two types of input may be requested. If the input is multiple choice, the desired choice is selected by pressing the 'Space' key. When the desired option is displayed, press the 'Enter' key to select that option. The other possible input type requires a numerical input (such as entering a frequency or data rate).

This type of input is followed by pressing the 'Enter' or carriage return key. An input can be aborted at any time by pressing the 'ESC' key. Invalid input keys cause an error message to be displayed on the terminal.

The Terminal Control Mode supports serial baud rates of 150, 300, 1200, 2400, 4800, 9600, 19200, and 38400. The connection must be set for 8 data bits, 1 stop bit and no parity (8,N,1). Three terminal emulations are supported: VT-100, WYSE 50, and ADDS-VP.

"\$" is used for setting the screen when the terminal is used for the first time the non-volatile memory is reset.

4.9.2 Modem Setup for Terminal Mode

Terminal Mode Communications and Protocol is set from the Front Panel Control by setting the "Control Mode" Parameter to "Terminal", and then setting the "Modem Port", "Term Baud" and "Emulation" Parameters as desired. Then a terminal is connected to Connector J5 on the Back Panel. All operating software for the Terminal Mode is contained within the DMD2050 Modem Internal Control Software.

A "break" signal on the communications line, pressing "ESC" on the terminal or Power On of the modem will initiate full screen terminal mode printing and redraw the full screen. The Terminal Mode displays the present status of all user parameters controlled and read by the processor, and offers a menu allowing change to any controlled parameter.

The Terminal Mode uses eight "Screens," each of which have the basic contents of the three modem monitor and control areas as set in the Front Panel matrix columns. This screen is used for setting the parameters of the Modulator, Demodulator, Event, Alarm, Latched Alarm, Drop Controls, Insert Controls, and Interface Areas.

4.10 Terminal Port User Interface

The Remote Port (J20) of the DMD2050 allows for complete control and monitoring of all DMD2050 parameters and functions via an RS-232 Serial Interface, or RS-485 for RLLP Protocol. 'Terminal Mode' can be entered from the front panel by selecting "System" and then "Control Mode" followed by "Terminal". The baud rate and evaluation type can be changed at the front panel by using the *System>Baud Rate* Menu.

The Terminal Control Mode is menu-driven and the allowable values for each item number will be shown. To change an item, type in its number followed by <ENTER>. If the parameter to be changed requires a numeric value, enter the number followed by <ENTER>. If the parameter is non-numeric, press <SPACE> to cycle through the list of available entries.



Items that do not have ID numbers are Status only and cannot be changed.

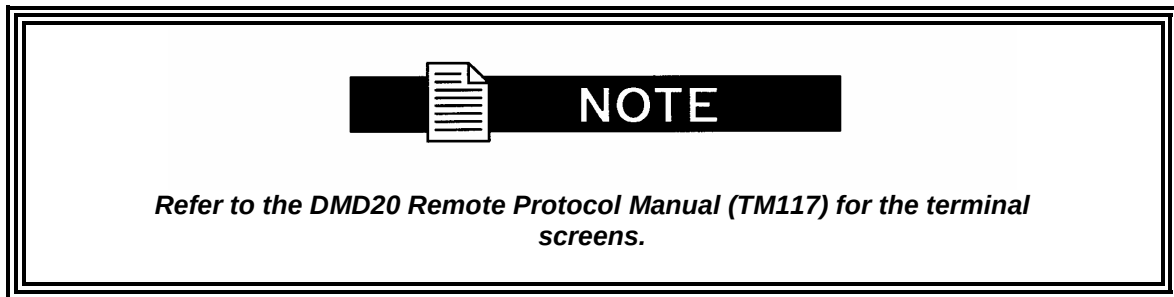
4.11 Connecting the Terminal

1. Connect the computer to the DMD2050 Remote Connector (J20) on the rear of the unit using the RS-232 Cable.
2. Enable the terminal by selecting Terminal Mode (located under the System - Control Mode Menu) from the front panel.
3. Verify that your emulation software is set to the following:

8 data bits
no parity
1 stop bit

Modify the DMD2050 selection, if necessary, to match the settings (the Front Panel 'SYSTEM' Sub-Menu contains all the Terminal Emulation Controls).

4.12 Terminal Screens





Rear Panel Interfaces

5

This section discusses the electrical interfaces available from the rear panel. All locations are as viewed from the rear of the unit unless otherwise specified.

5.0 DMD2050 Connections

All DMD2050 connections are made to labeled connectors located on the rear of the unit (refer to Figure 5-1). The connector definitions below are those on the DMD2050 unit. Any connection interfacing to the DMD2050 must be the appropriate mating connector.

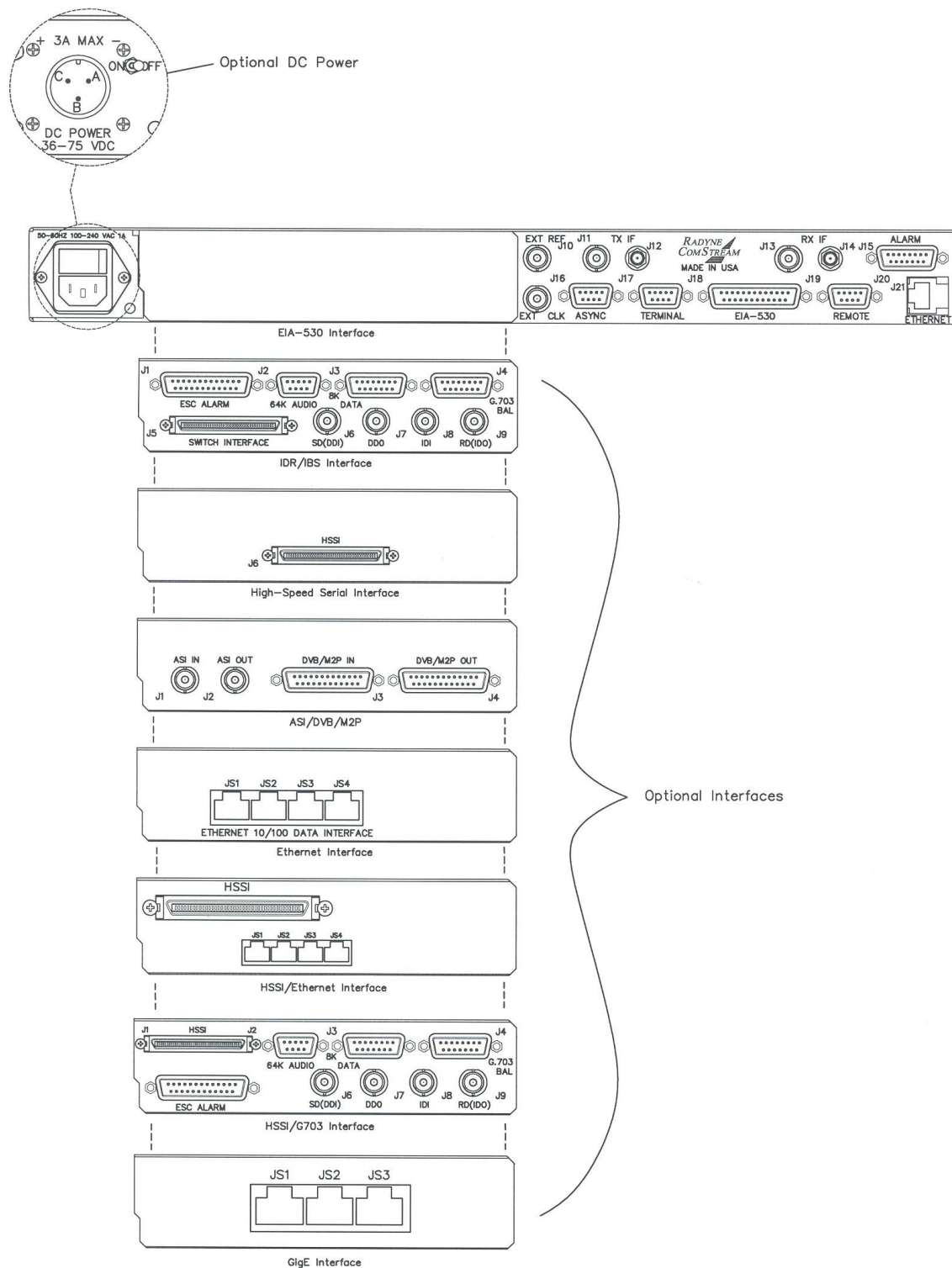


Figure 5-1. DMD2050 Universal Satellite Modem Rear Panel Configurations

5.1 Compact Flash

The compact flash slot is located on the right side as viewed from the rear of the unit. A 128 Mbit flash memory card stores all the modem M&C and operational data. It must be present when the modem is operating.

5.2 Power Input Modules

5.2.1 AC Power Input Module

AC Input Module (Figure 5-1) is located on the left side of the unit. Power applied to the port with the supplied power cable is 100 – 240 VAC, 50 – 60 Hz. Integrated into the Power Input Module is the Power On/Off Rocker Switch. Power consumption for the unit is 1A. A chassis ground connection (#10-32 threaded stud), is located to the lower right of the module.

5.2.2 DC Power Input/Switch

The Optional DC Power Input and Switch (Figure 5-1) is available for the DMD2050 products. The unit may be powered from a $48V \pm 5VDC$ source with a maximum unit power consumption of 3 A. Refer to Table 5-1 for pinouts.

Table 5-1. DC Power	
A	–
B	Ground
C	+

5.3 Chassis Connections (Standard)

5.3.1 EXT REF (J10)

The External Reference Port is a 50 Ohm Female BNC Connector and will accept the following frequencies: 1.0, 1.544, 2.0, 2.048, 5.0, and 10.0 MHz). Input level: 0 dBm $\pm 3dB$.

5.3.2 TX IF (J11)

The Transmit IF Output Port is a 50 Ohm Female TNC Connector that is used for 70/140 MHz IF. The power level is programmable from 0 to -25 dBm in 0.1 dBm steps. The IF Frequency can be programmed to 50 – 90 MHz or 100 – 180 MHz, in 1 Hz Steps.

5.3.3 TX L-Band IF (J12)

The Transmit IF Output Port is a 50 Ohm SMA Female Connector that can be used for L-Band IF. The power level is programmable from 0 to -25 dBm, in 0.1 dBm steps. The IF Frequency can be programmed to 950 – 2050 MHz, in 1 Hz Steps.

5.3.4 RX IF (J13)

The Receive IF Input Port is a 50 Ohm Female TNC Connector that is used for 70/140 MHz IF. If the customer orders the 70/140 MHz IF, the Transmit IF Output Port is a 50 Ohm Female TNC Connector.

5.3.5 RX L-Band IF (J14)

The Receive IF Input Port is a 50 Ohm SMA Female Connector that can be used for L-Band IF. The IF Frequency can be programmed from 950 to 2050 MHz in 1 Hz Steps.

5.3.6 ALARM (J15)

The Alarm Port is a 15-Pin Female “D” Connector. Refer to Table 5-2 for pinouts.

Table 5-2. ALARM Port 15-Pin Female “D” Connector (J15)			
Pin No.	Signal Name	Signal	Direction
1	Mod Fault - C	MF-C	No Direction
2	Mod Fault – NC	MF-NC	No Direction
3	Mod Fault – NO	MF-NO	No Direction
4	Demod Fault - C	DF-C	No Direction
5	Demod Fault – NC	DF-NC	No Direction
6	Demod Fault – NO	DF-NO	No Direction
7	Prompt - C	CEF-C	No Direction
8	Prompt – NC	CEF-NC	No Direction
9	Prompt – NO	CEF-NO	No Direction
10	Service – C	SP1-NO	No Direction
11	Service – NC	SP1-NC	No Direction
12	Service – NO	SP2-NO	No Direction
13	No Connect	SP2-NC	No Direction
14	AGC Out	AGC	No Direction
15	Ground	GND	---

5.3.7 EXT CLK (J16)

The External Clock Port is a 75 Ohm Female BNC Connector. It allows interfacing to an external clock source. Input level: .5 to 5 volts Peak-to-Peak.

5.3.8 ASYNC (J17)

The Asynchronous Data Interface Port is a 9-Pin Female “D” Connector. Refer to Table 5-3 for pinouts.

Table 5-3. ASYNC Port 9-Pin Female “D” Connector (J17)			
Pin No.	Signal Name	Signal	Direction
1	Receive Data B	RXD_B	Output
2	Receive Data A	RXD_A	Output
3	Transmit Data A	TXD_A	Input
4	Transmit Data B	TXD_B	Input
5	Ground	GND	---
6	No Connect	DSR	---
7	No Connect	RTS	---
8	No Connect	CTS	---
9	No Connect	NC	---

5.3.9 TERMINAL (J18)

Used for Radyne factory use only.

5.3.10 MIL-188-114A (J19)

The EIA-530 Port is an RS-422 Connection. It is a 25-Pin Female “D” Connector. Refer to Table 5-4 for pinouts.

Table 5-4. MIL-188-114A Port (RS-422) 25-Pin Female “D” Connector (J19)			
Pin No.	Signal Name	Signal	Direction
1	Shield	---	---
2	Send Data A (-)	SD-A	Input
3	Receive Data A (-)	RD-A	Output
4	Request To Send A (-)	RS-A	Input
5	Clear To Send A (-)	CS-A	Output
6	Data Mode A (-)	DM-A	Output
7	Signal Ground	SGND	---
8	Receiver Ready A (-)	RR-A	Output
9	Receive Timing B (+)	RT-B	Output
10	Receiver Ready B (+)	RR-B	Output
11	Terminal Timing B (+)	TT-B	Input
12	Send Timing B (+)	ST-B	Output

13	Clear T Send B (+)	CS-B	Output
14	Send Data B (+)	SD-B	Input
15	Send Timing A (-)	ST-A	Output
16	Receive Data B (+)	RD-B	Output
17	Receive Timing A (-)	RT-A	Output
18	Modulator Fault - Open Collector	MF	Output
19	Request To Send B (+)	RS-B	Input
20	Data Terminal Ready A (-)	TR-A	Input
21	Demodulator Fault	DF	Output
22	Data Mode B (+)	DM-B	Output
23	Data Terminal Ready B (+)	TR-B	Input
24	Terminal Timing A (-)	TT-A	Input
25	No Connect	---	---

5.3.11 REMOTE (J20)

The Remote Port is a RS-485 or RS-232 Connection for remote monitor and control of the modem. It is a 9-Pin Female “D” Connector. Refer to Table 5-5 for pinouts.

Table 5-5. Remote Port (RS-485 or RS-232) 9-Pin Female “D” Connector (J20)			
Pin No.	Signal Name	Signal	Direction
1	Transmit Data RS-485 (+)	TX-485-B	Output
2	Transmit Data RS-232	TXD-232	Output
3	Receive Data RS-232	RXD-232	Input
4	NC	NC	---
5	Ground	GND	---
6	Transmit Data RS-485 (-)	TX-485-A	Output
7	NC	No Connect	---
8	Receive Data RS-485 (+)	RX-485-B/CTS	Input
9	Receive Data RS-485 (-)	RX-485-A	Input

5.3.12 ETHERNET (J21)

The ETHERNET Port (J21) can be used for the Monitor & Control (M&C) Functions of the unit. The physical interface is a standard female RJ-45 Connector.

5.4 IDR/IBS Interface (Optional)

5.4.1 ESC ALARM (J1)

The ESC (Engineering Service Circuits) Alarms Port is a 25-Pin Female “D” Connector. Refer to Table 5-6 for pinouts.

Table 5-6. ESC ALARM Port 25-Pin Female “D” Connector (J1)			
Pin No.	Signal Name	Signal	Direction
1	Ground	GND	---
2	Backward Alarm Out - 1NO	ESCBWO 1NO	N/A
3	No Connection	NC	---
4	Backward Alarm Out - 2 NO	ESCBWO 2NO	N/A
5	No Connection	NC	---
6	Backward Alarm Out - 3 NO	ESCBWO 3NO	N/A
7	Ground	GND	---
8	Backward Alarm Out - 4 NO	ESCBWO 4NO	N/A
9	No Connection	NC	---
10	Backward Alarm In - 2	ESCBWI 2	Input
11	Backward Alarm In - 4	ESCBWI 4	Input
12	No Connection	NC	---
13	No Connection	NC	---
14	Backward Alarm Out - 1 C	ESCBWO 1C	N/A
15	Backward Alarm Out - 1 NC	ESCBWO 1NC	N/A
16	Backward Alarm Out - 2 C	ESCBWO 2C	N/A
17	Backward Alarm Out - 2 NC	ESCBWO 2NC	N/A
18	Backward Alarm Out - 3 C	ESCBWO 3C	N/A
19	Backward Alarm Out - 3 NC	ESCBWO 3NC	N/A
20	Backward Alarm Out - 4 C	ESCBWO 4C	N/A
21	Backward Alarm Out - 4 NC	ESCBWO 4NC	N/A
22	Backward Alarm In – 1	ESCBWI 1	Input
23	Backward Alarm In – 3	ESCBWI 3	Input
24	No Connection	NC	---
25	No Connection	NC	---

5.4.2 64K AUDIO (J2)

The 64K AUDIO Port allows for communications between Earth Stations. It is a 9-Pin Female “D” Connector that complies to IESS 308. Refer to Table 5-7 for pinouts.

Table 5-7. 64K AUDIO Port 9-Pin Female “D” Connector (J2)				
Pin No.	Signal Name	64K Mem	Audio Mode	Direction
1	Transmit Audio 1A	SD-A	ESCAUDTX 1A	Input
2	Receive Audio 1A	RD-A	ESCAUDRX 1A	Output
3	Ground	GND	GND	---
4	Transmit Audio 2B	ST-B	ESCAUDTX 2B	Input
5	Receive Audio 2B	RT-B	ESCAUDRX 2B	Output
6	Transmit Audio 1B	SD-B	ESCAUDTX 1B	Input
7	Receive Audio 1B	RD-B	ESCAUDRX 1B	Output
8	Transmit Audio 2A	ST-A	ESCAUDTX 2A	Input
9	Receive Audio 2A	RT-A	ESCAUDRX 2A	Output

5.4.3 8K DATA (J3)

The 8K Data Port allows for communications between Earth Stations. It is a 15-Pin Female “D” Connector that complies to IESS 308. Refer to Table 5-8 for pinouts.

Table 5-8. 8K DATA Port 15-Pin Female “D” Connector (J3)			
Pin No.	Signal Name	Signal	Direction
1	Receive Octet-B	ESCRXO-B	Output
2	Receive Clock-B	ESCRXC-B	Output
3	Receive Data-B	ESCRXD-B	Output
4	No Connection	NC	---
5	No Connection	NC	---
6	Transmit Data-A	ESCTXD-A	Input
7	Transmit Clock-A	ESCTXC-A	Output
8	Transmit Octet-A	ESCTXO-A	Output
9	Receive Octet-A	ESCRXO-A	Output
10	Receive Clock-A	ESCRXC-A	Output
11	Receive Data-A	ESCRXD-A	Output
12	Ground	GND	---
13	Transmit Data-B	ESCTXD-B	Input
14	Transmit Clock-B	ESCTXC-B	Output
15	Transmit Octet-B	ESCTXO-B	Output

5.4.4 G.703 BAL (J4)

The G.703 Interface Port (Balanced) is a 15-Pin Female “D” Connector. Refer to Table 5-9 for pinouts.

Table 5-9. G.703 BAL Port 15-Pin Female “D” Connector (J4)			
Pin No.	Signal Name	Signal	Direction
1	Send Data (-)	SD-A	Input
2	Ground	GND	---
3	Receive Data A (-)	RD-A	Output
4	Ground	GND	---
5	Drop Data Out (+)	DDO-B	Output
6	Insert Data In (+) EXC (+)	IDI-B	Input
7	External Clock A (-)	BAL EXC-A	Input
8	External Clock B (+)	BAL EXC-B	Input
9	Send Data (+)	SD-B	Input
10	No Connect	---	---
11	Receive Data B (+)	RD-B	Output
12	Drop Data Out (-)	DDO-A	Output
13	Insert Data In (-) EXC (-)	IDI-A	Input
14	Mod Fault	Mod Fault	Open Collector Output
15	Demod Fault	Demod Fault	Open Output Collector

5.4.5 SWITCH INTERFACE (J5)

The Switch Interface Port is a 68-Pin High-Density Female Connector. Refer to Table 5-10 for pinouts.

Table 5-10. SWITCH INTERFACE Port 68-Pin High-Density Female Connector (J5)			
Pin No.	Signal Name	Signal	Direction
1	G.703 Send Data Input A	G.703B SD-A	Input
2	Synchronous Data Send Data Input - A	SYNC SD-A	Input
3	IDR ESC Backward Alarm Out - 1 Common	ESCBWO 1C	No Direction
4	G.703 Insert Data Input – A	G.703B IDI-A	Input
5	Synchronous Data Send Timing Output – A	SYNC ST-A	Output

6	IDR ESC Backward Alarm Out - 1 Normally Open	ESCBWO 1NO	No Direction
7	Synchronous Data Terminal Timing Input – A	SYNC TT-A	Input
8	IDR ESC Backward Alarm Out - 2 Normally Closed	ESCBWO 2NC	No Direction
9	G.703 Drop Data Out A - Synchronous Data Receive Timing Output - A	DDO-A RT-A	Output
10	IDR ESC Backward Alarm Output - 3 Common	ESCBWO 3C	---
11	G.703 Insert Data Out A - Synchronous Data Receive Data A	IDO-A RD-A	Output
12	IDR ESC Backward Alarm Output - 3 Normally Open	ESCBWO 3NO	No Direction
13	External Clock Input - A	BAL EXC-A	No Connect
14	Ground	GND	---
15	IDR ESC Audio Input Channel 1A	ESCAUDTX 1A	Input
16	IDR ESC Audio Input Channel 2A	ESCAUDTX 2A	Input
17	IDR ESC Audio Output Channel 1A	ESCAUD RX 1A	Output
18	IDR ESC Audio Output Channel 2A	ESCAUD RX 2A	Output
19	IDR ESC Backward Alarm Input - 3	ESCBWI 3	Input
20	IBS ES Transmit Data A IDR ESC Backward Alarm Input 1	TXD-A BWI 1	Input
21	Mod Fault Open Collector Output	MOD FLT	Output Open Collector
22	IBS ES Receive Data Output - A	ES RXD-A	Output
23	IBS ES Data Set Ready (RS-232 Only)	ES DSR	No Connect
24	IDR ESC Transmit 8 Kbps Output Clock	ESCTXC-A	Output
25	IDR ESC Transmit 8 Kbps Output Data	ESCTXD-A	Input
26	IDR ESC Receive 8 Kbps Output Clock	ESCRXC-A	Output
27	IDR ESC Receive 8 Kbps Output Data	ESCRXD-A	Output
28	IDR ESC Backward Alarm Output - 4 Normally Closed	ESCBWO 4NC	---
29	IBS Transmit Octet Input - A	TXO-A	Input
30	Synchronous Data Mode A	SYNC DM-A	Output

31	Synchronous Data Clear to Send - A	SYNC CS-A	Output
32	IBS Receive Octet Output - A	RXO-A	Output
33	Synchronous Data Request to Send - A	SYNC RS-A	Input
34	Synchronous Data Receiver Ready - A	SYNC RR-A	Output
35	G.703 Send Data Input - B	G703B SD-B	Input
36	Synchronous Data Send Data Input - B	SYNC SD-B	Input
37	IDR ESC Backward Alarm Out - 1 Normally Closed	ESCBWO 1 NC	No Direction
38	G.703 Insert Data Input - B	G703B IDI-B	Input
39	Synchronous Data Send Timing Output - B	SYNC ST-B	Output
40	IDR ESC Backward Alarm Out - 2 Common	ESCBWO 2C	No Direction
41	Synchronous Data Terminal Timing – B	SYNC TT-B	Input
42	IDR ESC Backward Alarm Output - 2 Normally Open	ESCBWO 2NO	No Direction
43	G.703 Drop Data Out - B Synchronous Data Receive Timing – B	DDO-B RT-B	Output
44	IDR ESC Backward Alarm Out - 3 Normally Closed	ESCBWO 3NC	No Direction
45	G.703 Insert Data Out Synchronous Data	IDO-B RD-B	Output
46	IDR ESC Backward Alarm Out - 4 Common	ESCBWO 4C	No Direction
47	External Clock Input - B	BAL EXC-B	Input
48	Ground	GND	---
49	IDR ESC Audio Input Channel - 1B	ESCAUDTX 1B	Input
50	IDR ESC Audio Input Channel - 2B	ESCAUDTX 2B	Input
51	IDR ESC Audio Output Channel - 1B	ESCAUDRX 1B	Output
52	IDR ESC Audio Output Channel - 2B	ESCAUDRX 2B	Output

53	IDR ESC Backward Alarm Input - 4	ESCBWI 4	Input
54	IBS ES Transmit Data – B IDR ESC Backward Alarm Input - 2	TX-B BWI 2	Input
55	Demod Fault Open Collector Output	DMD FLT	Output Open Collector
56	IBS ES Receive Data Input - B	ES RXD-B	Output
57	Ground	GND	---
58	IDR ESC Transmit 8 Kbps Output Clock - B	ESCTXC-B	Output
59	IDR ESC Transmit 8 Kbps Output Data - B	ESCTXD-B	Input
60	IDR ESC Receive 8 Kbps Clock Output - B	ESCRXC-B	Output
61	IDR ESC Receive 8 Kbps Data Output - B	ESCRXD-B	Output
62	IDR ESC Backward Alarm Out - 4 Normally Open	ESCBWO 4NO	No Direction
63	IBS Transmit Octet Input - B	TXO-B	Input
64	Synchronous Data – Data Mode Out - B	SYNC DM-B	Output
65	Synchronous Data - Clear to Send - B	SYNC CS-B	Input
66	IBS Receive Octet Output - B	RXO-B	Output
67	Synchronous Data Request to Send – B	SYNC RS-B	Input
68	Synchronous Data Receiver Ready - B	SYNC RR-B	Output

5.4.6 SD (DDI) (J6)

The Send Data (Drop Data In) Port (Unbalanced) is a 75 Ohm Female BNC Connector.

5.4.7 DDO (J7)

The Drop Data Out Port (Unbalanced) is a 75 Ohm Female BNC Connector.

5.4.8 IDI (J8)

The Insert Data In Port (Unbalanced) is a 75 Ohm Female BNC Connector.

5.4.9 SD (IDO) (J9)

The Send Data (Insert Data Out) Port (Unbalanced) is a 75 Ohm Female BNC Connector.

5.5 High-Speed Serial Interface (HSSI) (Optional)

5.5.1 HSSI (J6)

The HSSI (High-Speed Serial Interface) (J6) complies with the HSSI Functional and Electrical Specifications. The physical interface is a 50-Pin SCSI-2 Type Connector. Electrical levels are ECL. The pinouts for this interface are listed in Table 5-11.

Table 5-11. J9 – HSSI (High-Speed Serial Interface) 50-Pin Connector				
Pin No. (+)	Pin No. (–)	Signal Name	Description	Direction
1	26	SG	Signal Ground	---
2	27	RT	Receive Timing	Output
3	28	CA	DCE Available	Output
4	29	RD	Receive Data	Output
6	31	ST	Send Timing (SCT)	Output
7	32	SG	Signal Ground	---
8	33	TA	DTE Available	Input
9	34	TT	Terminal Timing (SCTE)	Input
11	36	SD	Send Data	Input
13	38	SG	Signal Ground	---
14 - 18	39 – 43	5 Ancillary to DCE	Reserved	Input
19	44	SG	Signal Ground	---
20 - 23	45 - 48	4 Ancillary from DCE	Reserved	Output
24	49	TM	Test Mode	Output
25	N/A	MOD_FLT	Alarm	Output
50	N/A	DMD_FLT	Alarm	Output

5.6 ASI/DVB/M2P Interface (Optional)

5.6.1 ASI IN (J1)

The ASI IN Port (J1) is supported on the BNC Connector. The interface complies with DVB ASI Electrical Specifications.

5.6.2 ASI OUT (J2)

The ASI OUT Port (J2) is supported on the BNC Connector. The interface complies with DVB ASI Electrical Specifications.

5.6.3 DVB/M2P IN (J3)

DVB or M2P IN Port (J3) is supported on the DB-25 female connector. It complies with RS-422 Electrical Specifications. Refer to Table 5-12a for DVB and 5-12 b for M2P pinouts for this connector.

Table 5-12a. J3 – DVB In - 25-Pin Female		
Pin Number	Signal Name	Direction
1	CLK+	Input
14	CLK-	Input
2	SYSTEM GND	Input
15	SYSTEM GND	Input
3	D7+	Input
16	D7-	Input
4	D6+	Input
17	D6-	Input
5	D5+	Input
18	D5-	Input
6	D4+	Input
19	D4-	Input
7	D3+	Input
20	D3-	Input
8	D2+	Input
21	D2-	Input
9	D1+	Input
22	D1-	Input
10	D0+	Input
23	D0-	Input
11	DVALID+	Input

24	DVALID-	Input
12	PSYNC+	Input
25	PSYNC-	Input
13	Cable Shield	---

Table 5-12b. J3 – M2P In - 25-Pin Female		
Pin Number	Signal Name	Direction
1	OUTCLK+	Output
14	OUTCLK-	Output
2	CLK+	Input
15	CLK-	Input
3	SYNC+	Input
16	SYNC-	Input
4	VALID+	Input
17	VALID-	Input
5	D0+	Input
18	D0-	Input
6	D1+	Input
19	D1-	Input
7	D2+	Input
20	D2-	Input
8	D3+	Input
21	D3-	Input
9	D4+	Input
22	D4-	Input
10	D5+	Input
23	D5-	Input
11	D6+	Input
24	D6-	Input
12	D7+	Input
25	D7-	Input
13	Cable Shield	---

5.6.4 DVB/M2P OUT (J4)

The DVB or M2P OUT Port (J4) is also supported on the DB-25 Female Connector. It complies with RS-422 Electrical Specifications. Refer to Table 5-13a for DVB and 5-13 b for M2P pinouts for this connector.

Table 5-13a. J3 - DVB Out – 25-Pin Female ‘D’ Sub Connector		
Pin Number	Signal Name	Direction
1	CLK+	Output
14	CLK-	Output
2	SYSTEM GND	Output
15	SYSTEM GND	Output
3	D7+	Output
16	D7-	Output
4	D6+	Output
17	D6-	Output
5	D5+	Output
18	D5-	Output
6	D4+	Output
19	D4-	Output
7	D3+	Output
20	D3-	Output
8	D2+	Output
21	D2-	Output
9	D1+	Output
22	D1-	Output
10	D0+	Output
23	D0-	Output
11	DVALID+	Output
24	DVALID-	Output
12	PSYNC+	Output
25	PSYNC-	Output
13	Cable Shield	---

Table 5-13b. J3 - M2P Out – 25-Pin Female ‘D’ Sub Connector		
Pin Number	Signal Name	Direction
1	NC	Output
14	NC	Output
2	CLK+	Output
15	CLK-	Output
3	SYNC+	Output
16	SYNC-	Output
4	VALID+	Output
17	VALID-	Output
5	D0+	Output
18	D0-	Output
6	D1+	Output
19	D1-	Output
7	D2+	Output
20	D2-	Output
8	D3+	Output
21	D3-	Output
9	D4+	Output
22	D4-	Output
10	D5+	Output
23	D5-	Output
11	D6+	Output
24	D6-	Output
12	D7+	Output
25	D7-	Output
13	Cable Shield	---

5.7 Ethernet Data Interface (Optional)

The DMD20/DMD20 LBST Ethernet Data Interface provides four RJ-45, Auto-Crossover and Auto-Sensing, 10/100 Ethernet Data Ports. JS1 through JS4 may be referred to Port 1 through Port 4 respectively. Refer to Figures 5-1 and 5-2 for rear panel configurations.

JS1 is Port 1

JS2 is Port 2

JS3 is Port 3

JS4 is Port 4

Refer to Figure 5-1 and 5-2.

5.8 HSSI / G.703

The HSSI (High-Speed Serial Interface) (J1) complies with the HSSI Functional and Electrical Specifications. The physical interface is a 50-Pin SCSI-2 Type Connector. Electrical levels are ECL. The pinouts for this interface are listed in Table 5-14.

Table 5-14. J1 – HSSI (High-Speed Serial Interface) 50-Pin Connector				
Pin No. (+)	Pin No. (–)	Signal Name	Description	Direction
1	26	SG	Signal Ground	---
2	27	RT	Receive Timing	Output
3	28	CA	DCE Available	Output
4	29	RD	Receive Data	Output
6	31	ST	Send Timing (SCT)	Output
7	32	SG	Signal Ground	---
8	33	TA	DTE Available	Input
9	34	TT	Terminal Timing (SCTE)	Input
11	36	SD	Send Data	Input
13	38	SG	Signal Ground	---
14 - 18	39 – 43	5 Ancillary to DCE	Reserved	Input
19	44	SG	Signal Ground	---
20 - 23	45 - 48	4 Ancillary from DCE	Reserved	Output
24	49	TM	Test Mode	Output
25	N/A	MOD_FLT	Alarm	Output
50	N/A	DMD_FLT	Alarm	Output

5.8.1 64K AUDIO (J2)

The 64K AUDIO Port allows for communications between Earth Stations. It is a 9-Pin Female “D” Connector that complies to IESS 308. Refer to Table 5-15a for pinouts in audio mode and Table 5-15b for pinouts in 64k mode.

Table 5-15a. 64K AUDIO (In Audio Mode) Port 9-Pin Female “D” Connector (J2)			
Pin No.	Signal Name	Signal	Direction
1	Transmit Audio 1A	ESCAUDTX 1A	Input
2	Receive Audio 1A	ESCAUDRX 1A	Output
3	Ground	GND	---
4	Transmit Audio 2B	ESCAUDTX 2B	Input
5	Receive Audio 2B	ESCAUDRX 2B	Output
6	Transmit Audio 1B	ESCAUDTX 1B	Input
7	Receive Audio 1B	ESCAUDRX 1B	Output
8	Transmit Audio 2A	ESCAUDTX 2A	Input
9	Receive Audio 2A	ESCAUDRX 2A	Output

Table 5-15b. 64K AUDIO (In 64K Mode) Port 9-Pin Female “D” Connector (J2)			
Pin No.	Signal Name	Signal	Direction
1	Send Data A	SD-A	Input
2	Receive Data A	RD-A	Output
3	Ground	GND	---
4	Synchronous Data Send Timing B	ST-B	Input
5	Synchronous Data Receive Timing B	RT-B	Output
6	Send Data B	SD-B	Input
7	Receive Data B	RD-B	Output
8	Synchronous Data Send Timing A	ST-A	Input
9	Synchronous Data Receive Timing A	RT-A	Output

5.8.2 8K DATA (J3)

The 8K Data Port allows for communications between Earth Stations. It is a 15-Pin Female “D” Connector that complies to IESS 308. Refer to Table 5-16 for pinouts.

Table 5-16. 8K DATA Port 15-Pin Female “D” Connector (J3)			
Pin No.	Signal Name	Signal	Direction
1	Receive Octet-B	ESCRXO-B	Output
2	Receive Clock-B	ESCRXC-B	Output
3	Receive Data-B	ESCRXD-B	Output
4	No Connection	NC	---
5	No Connection	NC	---
6	Transmit Data-A	ESCTXD-A	Input
7	Transmit Clock-A	ESCTXC-A	Output
8	Transmit Octet-A	ESCTXO-A	Output
9	Receive Octet-A	ESCRXO-A	Output
10	Receive Clock-A	ESCRXC-A	Output
11	Receive Data-A	ESCRXD-A	Output
12	Ground	GND	---
13	Transmit Data-B	ESCTXD-B	Input
14	Transmit Clock-B	ESCTXC-B	Output
15	Transmit Octet-B	ESCTXO-B	Output

5.8.3 G.703 BAL (J4)

The G.703 Interface Port (Balanced) is a 15-Pin Female “D” Connector. Refer to Table 5-17 for pinouts.

Table 5-17. G.703 BAL Port 15-Pin Female “D” Connector (J4)			
Pin No.	Signal Name	Signal	Direction
1	Send Data (-)	SD-A	Input
2	Ground	GND	---
3	Receive Data A (-)	RD-A	Output
4	Ground	GND	---
5	Drop Data Out (+)	DDO-B	Output
6	Insert Data In (+) EXC (+)	IDI-B	Input
7	External Clock A (-)	BAL EXC-A	Input
8	External Clock B (+)	BAL EXC-B	Input
9	Send Data (+)	SD-B	Input

10	No Connection	---	---
11	Receive Data B (+)	RD-B	Output
12	Drop Data Out (-)	DDO-A	Output
13	Insert Data In (-) EXC (-)	IDI-A	Input
14	Mod Fault	MOD-FLT	Open Collector Output
15	Demod Fault	DMD-FLT	Open Collector Output

5.8.4 ESC ALARM (J5)

The ESC (Engineering Service Circuits) Alarms Port is a 25-Pin Female “D” Connector. Refer to Table 5-18 for pinouts.

Table 5-18. ESC ALARM Port 25-Pin Female “D” Connector (J1)			
Pin No.	Signal Name	Signal	Direction
1	Ground	GND	---
2	Backward Alarm Out - 1NO	ESCBWO 1NO	N/A
3	No Connection	NC	---
4	Backward Alarm Out - 2 NO	ESCBWO 2NO	N/A
5	No Connection	NC	---
6	Backward Alarm Out - 3 NO	ESCBWO 3NO	N/A
7	Ground	GND	---
8	Backward Alarm Out - 4 NO	ESCBWO 4NO	N/A
9	No Connection	NC	---
10	Backward Alarm In - 2	ESCBWI 2	Input
11	Backward Alarm In - 4	ESCBWI 4	Input
12	No Connection	NC	---
13	No Connection	NC	---
14	Backward Alarm Out - 1 C	ESCBWO 1C	N/A
15	Backward Alarm Out - 1 NC	ESCBWO 1NC	N/A
16	Backward Alarm Out - 2 C	ESCBWO 2C	N/A
17	Backward Alarm Out - 2 NC	ESCBWO 2NC	N/A
18	Backward Alarm Out - 3 C	ESCBWO 3C	N/A
19	Backward Alarm Out - 3 NC	ESCBWO 3NC	N/A
20	Backward Alarm Out - 4 C	ESCBWO 4C	N/A
21	Backward Alarm Out - 4 NC	ESCBWO 4NC	N/A
22	Backward Alarm In – 1	ESCBWI 1	Input

23	Backward Alarm In – 3	ESCBWI 3	Input
24	No Connection	NC	---
25	No Connection	NC	---

5.8.5 SD (DDI) (J6)

The Send Data (Drop Data In) Port (Unbalanced) is a 75-Ohm Female BNC Connector.

5.8.6 DDO (J7)

The Drop Data Out Port (Unbalanced) is a 75-Ohm Female BNC Connector.

5.8.7 IDI (J8)

The Insert Data In Port (Unbalanced) is a 75-Ohm Female BNC Connector.

5.8.8 SD (IDO) (J9)

The Send Data (Insert Data Out) Port (Unbalanced) is a 75-Ohm Female BNC Connector.

5.9 HSSI / Ethernet

5.9.1 HSSI (J1)

The HSSI (High-Speed Serial Interface) (J1) complies with the HSSI Functional and Electrical Specifications. The physical interface is a 50-Pin SCSI-2 Type Connector. Electrical levels are ECL. The pinouts for this interface are listed in Table 5-19.

Table 5-19. J1 – HSSI (High-Speed Serial Interface) 50-Pin Connector				
Pin No. (+)	Pin No. (–)	Signal Name	Description	Direction
1	26	SG	Signal Ground	---
2	27	RT	Receive Timing	Output
3	28	CA	DCE Available	Output
4	29	RD	Receive Data	Output
6	31	ST	Send Timing (SCT)	Output
7	32	SG	Signal Ground	---
8	33	TA	DTE Available	Input
9	34	TT	Terminal Timing (SCTE)	Input
11	36	SD	Send Data	Input
13	38	SG	Signal Ground	---
14 - 18	39 – 43	5 Ancillary to DCE	Reserved	Input
19	44	SG	Signal Ground	---

20 - 23	45 - 48	4 Ancillary from DCE	Reserved	Output
24	49	TM	Test Mode	Output
25	N/A	MOD_FLT	Alarm	Output
50	N/A	DMD_FLT	Alarm	Output

5.10 Ethernet Data Interface

The DMD20/DMD20 LBST Ethernet Data Interface provides four RJ-45, Auto-Crossover and Auto-Sensing, 10/100 Ethernet Data Ports. JS1 through JS4 may be referred to Port 1 through Port 4 respectively. Refer to Figures 5-1 and 5-2 for rear panel configurations.

JS1 is Port 1
JS2 is Port 2
JS3 is Port 3
JS4 is Port 4

Refer to Figure 5-1 and 5-2.

5.11 GigE Interface

<FUTURE OPTION>

Maintenance and Troubleshooting

6

This section discusses unit maintenance and troubleshooting for the DMD2050 Universal Satellite Modem.



CAUTION!!

*The DMD2050 contains a Lithium Battery. **DANGER OF EXPLOSION** exists if the battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the manufacturer. Dispose of used batteries in accordance with local and national regulations.*

6.0 Periodic Maintenance

There is no external fuse on the DMD2050. The fuse is located on the power supply assembly inside the case, and replacement is not intended in the field.

6.0.1 Clock Adjustment

The DMD2050 allows for internal VCO speed adjustment from the front panel. Clock adjustment should be performed only when an internal clock source has insufficient accuracy for the custom modem application.

6.1 Troubleshooting

Should a unit be suspected of a defect in field operations after all interface signals are verified, the correct procedure is to replace the unit with another known working DMD2050. If this does not cure the problem, wiring or power should be suspect.

The following is a brief list of possible problems that could be caused by failures of the modem or by improper setup and configuration for the type of service. The list is arranged by possible symptoms exhibited by the modem.

Symptom	Possible Cause
The Modem will not acquire the incoming carrier:	There is an improper receive input to modem.
	The Receive Carrier Level is too low.
	The Receive Carrier Frequency is outside of the acquisition range.
	The Transmit Carrier is incompatible.
	Modem is in Test Mode.
The Async Port is not configured correctly.	The switches may not be set in the correct positions.

6.1.1 Alarm Faults

6.1.1.1 Major Tx Alarms

Alarm	Possible Cause
FPGA CFG	Indicates a transmit FPGA hardware failure.
DSP CFG	Indicates a transmit FPGA failure.
SCT Clock PLL	Indicates that the Tx SCT Clock PLL is not locked. This alarm will flash on during certain modem parameter changes. A solid indication points toward a configuration problem within the modem.
SYM Clock PLL	Indicates that the Tx Symbol Clock PLL is not locked. This alarm will flash on during certain modem parameter changes. A solid indication points toward a problem with the incoming clock to the modem (SCTE).
LB Synth PLL	Indicates that the Tx L-Band Synthesizer is not locked. This alarm will flash on during certain modem parameter changes. A solid indication points toward a configuration problem within the modem.
IF Synth PLL	Indicates that the Tx IF Synthesizer is not locked. This alarm will flash on during certain modem parameter changes. A solid indication points toward a configuration problem within the modem.
Ethernet WAN	Indicates that the WAN Port is down.

6.1.1.2 Major Rx Alarms

Alarm	Possible Cause
FPGA CFG	Indicates a receive FPGA hardware failure.
DSP CFG	Indicates a receive DSP failure.
SIGNAL LOCK	Indicates that the demod is unable to lock to a signal.
FRAME LOCK	Indicates that the Framing Unit is unable to find the expected framing pattern.
MULTIFRAME LOCK	Indicates that the Framing Unit is unable to find the expected framing pattern.
LB SYNTH PLL	Indicates that the Rx L-Band Synthesizer is not locked. This alarm will flash on during certain modem parameter changes. A solid indication points toward a configuration problem within the modem.
IF SYNTH PLL	Indicates that the Rx IF Synthesizer is not locked. This alarm will flash on during certain modem parameter changes. A solid indication points toward a configuration problem within the modem.
Ethernet WAN	Indicates that the WAN Port is down.

6.1.1.3 Minor Tx Alarms

Alarm	Possible Cause
TERR CLK ACT	Indicates no Terrestrial Clock activity.
TERR DATA ACT	Indicates no Tx Data activity.
TX TERR AIS	Indicates that AIS has been detected in the Tx Data Stream.
DnI FRAME LOCK	When running Drop Mode, indicates that the framing unit is unable to find the exported terrestrial framing pattern.
DnI M-FRAME LOCK	When running Drop Mode, indicates that the framing unit is unable to find the exported terrestrial framing pattern.
TX DVB FRAME LOCK	Indicates that the Tx Input Data Stream Framing does not match the user selected Tx Terr Framing. Incorrect Tx Terr Framing selected. Incorrectly framed Tx Input Data Stream.

6.1.1.4 Minor Rx Alarms

Alarm	Possible Cause
BUFF UNDERFLOW	Indicates that a Doppler Buffer underflow has occurred.
BUFF NEAR EMPTY	Indicates that the Doppler Buffer is about to underflow.
BUFF NEAR FULL	Indicates that the Doppler Buffer is about to overflow.
BUFF OVERFLOW	Indicates that a Doppler Buffer overflow has occurred.
RX DATA ACTIVITY	Indicates that there is no Rx Data activity. For the Ethernet Interface, indicates that no Ethernet port is active (no cable is plugged in).
SAT AIS	Indicates that AIS has been detected in the receive satellite data stream.
DnI FRAME LOCK	Indicates if drop/insert data is frame locked.
DnI M-FRAME LOCK	Indicates if drop/insert data has multiframe lock.
INSERT CRC	Indicates if the Circular Redundancy Check is passing in PCM-30C and PCM-31C Modes.
T1/E1 SIGNALING	Indicates that the T1/E1 Signal is not locked.
IFEC LOCK	Indicates that the Inner Codec is not locked.
OFEC LOCK	Indicates that the Reed-Solomon Decoder is not locked.
INTERLEAVER	Indicates that the Reed-Solomon Interleaver is not synchronized.
EBNO (dB)	Indicates that the Eb/No is outside of limits.
IBS BER	Indicates that there are more than one in 1000 bits in error in IBS mode.
RX DVB FRAME LOCK	Indicates that the Rx Satellite Data Stream Framing is not DVB.

6.1.1.5 Drop and Insert Alarms

Alarm	Possible Cause
Multiframe Lock	The insert framer is not in sync.
CRC Lock	An Insert CRC Fault occurred. Valid in T1-ESF, PCM-30, or PCM-30C Modes.
T1 Signaling	An Insert T1 Yellow Fault occurred. Valid in T1-ESF, T1D4, or SCL-96 Modes.
E1 FAS (E1 Frame Acquisition Sync)	An E1 FAS Fault occurred. Valid in PCM-30, or PCM-30C, PCM-31, or PCM-31C Modes.
E1 MFAS (E1 Multi-Frame Acquisition Sync)	An E1 MFAS Fault occurred. Valid in PCM-30, or PCM-30C, PCM-31, or PCM-31C Modes.

6.1.1.6 Common Major Alarms

Alarm	Possible Cause
TERR FPGA CFG	Indicates an Interface Card FPGA configuration failure probably caused by a missing, or wrong file.
CODEC FPGA CFG	Indicates Turbo Codec Card FPGA configuration failure probably caused by a missing, or wrong file.
+1.5V RX SUPPLY	Displays the measured voltage of the 1.5 Volt Rx power bus located inside the modem.
+1.5V TX SUPPLY	Displays the measured voltage of the 1.5 Volt Tx power bus located inside the modem.
+3.3V SUPPLY	Displays the measured voltage of the +3.3 Volt power bus located inside the modem.
+5V SUPPLY	Displays the measured voltage of the +5 Volt power bus located inside the modem.
+12V SUPPLY	Displays the measured voltage of the +12 Volt power bus located inside the modem.
+20V SUPPLY	Displays the measured voltage of the +20 Volt power bus located inside the modem.
EXT CLOCK ACT	Indicates that the External Clock is not active.
EXT REF ACT	Indicates no activity on the External Reference.
EXT REF LOCK	Indicates that the External Reference PLL is not locked.

6.1.2 Alarm Masks

The DMD2050 performs a high degree of self-monitoring and fault isolation. The alarms for these faults are separated into the following three categories:

- Active Alarms
- Common Equipment Alarms
- Backward Alarms

A feature exists that allows the user to 'Mask' out certain alarms as explained below. Alarms that are recorded in the event buffer are the same as the alarm buffer.



When an alarm is masked, the Front Panel LEDs and the Fault Relays do not get asserted, but the Alarm will still be displayed. This feature is very helpful during debugging or to lock out a failure of which the user is already aware.

6.1.2.1 Active Alarms

6.1.2.1.1 Major Alarms

Major Alarms indicate a modem hardware failure. Major Alarms may flash briefly during modem configuration changes and during power-up but should not stay illuminated. Alarms are grouped into Transmit and Receive Alarms - Transmit and Receive are completely independent.

6.1.2.1.2 Minor Alarms

Minor Alarms indicate that a problem may persist outside the modem such as loss of Terrestrial Clock, loss of terrestrial data activity, or a detected transmit or receive AIS condition. Alarms are grouped into Transmit and Receive Alarms - Transmit and Receive are completely independent.

6.1.2.1.3 Common Equipment Faults

Common equipment faults indicate hardware or configuration problems in the modem that effect both transmit and receive operation. Most common faults indicate a hardware failure within the modem, such as a bad power supply. Common faults for the External Reference and External Clock indicate a bad modem configuration, not a hardware failure.

6.1.2.2 Latched Alarms

Latched Alarms are used to catch intermittent failures. If a fault occurs, the fault indication will be latched even if the alarm goes away. After the modem is configured and running, it is recommended that the Latched Alarms be cleared as a final step.

6.1.2.3 Backward Alarms

Backward Alarms are alarms that are fed back to or received from the other end of the satellite link. In IBS Mode (including Drop & Insert), Backward Alarm 1 is the only one used. It would be received if the distant end demod drops lock.

6.2 IBS Fault Conditions and Actions

Figure 6-10 and Table 6-4 illustrate the IBS Fault Conditions and Actions to be taken at the Earth Station, at the Terrestrial Data Stream, and the Satellite. These faults include those detected on the Terrestrial link and those detected from the satellite.

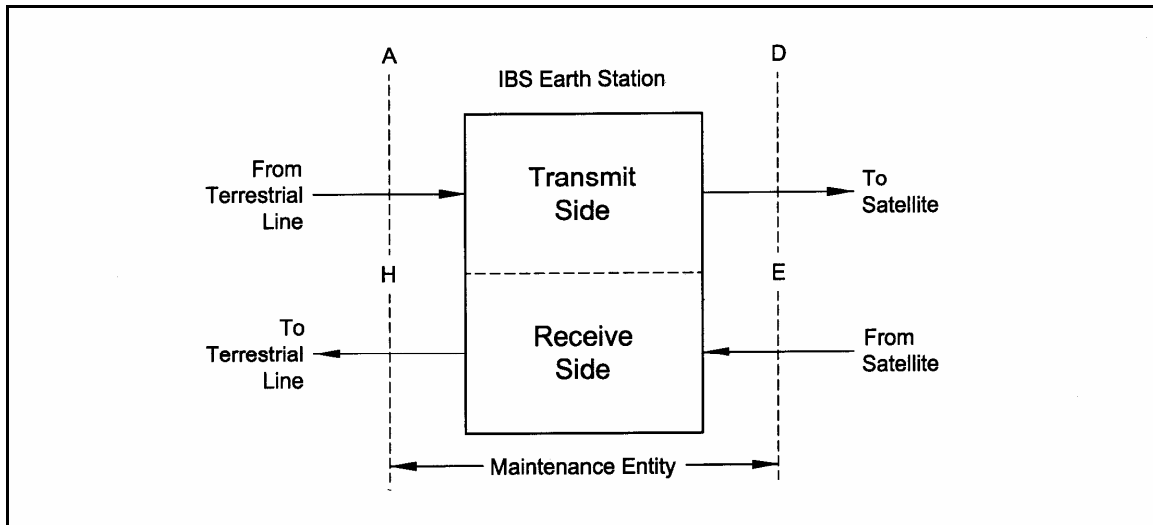


Figure 6-10. IBS Alarm Concept

Table 6-4. IBS Fault Conditions and Actions (includes Drop and Insert)			
Fault Detected on Terrestrial Link (Across Interface A)	Action In Earth Station	Action to Terrestrial (Across Interface H)	Action to Satellite (Across Interface D)
FA1 - Loss of Terrestrial Input	AS1, 2 - IBS Prompt, Service Alarm	AH2 - '1' in Bit 3 of NFAS TSO, Yellow Alarm	AD1 - AIS in Relevant TSs
FA2 - Loss of Terrestrial Signaling	AS1 - - IBS Prompt Alarm	AH2 - '1' in Bit 3 of NFAS TSO, Yellow Alarm	AD3 - '1111' in Relevant TS16's
FA3 - Loss of Terrestrial Frame	AS1 - - IBS Prompt Alarm	AH2 - '1' in Bit 3 of NFAS TSO, Yellow Alarm	AD1 - AIS in Relevant TSs
FA4 - Loss of Terrestrial Multiframe	AS1 - IBS Prompt Alarm	AH2 - '1' in Bit 3 of NFAS TSO, Yellow Alarm	AD3 - '1111' in Relevant TS16's
FA5 - BER of 1×10^{-3} or Greater on Terrestrial Input	AS1 - IBS Prompt Alarm	AH2 - '1' in Bit 3 of NFAS TSO, Yellow Alarm	AD1 - AIS in Relevant TSs
FA6 - Alarm Indication Received on Terrestrial Input	---	---	AD2 - '1' in Bit 3 of Byte 32
Fault Detected From Satellite (Across Interface E)			
FA1 - Loss of Satellite Signal Input	AS1, 2 - IBS Prompt, Service Alarm	AH1, 3 - AIS in TSs, '1111' in TS16	AD2 - '1' in Bit 3 of Byte 32
FA2 - Loss of Satellite Frame	AS1, 2 - IBS Prompt, Service Alarm	AH1, 3 - AIS in TSs, '1111' in TS16	AD2 - '1' in Bit 3 of Byte 32
FA3 - Loss of Satellite Multiframe	AS1, 2 - IBS Prompt, Service Alarm	AH1, 3 - AIS in TSs, '1111' in TS16	AD2 - '1' in Bit 3 of Byte 32
FA4 - BER of 1×10^{-3} or Greater From Satellite Input	AS1, 2 - IBS Prompt, Service Alarm	AH1, 3 - AIS in TSs, '1111' in TS16	AD2 - '1' in Bit 3 of Byte 32
FA5 - Alarm Indication Received From Satellite Input	AS2 - IBS Service Alarm	AH2 - '1' in Bit 3 of NFAS TSO, Yellow Alarm	---

Technical Specifications

7

7.0 Data Rates

BPSK Uncoded	4.8 Kbps to 10.0 Mbps
1/2 Rate BPSK	2.4 Kbps to 10.0 Mbps
3/4 Rate BPSK	3.6 Kbps to 10.0 Mbps
7/8 Rate BPSK	4.2 Kbps to 10.0 Mbps
QPSK Uncoded	9.6 Kbps to 10.0 Mbps
1/2 Rate QPSK	4.8 Kbps to 20.0 Mbps
3/4 Rate QPSK	7.2 Kbps to 20.0 Mbps
7/8 Rate QPSK	8.4 Kbps to 20.0 Mbps
Rate 2/3 8PSK	9.6 Mbps to 52.0 Mbps
3/4 Rate 16QAM	14.4 Mbps to 52.0 Mbps
7/8 Rate 16QAM	16.84 Mbps to 52.0 Mbps

7.1 Modulator

Modulation	BPSK, QPSK, and OQPSK, 8PSK, 16QAM
IF Tuning Range	50 to 90, 100 to 180 MHz in 1 Hz Steps
L-Band Tuning Range	950 to 2050 MHz in 1 Hz Steps
Impedance	IF, 50 Ohm (75 Ohm Optional) L-Band, 50 Ohm
Connector	TNC, 50 Ohm SMA, 50 Ohm, L-Band
Return Loss	IF, 1.5:1 L-Band, 2.0:1
Output Power	0 to -25 dB
Output Stability	IF, ± 0.5 dB Over Frequency and Temperature L-Band, ± 1.0 dB Over Frequency and Temperature
Output Spectrum	Selectable and Meets MIL-188-165A or IESS 308/309/ 310 Power Spectral Mask
Spurious	-55 dBc In-Band (50 to 90 MHz, 100 to 180 MHz, 950 to 2050 MHz) -45 dBc Out-of-Band
On/Off Power Ratio	>60 dB
Scrambler	OM-73, CCITT V.35 or IBS
FEC	Viterbi, K = 7 at 1/2, 3/4 and 7/8 2/3 Rate Trellis Turbo Product Code (Optional) BPSK 21/44 QPSK/OQPSK 1/2, 3/4, 7/8 <20Mbps QPSK/OQPSK .750, .875 >20Mbps 8PSK/16QAM 3/4, 7/8 <20Mbps 8PSK/16QAM .750, .875 >20Mbps Legacy Turbo Rates: 0.495, 0.793
Outer Encoder Options	Reed-Solomon INTELSAT (DVB Optional) Custom (N, K) Reed-Solomon
Data Clock Source	Internal, External, Rx Recovered
Internal Stability	5×10^{-8}

7.2 Demodulator

Demodulation	BPSK, QPSK, and OQPSK, 8PSK, 16QAM
IF Tuning Range	50 to 90, 100 to 180 MHz in 1 Hz Steps
L-Band Tuning Range	950 to 2050 MHz in 1 Hz Steps
Impedance	IF, 50 Ohm (75 Ohm Optional) L-Band, 50 Ohm
Connector	TNC, 50 Ohm SMA, 50 Ohm, L-Band
Return Loss	IF, 1.5:1 L-Band, 2.0:1
Spectrum	Selectable and Meets MIL-188-165A or INTELSAT IESS 308/309/310 Compliant
Input Level	-55 to +10 dBm
Total Input Power	+20 dBm or +40 dBc (the Lesser)
FEC	Viterbi, K = 7 at 1/2, 3/4 and 7/8 Rate, Rate Sequential 1/2 Rate, 3/4, 7/8 (Optional) Trellis 2/3 Turbo Product Code (Optional) BPSK 21/44 QPSK/OQPSK 1/2, 3/4, 7/8 <20Mbps QPSK/OQPSK .750, .875 >20Mbps 8PSK/16QAM 3/4, 7/8 <20Mbps 8PSK/16QAM .750, .875 >20Mbps Legacy Turbo Rates: 0.495, 0.793
Decoder Options	Reed-Solomon INTELSAT (DVB Optional) Custom (N, K) Reed-Solomon
Descrambler	OM-73, CCITT V.35 or IBS
Acquisition Range	Programmable ± 1 kHz to ± 255 kHz
Reacquisition Range	Programmable ± 1 Hz to 25000 Hz
Sweep Delay Value	0 to 6000 seconds in 100 msec Steps

7.3 Plesiochronous Buffer

Size	0 msec to 64 msec
Centering	Automatic on Underflow/Overflow
Centering Modes	IBS: Integral Number of Frames IDR: Integral Number of Multi-Frames
Clock	Transmit, External, Rx Recovered or SCT (Internal)

7.4 Monitor and Control

Ethernet 10 Base-T/Web Browser, Remote RS-485/Terminal RS-232/DMD15 Protocol Compatible

7.5 DMD2050 Drop and Insert (Optional)

Terrestrial Data	1.544 Mbps or 2.048 Mbps, G.732/733
Line Coding	AMI or B8ZS for T1 and HDB3 for E1
Framing	D4, ESF and PCM-30 (PCM-30C) or PCM-31 (PCM-31C) for E1
Time Slot Selection	n x 64 Contiguous or Arbitrary Blocks for Drop or Insert.
Time Slots	TS1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 16, 20, 24, 30, 31
Data Rates	64, 128, 192, 256, 320, 384, 512, 640, 768, 960, 1024, 1280, 1536, 1920 Kbps
Efficient D&I	Closed Network, Satellite Overhead 0.4%

Time Slots 1-31 Any combination

7.6 Terrestrial Interfaces

MIL-188-114A, DVB ASI/SPI, HSSI, Ethernet 4 Port 10/100 Base-T

7.7 IDR/ESC Interface (Optional)

G.703 T1 (100)	1.544 Mbps, 100 Ohm Balanced, AMI and B8ZS
G.703 E1	2.048 Mbps, 75 Ohm Unbalanced and 120 Ohm Balanced, HDB3
G.703 T2 (100)	6.312 Mbps, 75 Ohm Unbalanced and 110 Ohm Balanced, B8ZS and B6ZS
G.703 E2	8.448 Mbps, 75 Ohm BNC, Unbalanced, HDB3

7.8 IBS/Synchronous Interface (Standard)

MIL-188-144A/RS-422/-530	All Rates, Differential, Clock/Data, DCE
ITU V.35	All Rates, Differential, Clock/Data, DCE
RS-232	(DCE up to 200 Kbps)

7.9 High-Speed Serial Interface (HSSI)

HSSI: HSSI, Serial, 50-Pin SCSI-2 Type Connector (Female)

7.10 ASI

ASI/RS-422 Parallel: ASI, Serial, 75 Ohm BNC (Female)
DVB/M2P, Parallel, RS-422, DB-25 (Female)

ASI/LVDS Parallel: ASI, Serial, 75 Ohm BNC (Female)
DVB/M2P, Parallel, LVDS, DB-25 (Female)

7.11 DVB/M2P

DVB/M2P: DB-25 Female Connector. It complies with RS-422 Electrical Specifications.

7.12 Ethernet Data Interface (Optional)

Ethernet Data Interface Four RJ-45, Auto-Crossover, Auto-Sensing, 10/100 Ethernet Data Ports. Complies with IEEE 802.3 and IEEE 802.3u.

7.13 HSSI / G703

HSSI	High-Speed Serial Interface, 50-pin SCSI-2 Type Connector (Female)
G.703 T1 (DSX1)	1.544 Mbps, 100-Ohm Balanced, AMI and B8ZS
G.703 E1	2.048 Mbps, 75-Ohm Unbalanced and 120-Ohm Balanced, HDB3
G.703 T2 (DSX2)	6.312 Mbps, 75-Ohm Unbalanced and 110-Ohm Balanced, B8ZS and B6ZS
G.703 E2	8.448 Mbps, 75-Ohm BNC, Unbalanced, HDB3
	Note: Does not support backward alarms

7.14 HSSI /ETHERNET

HSSI	HSSI, High-Speed Serial Interface, 50-pin SCSI-2 Type Connector (Female)
Ethernet Data Interface	Four RJ-45, Auto-Crossover, Auto-Sensing, 10/100 Ethernet Data Ports. Complies with IEEE 802.3 and IEEE 802.3u.

7.15 Environmental

Prime Power	100 to 240 VAC, 50 to 60 Hz, 250 Watts Maximum 48 VDC (Optional)
Operating Temperature	-10 to +60°C, 95% Humidity, Non-Condensing
Storage Temperature	-20 to 70°C, 99% humidity, Non-Condensing

7.16 Physical

Size	19" W x 19.25" D x 1.75" H (48.26 x 48.89 x 4.45 cm)
Weight	6.5 Pounds (3.0 Kg)

7.17 DMD2050 Data Rate Limits

7.17.1 Non-DVB

Modulation	Code Rate	Min Data Rate	Max Data Rate
BPSK	NONE	4800	10000000
BPSK	VIT 1/2	2400	10000000
BPSK	VIT 3/4	3600	10000000
BPSK	VIT 7/8	4200	10000000
BPSK	SEQ 1/2	2400	2048000
BPSK	SEQ 3/4	3600	2048000
BPSK	SEQ 7/8	4200	2048000
BPSK	TPC 21/44	2400	4772727
BPSK	TPC 3/4	4800	7500000
BPSK	TPC 7/8	4800	8750000
BPSK	TPC .495	2376	4950000
BPSK	TPC .793	3806	7930000
QPSK	NONE	9600	20000000
QPSK	VIT 1/2	4800	20000000
QPSK	VIT 3/4	7200	20000000
QPSK	VIT 7/8	8400	20000000
QPSK	SEQ 1/2	4800	2048000
QPSK	SEQ 3/4	7200	2048000
QPSK	SEQ 7/8	8400	2048000
QPSK	TPC 1/2	4582	9545454
QPSK	TPC 3/4	7200	15000000
QPSK	TPC 7/8	8400	17500000
QPSK	TPC .495	4752	6312000
QPSK	TPC .793	7612	6312000
OQPSK	NONE	9600	20000000
OQPSK	VIT 1/2	4800	20000000
OQPSK	VIT 3/4	7200	20000000
OQPSK	VIT 7/8	8400	20000000
OQPSK	SEQ 1/2	4800	2048000
OQPSK	SEQ 3/4	7200	2048000
OQPSK	SEQ 7/8	8400	2048000
OQPSK	TPC 1/2	4582	9545454

OQPSK	TPC 3/4	7200	15000000
OQPSK	TPC 7/8	8400	17500000
OQPSK	TPC .495	4752	6312000
OQPSK	TPC .793	7612	6312000
8PSK	TRE 2/3	9600	52000000
8PSK	TPC 3/4	10800	20000000
8PSK	TPC 7/8	12600	20000000
8PSK	TPC .495	9504	6312000
8PSK	TPC .793	15225	6312000
8PSK	TPC .750	20000001	52000000
8PSK	TPC .875	20000001	52000000
16QAM	VIT 3/4	14400	20000000
16QAM	VIT 7/8	16800	20000000
16QAM	TPC 3/4	14400	20000000
16QAM	TPC 7/8	16800	20000000
16QAM	TPC .495	9504	6312000
16QAM	TPC .793	15225	6312000
16QAM	TPC .750	20000001	52000000
16QAM	TPC .875	20000001	52000000

7.17.2 DVB

187 Mode			
Modulation	Code Rate	Min Data Rate	Max Data Rate
BPSK	VIT 1/2	2400	4583333
BPSK	VIT 2/3	2934	6111111
BPSK	VIT 3/4	3300	6875000
BPSK	VIT 5/6	3667	7638888
BPSK	VIT 7/8	3850	8020833
QPSK	VIT 1/2	4400	9166666
QPSK	VIT 2/3	5867	12222222
QPSK	VIT 3/4	6600	13750000
QPSK	VIT 5/6	7334	15277777
QPSK	VIT 7/8	7700	16041666
8PSK	TRE 2/3	8800	18333333
8PSK	TRE 5/6	11000	20000000

8PSK	TRE 8/9	11550	20000000
16QAM	TRE 3/4	13200	20000000
16QAM	TRE 7/8	15400	20000000

188 Mode			
Modulation	Code Rate	Min Data Rate	Max Data Rate
BPSK	VIT 1/2	2400	4607843
BPSK	VIT 2/3	2950	6143790
BPSK	VIT 3/4	3318	6911764
BPSK	VIT 5/6	3687	7679738
BPSK	VIT 7/8	3871	8063725
QPSK	VIT 1/2	4424	9215686
QPSK	VIT 2/3	5899	12287581
QPSK	VIT 3/4	6636	13823529
QPSK	VIT 5/6	7373	15359476
QPSK	VIT 7/8	7742	16127450
8PSK	TRE 2/3	8848	18431372
8PSK	TRE 5/6	11059	20000000
8PSK	TRE 8/9	11797	20000000
16QAM	TRE 3/4	13271	20000000
16QAM	TRE 7/8	15483	20000000

204 Mode			
Modulation	Code Rate	Min Data Rate	Max Data Rate
BPSK	VIT 1/2	2400	5000000
BPSK	VIT 2/3	3200	6666666
BPSK	VIT 3/4	3600	7500000
BPSK	VIT 5/6	4000	8333333
BPSK	VIT 7/8	4200	8750000
QPSK	VIT 1/2	4800	10000000
QPSK	VIT 2/3	6400	13333333
QPSK	VIT 3/4	7200	15000000
QPSK	VIT 5/6	8000	16666666

QPSK	VIT 7/8	8400	17500000
8PSK	TRE 2/3	9600	20000000
8PSK	TRE 5/6	12000	20000000
8PSK	TRE 8/9	12800	20000000
16QAM	TRE 3/4	14400	20000000
16QAM	TRE 7/8	16800	20000000

7.18 DMD2050 BER Specifications

7.18.1 BER Performance (Viterbi)

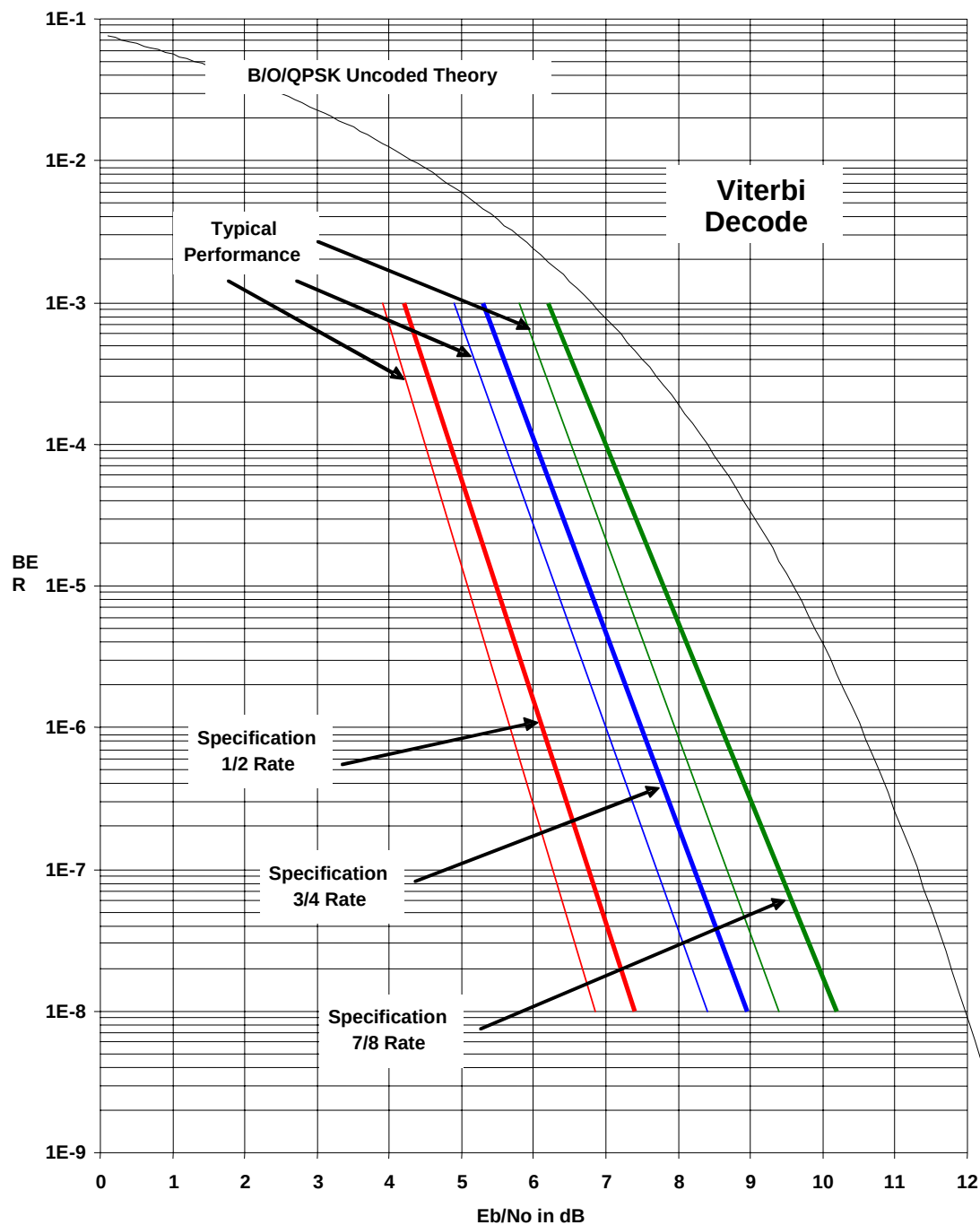


Figure 7-1. B/O/QPSK BER Performance (Viterbi)

Note: E_b/N_0 values include the effect of using Differential Decoding and V.35 descrambling.

7.18.2 BER Performance (Sequential)

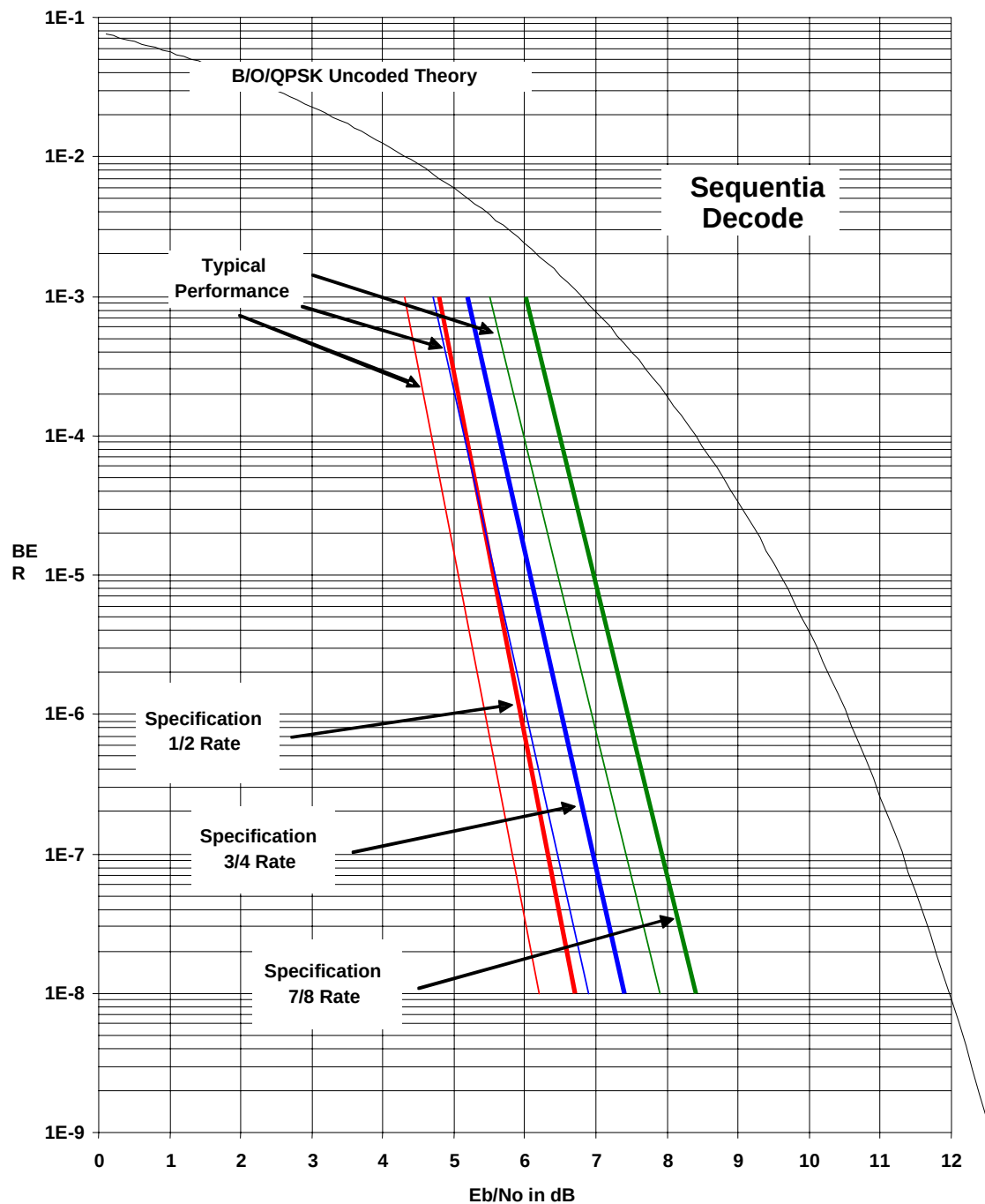


Figure 7-2. B/O/QPSK BER Performance (Sequential)

Note: E_b/N_0 values include the effect of using Differential Decoding and V.35 descrambling.

7.18.3 BER Performance (Viterbi with Reed-Solomon)

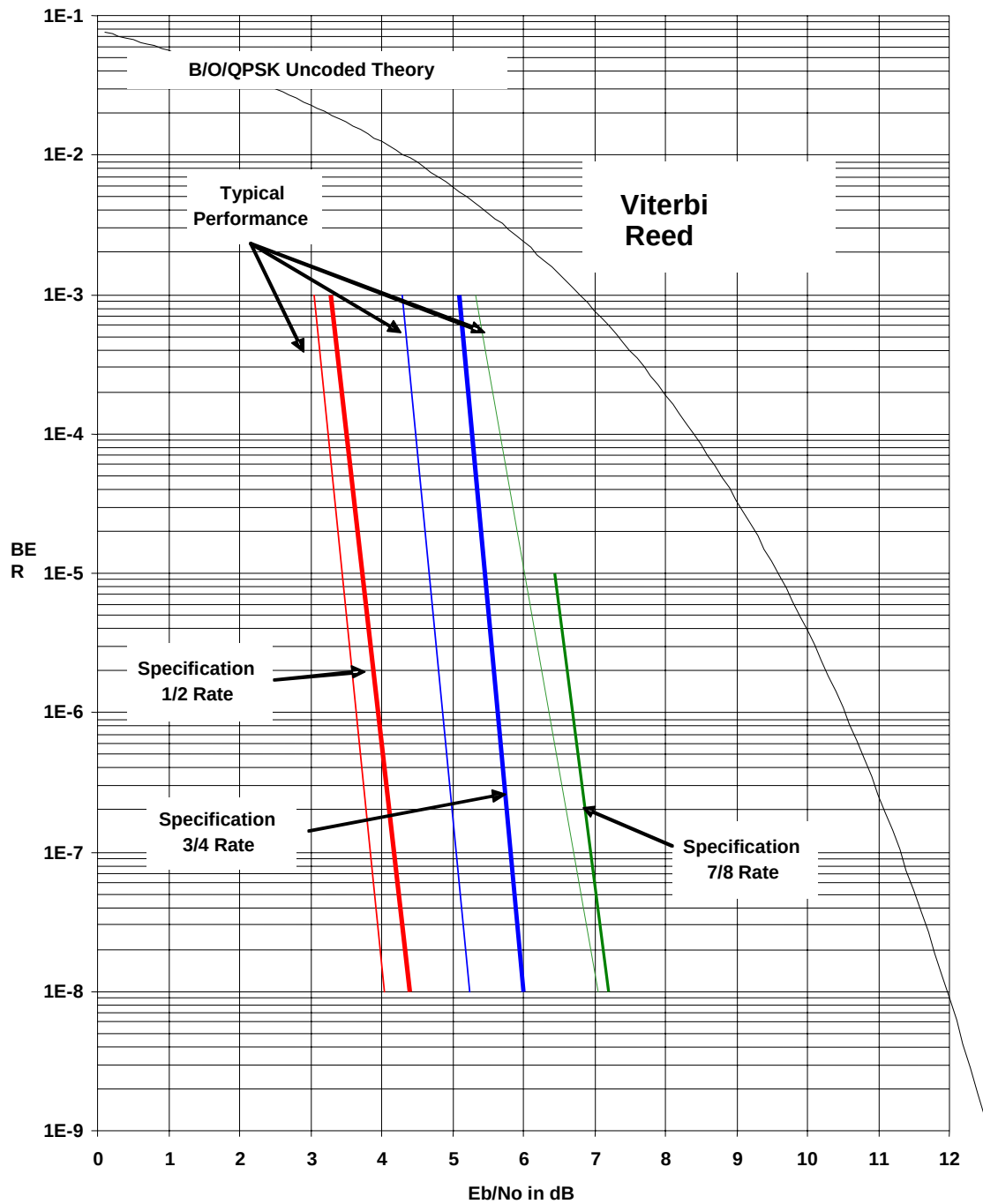


Figure 7-3. B/O/QPSK BER Performance (Viterbi - w/RS)

Note: E_b/N_0 values include the effect of using Differential Decoding.

7.18.4 BER Performance (Turbo)

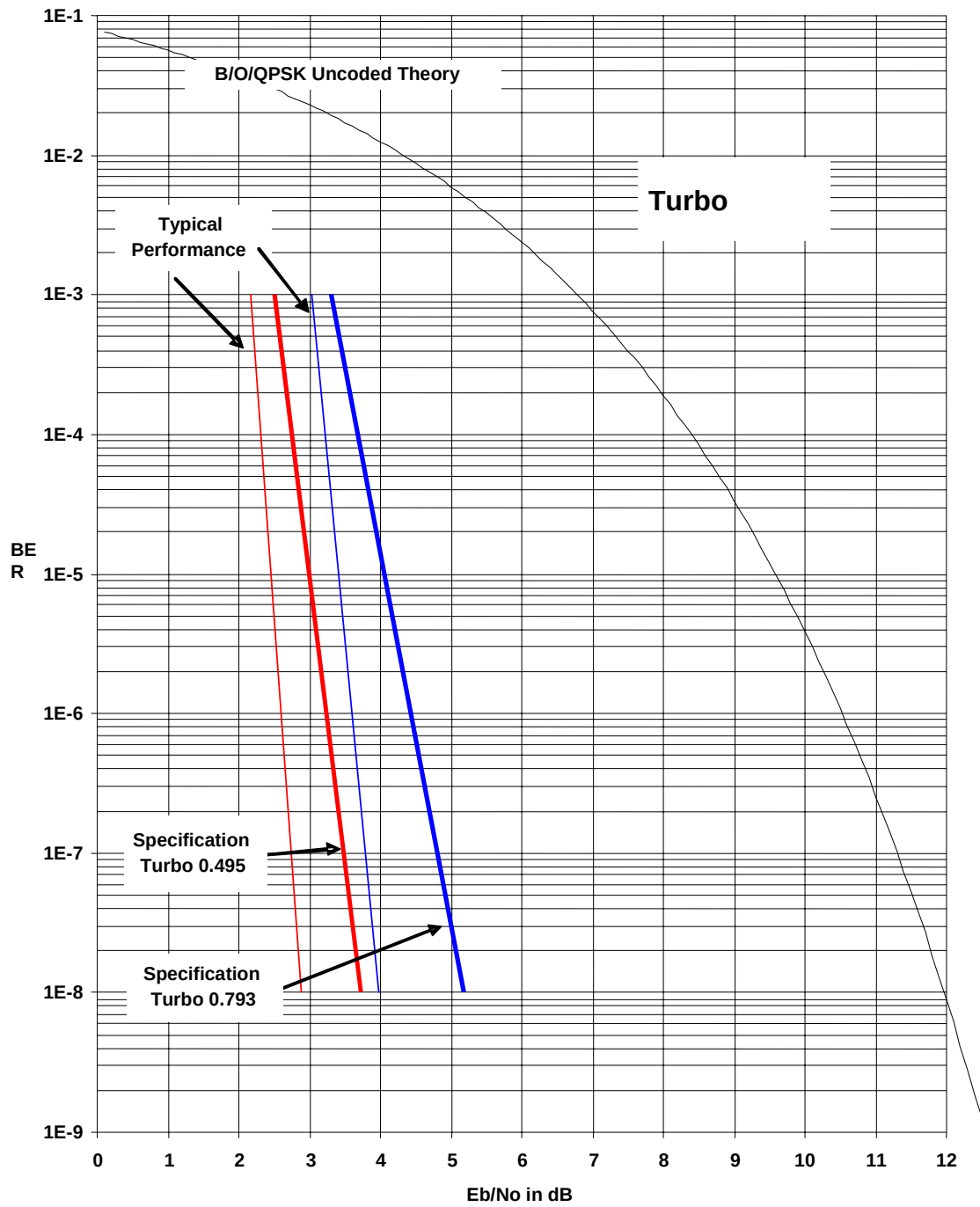


Figure 7-4. B/O/QPSK BER Performance (Turbo)

Note: E_b/N_0 values include the effect of using interleaving and maximum iterations.

7.18.5 BER Performance (8PSK Trellis)

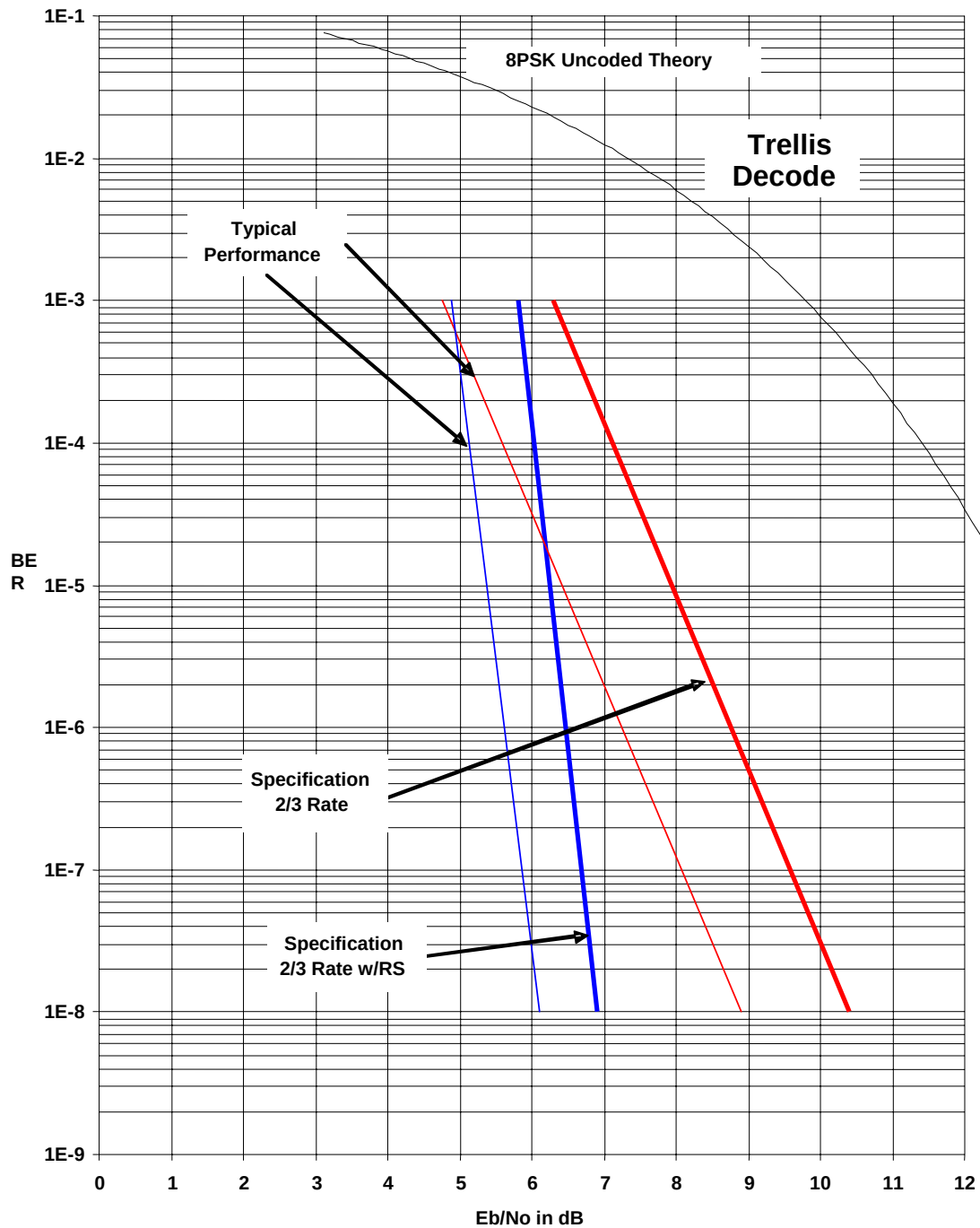


Figure 7-5. 8PSK BER Performance (Trellis)

Note: E_b/N_0 values include the effect of using Differential Decoding and V.35 Descrambling.

7.18.6 BER Performance (8PSK Turbo)

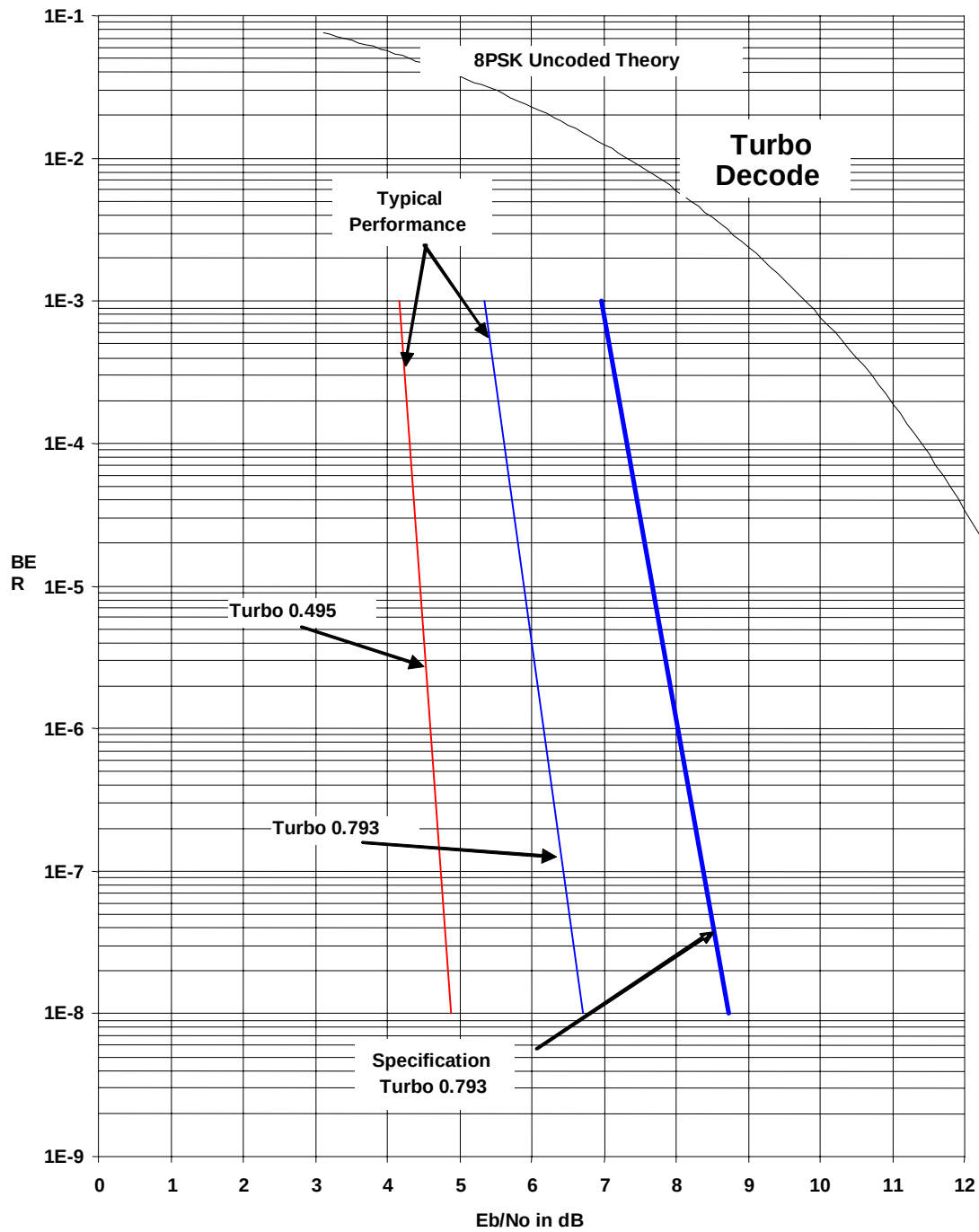


Figure 7-6. 8PSK BER Performance (Turbo)

Note: E_b/N_0 values include the effect of using interleaving and maximum iterations.

7.18.7 BER Performance (16QAM Viterbi)

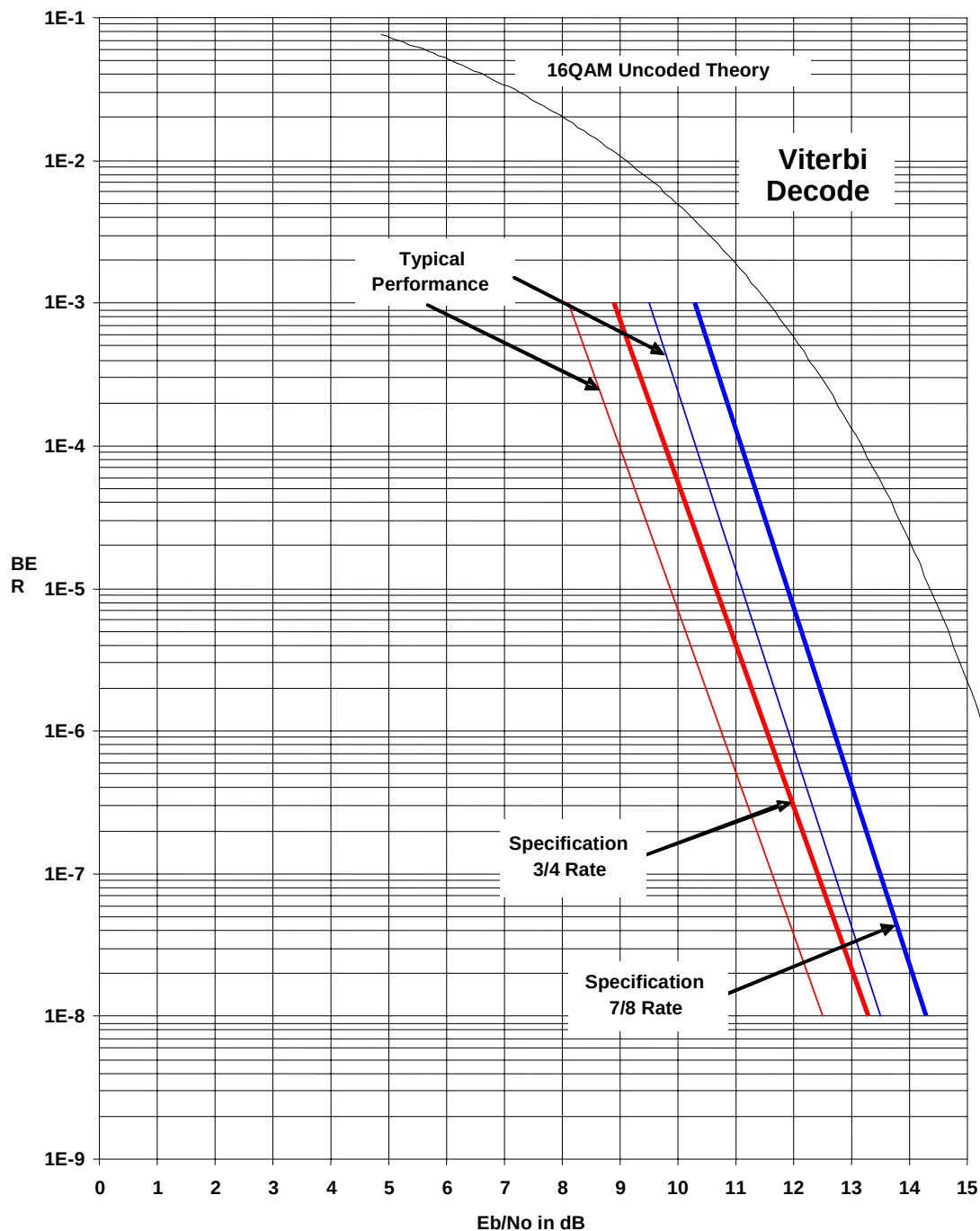


Figure 7-7. 16QAM BER Performance (Viterbi)

Note: E_b/N_0 values include the effect of using Differential Decoding and V.35 Descrambling.

7.18.8 BER Performance (16QAM Viterbi with Reed-Solomon)

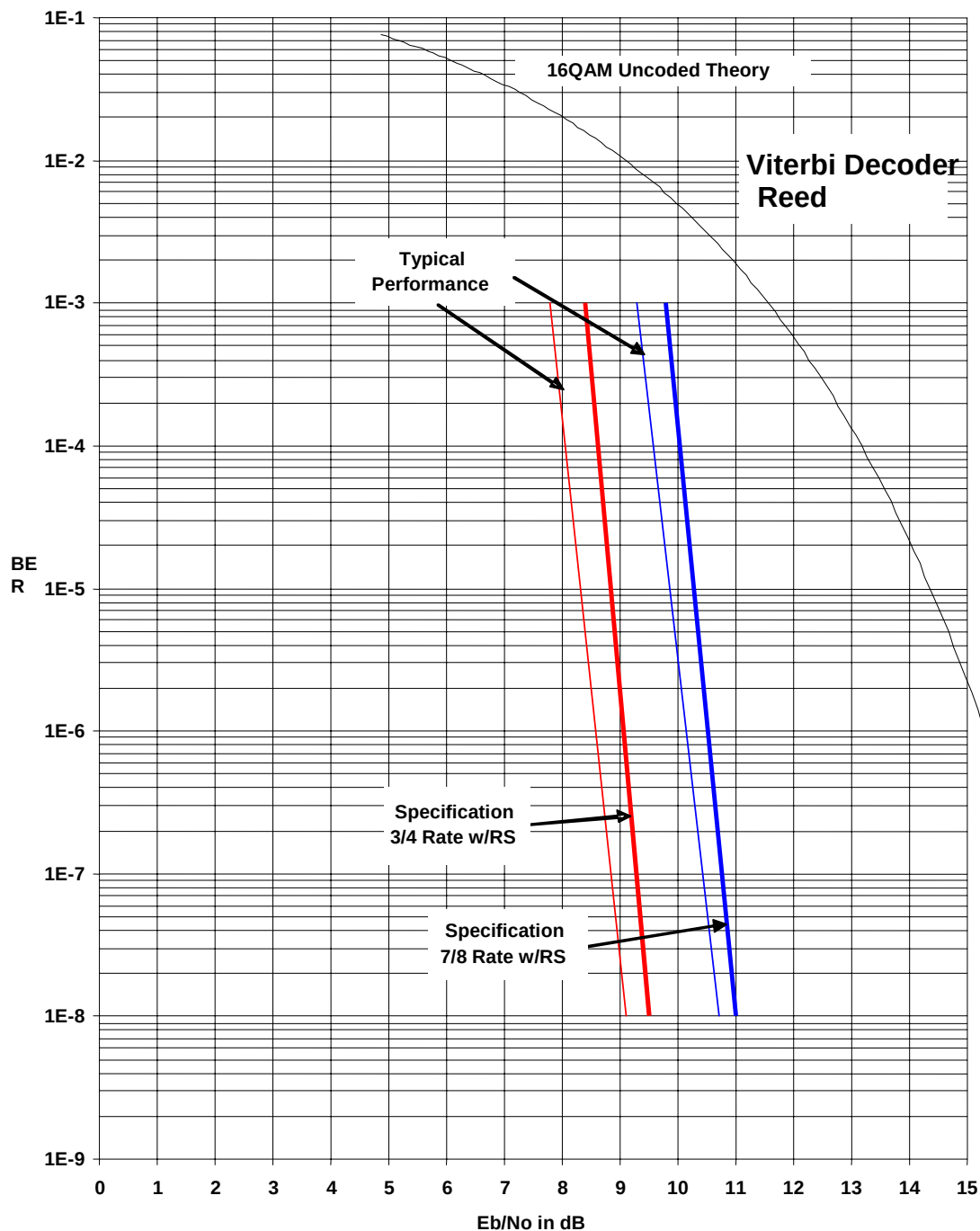


Figure 7-8. 16QAM BER Performance (Viterbi w/RS)

Note: E_b/N_0 values include the effect of using Differential Decoding.

7.18.9 BER Performance (16QAM Turbo)

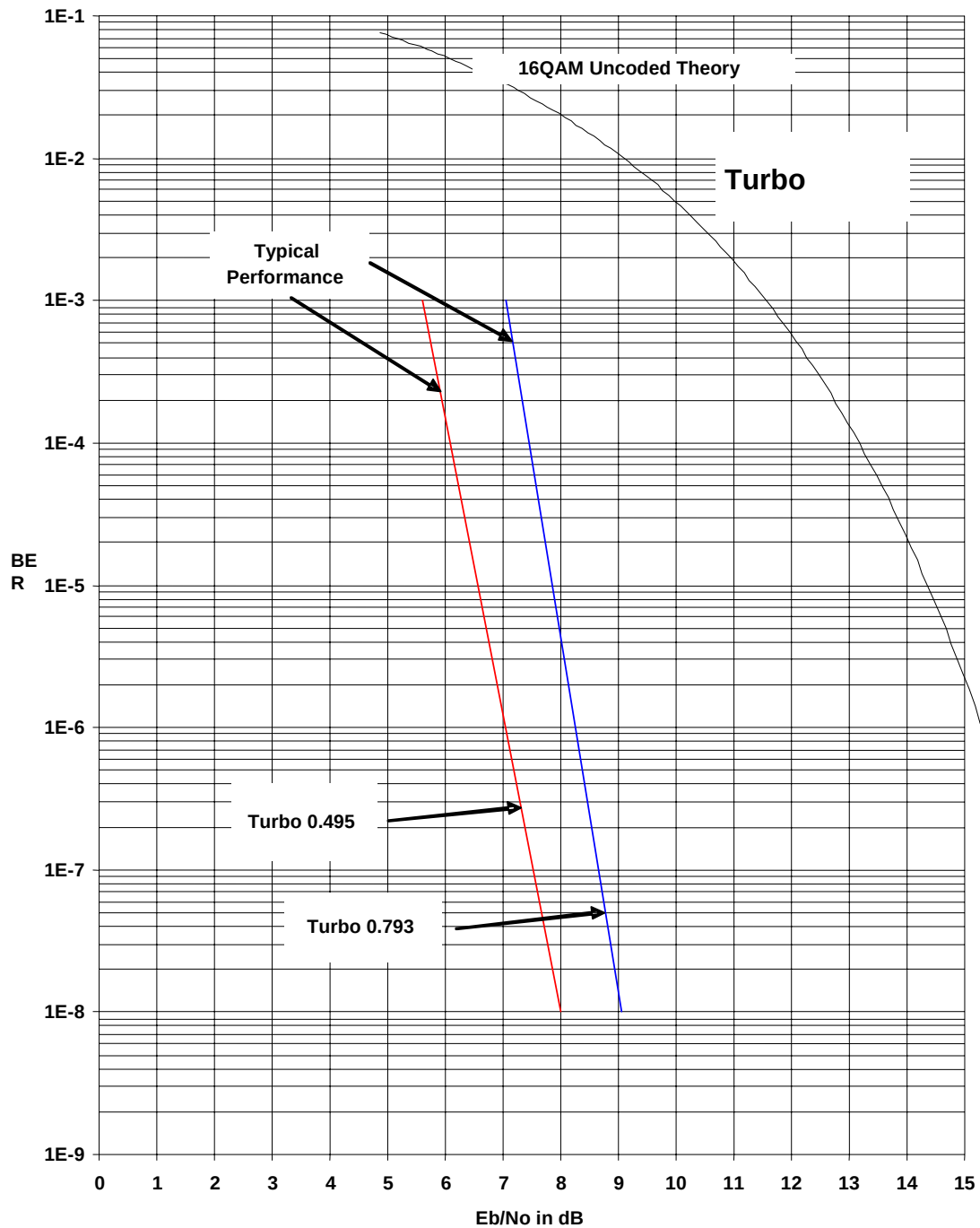


Figure 7-9. 16QAM BER Performance (Turbo)

Note: E_b/N_0 values include the effect of using interleaving and maximum iterations.

7.18.10 BER Performance ((O)QPSK Turbo)

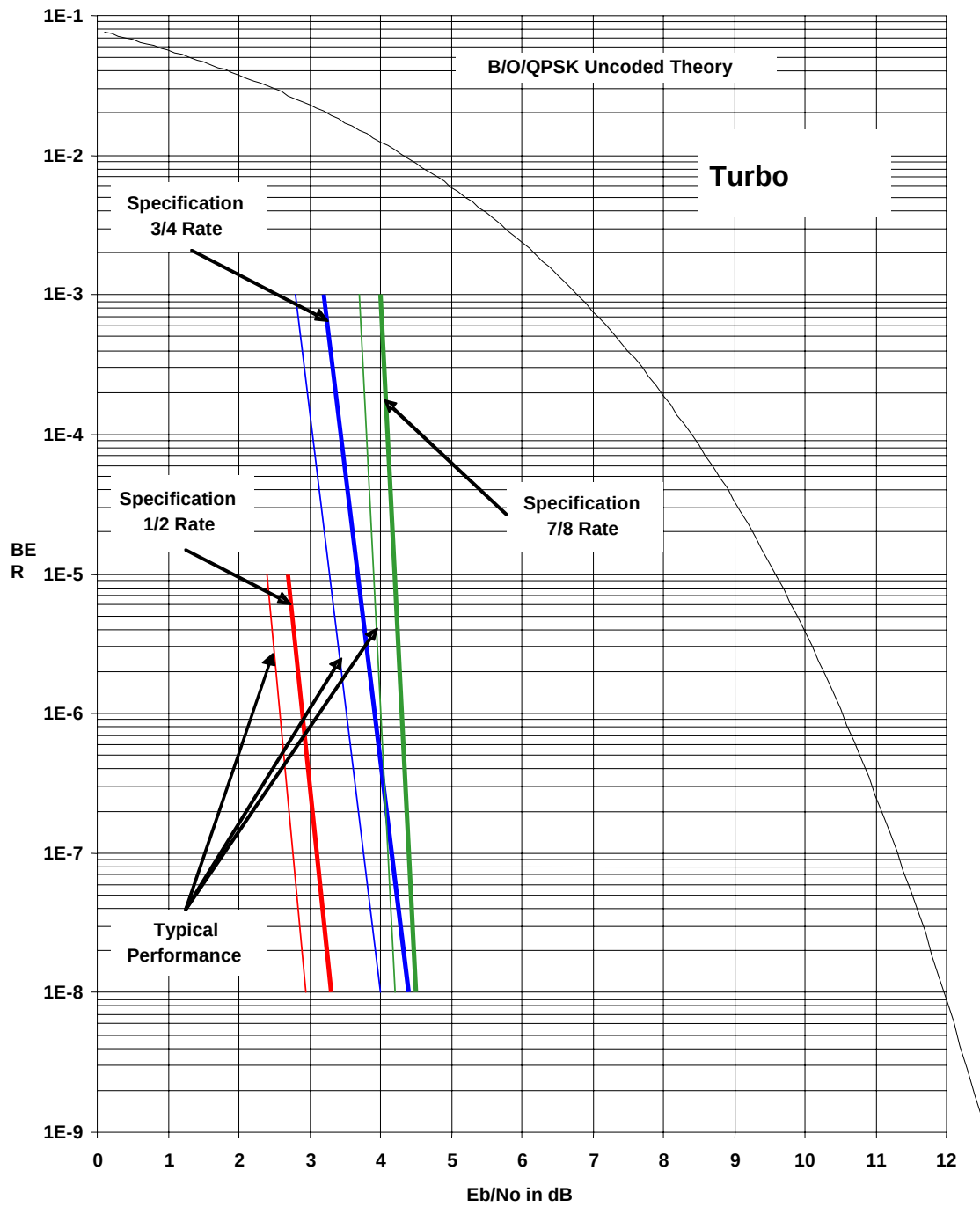


Figure 7-10. (O)QPSK BER Performance (Turbo)

7.18.11 BER Performance (8PSK Turbo)

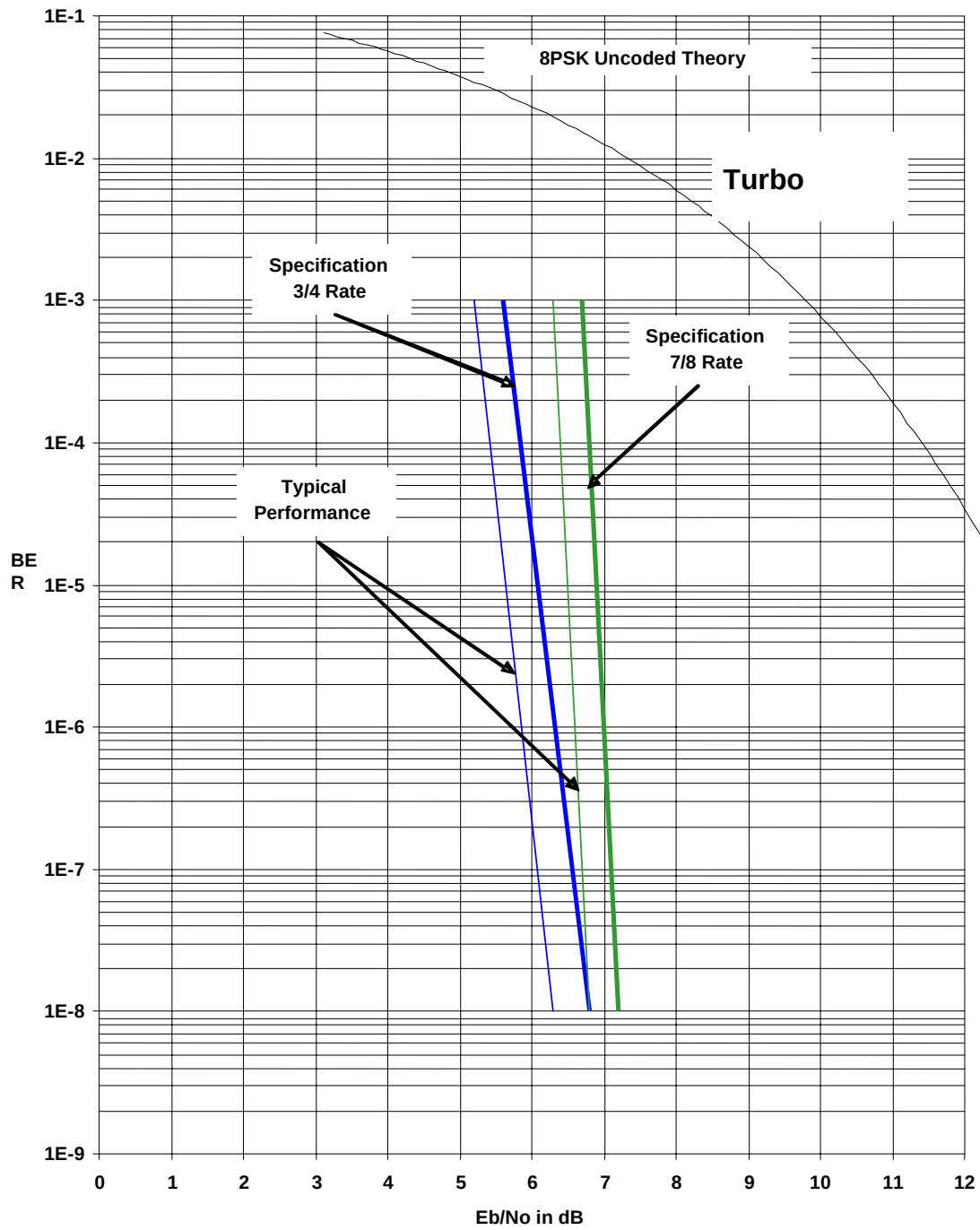


Figure 7-11. 8PSK BER Performance (Turbo)

7.18.12 BER Performance (16QAM Turbo)

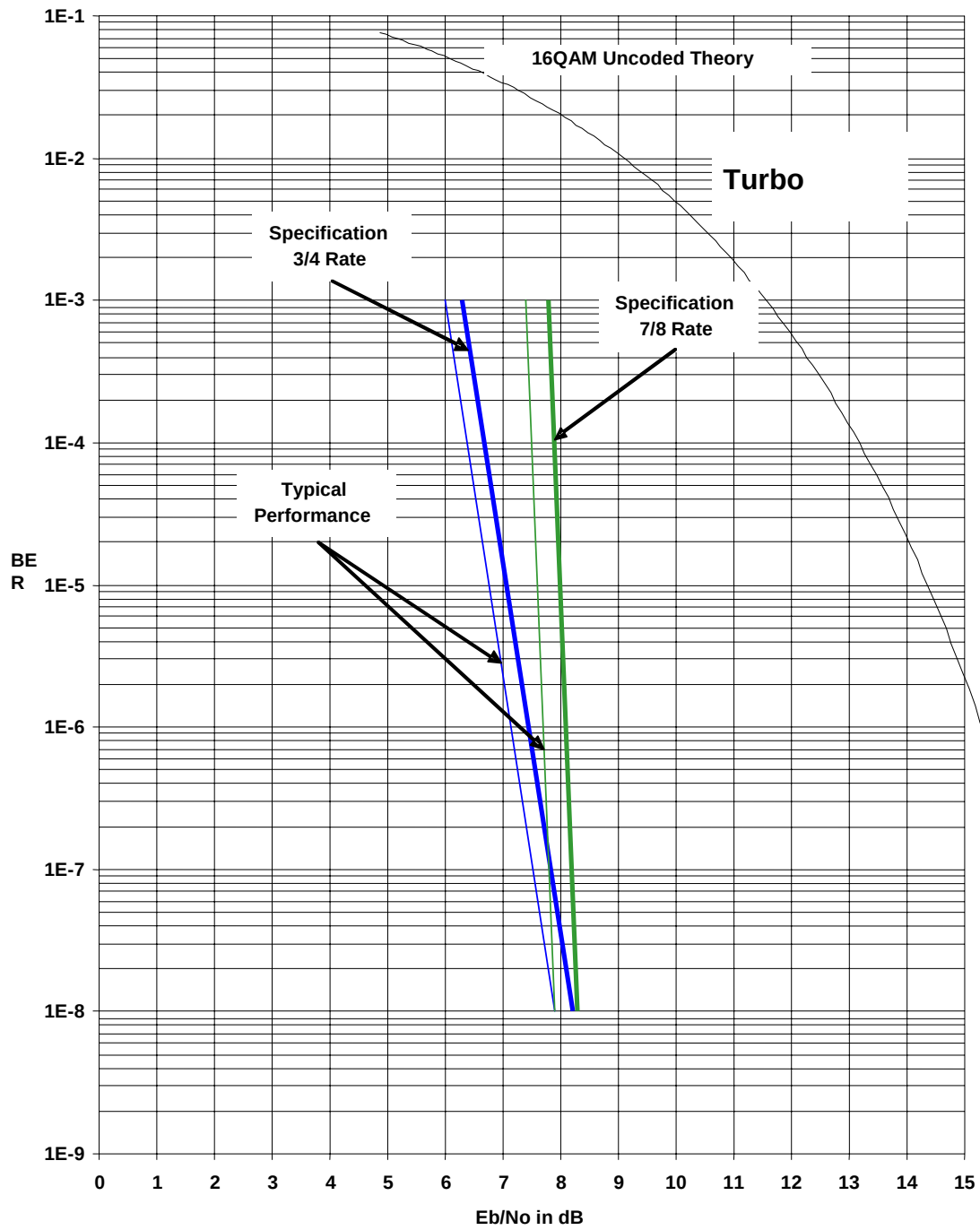


Figure 7-12. 16 QAM BER Performance (Turbo)

Table 7-1 - B/O/QPSK BER Performance (Viterbi)						
BER	Specification			Typical		
	1/2 Rate	3/4 Rate	7/8 Rate	1/2 Rate	3/4 Rate	7/8 Rate
1E-3	4.2 dB	5.3 dB	6.2 dB	3.9 dB	4.9 dB	5.8 dB
1E-4	4.8 dB	6.1 dB	7.1 dB	4.5 dB	5.6 dB	6.5 dB
1E-5	5.5 dB	6.8 dB	7.9 dB	5.1 dB	6.3 dB	7.2 dB
1E-6	6.1 dB	7.6 dB	8.6 dB	5.7 dB	7 dB	7.9 dB
1E-7	6.7 dB	8.3 dB	9.3 dB	6.2 dB	7.7 dB	8.6 dB
1E-8	7.4 dB	8.9 dB	10.2 dB	6.8 dB	8.4 dB	9.4 dB
1E-9	8.2 dB	9.7 dB	11 dB	7.4 dB	9.1 dB	10.0 dB
1E-10	9 dB	10.3 dB	11.7 dB	8.1 dB	9.8 dB	10.5 dB

Table 7-2 - B/O/QPSK BER Performance (Sequential)						
BER	Specification			Typical		
	1/2 Rate	3/4 Rate	7/8 Rate	1/2 Rate	3/4 Rate	7/8 Rate
1E-3	4.8 dB	5.2 dB	6 dB	4.3 dB	4.7 dB	5.5 dB
1E-4	5.2 dB	5.7 dB	6.4 dB	4.7 dB	5.2 dB	5.9 dB
1E-5	5.6 dB	6.1 dB	6.9 dB	5.1 dB	5.6 dB	6.4 dB
1E-6	5.9 dB	6.5 dB	7.4 dB	5.4 dB	6.1 dB	6.9 dB
1E-7	6.3 dB	7 dB	7.9 dB	5.8 dB	6.5 dB	7.4 dB
1E-8	6.7 dB	7.4 dB	8.4 dB	6.2 dB	6.9 dB	7.9 dB
1E-9	7.1 dB	7.8 dB	8.9 dB	6.6 dB	7.4 dB	8.4 dB
1E-10	7.4 dB	8.3 dB	9.4 dB	6.9 dB	7.8 dB	8.9 dB

Table 7-3 - B/O/QPSK BER Performance (Viterbi - w/RS)						
BER	Specification			Typical		
	1/2 Rate	3/4 Rate	7/8 Rate	1/2 Rate	3/4 Rate	7/8 Rate
1E-3	3.3 dB	5.1 dB	-	3 dB	4.3 dB	5.3 dB
1E-4	3.5 dB	5.3 dB	-	3.2 dB	4.5 dB	5.7 dB
1E-5	3.8 dB	5.4 dB	6.5 dB	3.4 dB	4.7 dB	6 dB
1E-6	4.1 dB	5.6 dB	6.7 dB	3.6 dB	4.9 dB	6.4 dB
1E-7	4.2 dB	5.8 dB	6.9 dB	3.8 dB	5.1 dB	6.7 dB
1E-8	4.4 dB	6 dB	7.2 dB	4 dB	5.3 dB	7.1 dB
1E-9	4.7 dB	6.1 dB	7.5 dB	4.2 dB	5.4 dB	7.4 dB
1E-10	5 dB	6.3 dB	7.8 dB	4.4 dB	5.6 dB	7.7 dB

Table 7-4 - B/O/QPSK BER Performance (Turbo)				
BER	Specification		Typical	
	Turbo 0.495	Turbo 0.793	Turbo 0.495	Turbo 0.793
1E-3	2.5 dB	3.3 dB	2.2 dB	3 dB
1E-4	2.7 dB	3.7 dB	2.3 dB	3.2 dB
1E-5	3 dB	4.1 dB	2.5 dB	3.4 dB
1E-6	3.2 dB	4.4 dB	2.6 dB	3.6 dB
1E-7	3.5 dB	4.8 dB	2.7 dB	3.8 dB
1E-8	3.7 dB	5.2 dB	2.9 dB	4 dB
1E-9	4 dB	5.6 dB	3 dB	4.2 dB
1E-10	4.2 dB	5.9 dB	3.2 dB	4.4 dB

Table 7-5 - 8PSK BER Performance (Trellis)				
BER	Specification		Typical	
	2/3 Rate	2/3 Rate w/RS	2/3 Rate	2/3 Rate w/RS
1E-3	6.3 dB	5.8 dB	4.8 dB	4.9 dB
1E-4	7.3 dB	6.1 dB	5.6 dB	5.1 dB
1E-5	8.2 dB	6.3 dB	6.4 dB	5.4 dB
1E-6	9 dB	6.5 dB	7.2 dB	5.6 dB
1E-7	9.8 dB	6.7 dB	8.1 dB	5.8 dB
1E-8	10.4 dB	6.9 dB	8.9 dB	6.1 dB
1E-9	11.1 dB	7.1 dB	9.7 dB	6.3 dB
1E-10	11.9 dB	7.3 dB	10.5 dB	6.6 dB

Table 7-6 - 8PSK BER Performance (Turbo)				
BER	Specification		Typical	
	Turbo 0.495	Turbo 0.793	Turbo 0.495	Turbo 0.793
1E-3	-	7 dB	4.2 dB	5.4 dB
1E-4	-	7.3 dB	4.3 dB	5.6 dB
1E-5	-	7.7 dB	4.5 dB	5.9 dB
1E-6	-	8 dB	4.6 dB	6.2 dB
1E-7	-	8.4 dB	4.7 dB	6.4 dB
1E-8	-	8.7 dB	4.9 dB	6.7 dB
1E-9	-	9.1 dB	5 dB	7 dB
1E-10	-	9.5 dB	5.2 dB	7.3 dB

Table 7-7 - 16QAM BER Performance (Viterbi)				
BER	Specification		Typical	
	3/4 Rate	7/8 Rate	3/4 Rate	7/8 Rate
1E-3	8.9 dB	10.3 dB	8.1 dB	9.5 dB
1E-4	9.8 dB	11.1 dB	9 dB	10.3 dB
1E-5	10.7 dB	11.9 dB	9.9 dB	11.1 dB
1E-6	11.5 dB	12.7 dB	10.7 dB	11.9 dB
1E-7	12.4 dB	13.5 dB	11.6 dB	12.7 dB
1E-8	13.3 dB	14.3 dB	12.5 dB	13.5 dB
1E-9	14.2 dB	15.1 dB	13.4 dB	14.3 dB
1E-10	15 dB	15.9 dB	14.2 dB	15.1 dB

Table 7-8 - 16QAM BER Performance (Viterbi w/RS)				
BER	Specification		Typical	
	3/4 Rate	7/8 Rate	3/4 Rate	7/8 Rate
1E-3	8.4 dB	9.8 dB	7.8 dB	9.3 dB
1E-4	8.6 dB	10.1 dB	8.1 dB	9.6 dB
1E-5	8.9 dB	10.3 dB	8.3 dB	9.9 dB
1E-6	9.1 dB	10.5 dB	8.6 dB	10.2 dB
1E-7	9.3 dB	10.8 dB	8.8 dB	10.4 dB
1E-8	9.5 dB	11.0 dB	9.1 dB	10.7 dB
1E-9	9.8 dB	11.3 dB	9.3 dB	11.0 dB
1E-10	10.0 dB	11.5 dB	9.6 dB	11.3 dB

Table 7-9 - 16QAM BER Performance (Turbo)				
BER	Specification		Typical	
	Turbo 0.495	Turbo 0.793	Turbo 0.495	Turbo 0.793
1E-3	-	-	5.6 dB	7 dB
1E-4	-	-	6.1 dB	7.4 dB
1E-5	-	-	6.6 dB	7.8 dB
1E-6	-	-	7 dB	8.2 dB
1E-7	-	-	7.5 dB	8.6 dB
1E-8	-	-	8 dB	9 dB
1E-9	-	-	8.5 dB	9.4 dB
1E-10	-	-	9 dB	9.9 dB

Table 7-10 - (O)QPSK BER Performance (Turbo)						
BER	Specification			Typical		
	1/2 Rate	3/4 Rate	7/8 Rate	1/2 Rate	3/4 Rate	7/8 Rate
1E-3	-	3.2 dB	4 dB	-	2.8 dB	3.7 dB
1E-4	-	3.4 dB	4.1 dB	-	3 dB	3.8 dB
1E-5	2.7 dB	3.6 dB	4.2 dB	2.4 dB	3.2 dB	3.9 dB
1E-6	2.9 dB	3.8 dB	4.3 dB	2.6 dB	3.4 dB	4 dB
1E-7	3.1 dB	4.1 dB	4.4 dB	2.8 dB	3.7 dB	4.1 dB
1E-8	3.3 dB	4.4 dB	4.5 dB	3 dB	4 dB	4.2 dB

Table 7-11 - 8PSK BER Performance (Turbo)				
BER	Specification		Typical	
	3/4 Rate	7/8 Rate	3/4 Rate	7/8 Rate
1E-3	5.6 dB	6.7 dB	5.2 dB	6.3 dB
1E-4	5.8 dB	6.8 dB	5.4 dB	6.4 dB
1E-5	6 dB	6.9 dB	5.6 dB	6.5 dB
1E-6	6.3 dB	7 dB	5.8 dB	6.6 dB
1E-7	6.5 dB	7.1 dB	6.0 dB	6.7 dB
1E-8	6.7 dB	7.2 dB	6.3 dB	6.8 dB

Table 7-12 - 16 QAM BER Performance (Turbo)				
BER	Specification		Typical	
	3/4 Rate	7/8 Rate	3/4 Rate	7/8 Rate
1E-3	6.3 dB	7.8 dB	6 dB	7.4 dB
1E-4	6.7 dB	7.9 dB	6.4 dB	7.5 dB
1E-5	7 dB	8 dB	6.7 dB	7.6 dB
1E-6	7.4 dB	8.1 dB	7.1 dB	7.7 dB
1E-7	7.8 dB	8.2 dB	7.5 dB	7.8 dB
1E-8	8.2 dB	8.3 dB	7.9 dB	7.9 dB

Table 7-13. Open Network Performance				
BER	Specification		Typical	
	IBS	IDR	IBS	IDR
	1/2 Rate	3/4 Rate	3/4 Rate	3/4 Rate
1E-3	4.1 dB	5.2 dB	4.2 dB	4.35 dB
1E-4	4.6 dB	6.0 dB	4.9 dB	5.25 dB
1E-5	5.3 dB	6.7 dB	5.6 dB	5.9 dB
1E-6	6.0 dB	7.5 dB	6.3 dB	6.6 dB
1E-7	6.6 dB	8.2 dB	6.9 dB	7.3 dB
1E-8	7.1 dB	8.7 dB	7.5 dB	7.8 dB

7.16.13 ACG Output Voltage

the AGC Output Voltage is a function of the Input Power Level in dBm. The AGC Output Voltage is found on the Alarm connector Pin 14 of J15.

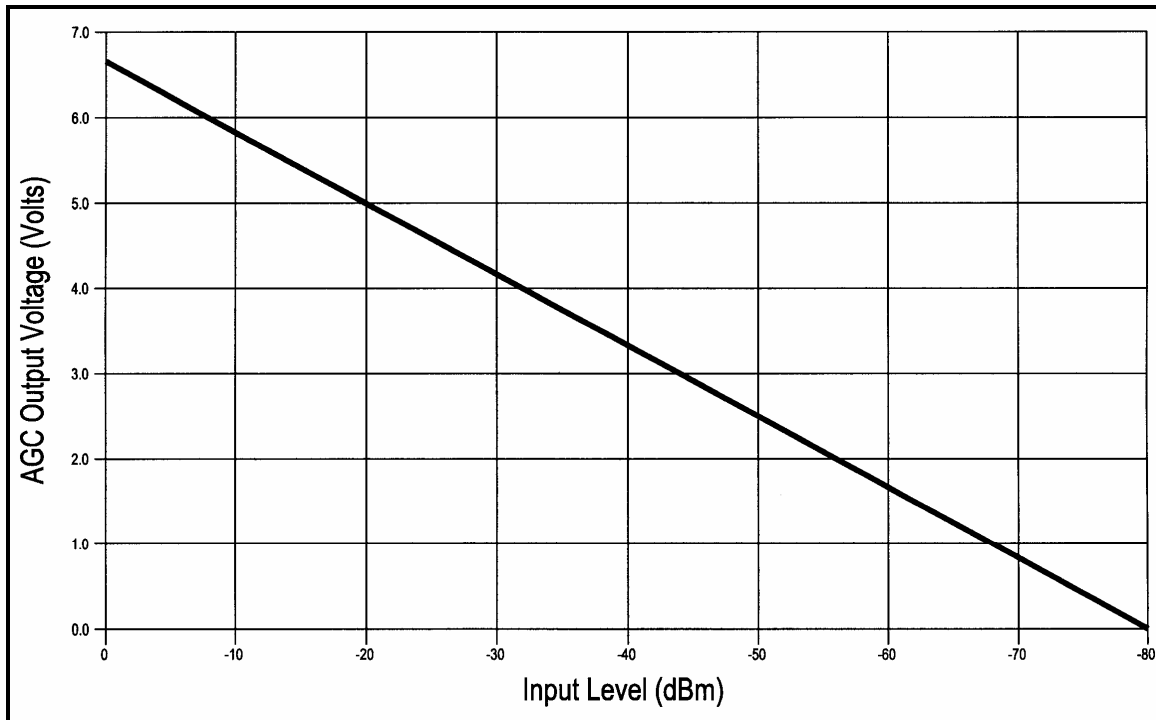


Figure 7-13. AGC Voltage Monitor



Product Options

A

A.0 Hardware Options

The following enhanced interface cards are available.

A.0.1 G.703/IDR ESC Interface

The DMD2050 can be equipped with a G.703/IDR ESC Interface.

A.0.2 Turbo Card

The DMD2050 can be equipped with an optional Turbo Codec Outer Code. This option must be installed at the factory and may require other options.

A.0.3 Internal High-Stability

The DMD2050 can be equipped with a 5×10^{-8} or better Stability Frequency Reference as an add-on enhancement. This is a factory upgrade only.

A.0.4 DC Input Prime Power

Allows for an optional DC Input Power Source (**standard unit only**).

A.0.5 ASI/RS-422 Parallel

ASI, Serial, BNC (Female)
DVB/M2P, Parallel, RS-422, DB-25 (Female)

A.0.6 ASI/LVDS Parallel

ASI, Serial, BNC (Female)
DVB/M2P, Parallel, LVDS, DB-25 (Female)

A.0.7 HSSI

High-Speed Serial Interface 50-Pin SCSI-2 Type Connector. Complies with Cisco Systems in HSSI Design Specification, Revision 3.0.

A.0.8 Ethernet Data Interface

Four RJ-45, Auto-Crossover, Auto-Sensing, 10/100 Ethernet Data Ports. Complies with IEEE 802.3 and IEEE 802.3u.

A.0.9 Sequential Interface

The DMD2050 is equipped with a Sequential option.

A.0.10 HSSI / G703

High-Speed Serial Interface 50-pin SCSI-2 Type Connector. Complies with Cisco Systems in HSSI Design Specification, Revision 3.0. The G703 interface supports T1, E1, T2, E2 rates balanced or unbalanced. It does not support backwards alarms.

A.0.11 HSSI / Ethernet

High-Speed Serial Interface 50-pin SCSI-2 Type Connector. Complies with Cisco Systems in HSSI Design Specification, Revision 3.0. Four RJ-45, Auto-Crossover, Auto-Sensing, 10/100 Ethernet Data Ports. Complies with IEEE 802.3 and IEEE 802.3u

A.0.12 AS/5167 Super Card (Variable Reed-Solomon)

The DMD2050 can be equipped with an optional AS/5167 Super Card. This card allows variable Reed-Solomon rates as well as Turbo Codec and Sequential Codec Outer Code. This option must be installed at the factory and may require other options.

A.0.13 Custom Reed-Solomon

The DMD2050 offers custom Reed-Solomon N+K values as an optional hardware upgrade.

A.0.14 DC Input Power

The DMD2050 offers 48 VDC input power.

A.1 Customized Options

The DMD2050 may be customized for specific customer requirements. Most modifications or customization can be accomplished by means of firmware/software modifications.

The following are examples of the types of customization available to the user:

- Customized Data Rates.
- Customized Scrambler/Descramblers.
- Customized Overhead Framing Structures.
- Customized Modulation Formats.
- Customized Uses for the ES-ES Overhead Channel.
- Drop and Insert.

Contact the Radyne Inc. Customer Service or Sales Department at (602) 437-9620 for all requests.



Front Panel Upgrade Procedure

B

B.0 Introduction

The DMD2050 Universal Satellite Modem offers the ability to perform field upgrades of the modem's feature set quickly and easily from the front panel. Purchased upgrades will become part of the modems permanent configuration. Demonstration upgrades will enable the optional features for a 30-day evaluation period.

B.1 Required Equipment

The DMD2050 Universal Satellite Modem is the only equipment required for this procedure.

B.2 Upgrade Procedure

The following paragraphs describe the procedure for permanently updating the feature set of the DMD2050 Universal Satellite Modem

1. The following steps allow users to quickly determine from the front panel whether or not the desired feature(s) are supported by the hardware currently installed in the modem.
 - a. From the modem's Main Menu, scroll right to the SYSTEM Menu.
 - b. Scroll down.
 - c. Scroll right to the HW/FW CONFIG Menu.
 - d. Scroll down
 - e. Scroll right to the FEATURES Menu. The second line will display NORMAL.
 - f. Scroll down.
 - g. Scroll right to the UPGRADE LIST Menu.
 - h. Scroll down.
 - i. Scroll right through the available list of options.

The top line identifies the options and the second line identifies the following options status:

INSTALLED indicates that the option is already available as part of the modems feature set.

HW & KEY REQ indicates that additional hardware is required to support the option. Contact your Radyne sales representative for more information regarding the required hardware upgrade.

KEY CODE REQ indicates that the desired option is available as a front panel upgrade.

2. Contact Radyne with the Unit ID and Desired Upgrades. The modem's Unit ID can be found on the front panel as follows:
 - a. From the modem's Main Menu, scroll right to the SYSTEM Menu.
 - b. Scroll down.
 - c. Scroll right to the HW/FW CONFIG Menu.
 - d. Scroll down.
 - e. Scroll right to the FEATURES Menu. The second line will display NORMAL
 - f. Scroll down

The value displayed on the top line of this menu is the 12-digit Unit ID. It is displayed as on the front panel of the modem as 3 sets of 4 digits in a dot-delineated format as follows:

1 2 3 4 . 1 2 3 4 . 1 2 3 4

Your Radyne sales representative will ask you for this number along with your desired feature set upgrades when placing your order.

3. Once your order has been processed, you will be issued a 12-digit feature set upgrade code. This code is only good on the modem for which it was ordered. To enter this code from the front panel, perform the following:
 - a. From the modem's Main Menu, scroll right to the SYSTEM Menu.
 - b. Scroll down.
 - c. Scroll right to the HW/FW CONFIG Menu.
 - d. Scroll down.
 - e. Scroll right to the FEATURES Menu. The second line will display NORMAL.
 - f. Scroll down.

The value displayed on the top line of this menu is the 12-digit Unit ID. It is displayed on the front panel of the modem as 3 sets of 4 digits in a dot-delineated format indicated in Step 2. The second line is the data entry area and is displayed as 3 sets of 4 underscores in a dot-delineated format.

- g. Press <ENTER>. A cursor will begin flashing in the data entry area.
- h. Using the numeric keypad, enter your 12-digit upgrade code.
- i. Press <ENTER>.

If the code entered is correct, the display will display **CODE ACCEPTED**, otherwise the display will display **INVALID CODE**.



NOTE

Care should be taken to insure that the upgrade code is entered properly. After three unsuccessful attempts to enter a code, the front panel upgrade and demonstration capability will be locked out and it will be necessary to cycle power on the modem in order to continue.

B.3 Demonstration Procedure

The procedure for enabling a 30-day demo of the DMD2050 options is similar to the procedure used for permanently updating the modems feature set. The one big difference being that at the end of 30 days, the demo features will automatically be disabled and the modem will revert back to its permanent configuration.



NOTE

At the end of the demonstration period, when the modem reverts back to its permanent configuration an interrupt in traffic will occur, regardless of whether or not a demo enabled features was being run at the time. In addition, operator intervention may be required to restore the data paths. In order to avoid this interruption in service, the user can cancel the demonstration at any time by following the instructions outlined in the section on "Canceling Demonstration Mode."

1. The following steps allow users to quickly determine from the front panel whether or not the desired feature(s) are supported by the hardware currently installed in the modem.
 - a. From the modem's Main Menu, scroll right to the SYSTEM Menu.
 - b. Scroll down.
 - c. Scroll right to the HW/FW CONFIG Menu.
 - d. Scroll down.
 - e. Scroll right to the FEATURES Menu. The second line will display NORMAL.
 - f. Scroll down.
 - g. Scroll right to the UPGRADE LIST Menu.
 - h. Scroll down.
 - i. Scroll right through the available list of options.

The top line identifies the options and the second line identifies the options status.

INSTALLED indicates that the option is already available as part of the modems feature set.

HW & KEY REQ indicates that additional hardware is required to support the option. Contact your Radyne sales representative for more information regarding the required hardware upgrade.

KEY CODE REQ indicates that the desired option can be enabled as a demonstration from the front panel.

2. Contact Radyne with the Unit ID and Request a Demonstration.

The modem's Unit ID can be found on the front panel as follows:

- a. From the modem's Main Menu, scroll right to the SYSTEM Menu.
- b. Scroll down.
- c. Scroll right to the HW/FW CONFIG Menu.
- d. Scroll down.
- e. Scroll right to the FEATURES Menu. The second line will display NORMAL.
- f. Scroll down.

The value displayed on the top line of this menu is the 12-digit Unit ID. It is displayed as on the front panel of the modem as 3 sets of 4 digits in a dot-delineated format as follows:

1 2 3 4 . 1 2 3 4 . 1 2 3 4

Your Radyne sales representative will ask you for this number along with the features you wish to demo.

3. Once your order has been processed, you will be issued a 12-digit demonstration code. This code can only be used one time and it is only good on the modem for which it was originally requested. To enter this code from the front panel, perform the following:

- a. From the modem's Main Menu, scroll right to the SYSTEM Menu.
- b. Scroll down.
- c. Scroll right to the HW/FW CONFIG Menu.
- d. Scroll down.
- e. Scroll right to the FEATURES Menu. The second line will display NORMAL.
- f. Scroll down.

The value displayed on the top line of this menu is the 12-digit Unit ID. It is displayed on the front panel of the modem as 3 sets of 4 digits. The second line is the data entry area and is displayed as 3 sets of 4 underscores in a dot-delineated format.

Press <ENTER> and a cursor will begin flashing in the data entry area

Using the numeric keypad, enter your 12-digit demonstration code
Press <ENTER>.

If the code entered is correct, the display will display CODE ACCEPTED, otherwise the display will read INVALID CODE



NOTE

Care should be taken to insure that the demonstration code is entered properly. After three unsuccessful attempts to enter a code, the front panel upgrade and demonstration capability will be locked out and it will be necessary to cycle power on the modem in order to continue.

B.3.3 Running in Demonstration Mode

Because of the possible interruption in traffic when the demonstration mode expires, several indicators are used to inform an operator that the modem is indeed, operating in demonstration mode. The most obvious of these are the flashing LEDs.

When the modem is running in demonstration mode, the Mod and Demod Test LEDs will be flashing. The flashing will occur even when the modem is placed in a test mode where the LED would normally be lit continuously.

A second indication can be found on the Features Menu as follows:

1. From the modem's Main Menu, scroll right to the SYSTEM Menu.
2. Scroll down.
3. Scroll right to the HW/FW CONFIG Menu.
4. Scroll down.
5. Scroll right to the FEATURES Menu. The second line will display DEMO.

A third indication can be found in the upgrade list as follows:

1. From the FEATURES Menu.
2. Scroll down.
3. Scroll right to the UPGRADE LIST Menu.
4. Scroll down.
5. Scroll right through the available list of options.

The top line identifies the options and the second line identifies the options status.

DEMO MODE indicates that the option is has been temporarily activated and is now available for evaluation as part of the modems feature set.

At the end of the demonstration period, the modem will revert back to its permanent configuration. When it does, an interrupt in traffic will occur, regardless of whether or not a demo enabled features was being run at the time. In addition, operator intervention may be required to restore the data paths. In order to avoid this interruption in service, the user can cancel the demonstration at any time by following the instructions outlined in the section on Canceling Demonstration Mode.

B.3.4 Canceling Demonstration Mode

At any time, a demonstration may be canceled and have the modem return to its normal operation. Once the demonstration has been canceled, it cannot be restarted using the old demonstration code. In order to restart a demonstration, it will be necessary to obtain a new demonstration code.

To cancel a demonstration from the front panel, perform the following:

1. From the modem's Main Menu, scroll right to the SYSTEM Menu..
2. Scroll down.
3. Scroll right to the HW/FW CONFIG Menu.
4. Scroll down.
5. Scroll right to the FEATURES Menu. The second line will display DEMO.
6. Scroll down.

The value displayed on the top line of this menu is the 12-digit Unit ID. It is displayed on the front panel of the modem as 3 sets of 4 digits in a dot-delineated format indicated in section 4.2. The second line is the data entry area and is displayed as 3 sets of 4 underscores in a dot-delineated format.

Press <ENTER> and a cursor will begin flashing in the data entry area

Using the numeric keypad, enter 0000 0000 0000

Press <ENTER>.

The modem will immediately terminate the demonstration and the feature set will revert back to the permanent configuration.

The Mod and Demod Test LED's will stop flashing.

Carrier Control

C

C.0 States

The DMD2050 transmitter will turn off the carrier output automatically when the modem determines there is a major alarm. This is done to prevent the carrier from outputting an unknown spectrum and possibly disturbing adjacent carriers. This automatic drop of the carrier can be overridden by masking the alarm that is causing the fault. This will keep the modulator output spectrum transmitting, even when the fault occurs. The following Carrier Control states are available:

- Carrier **OFF**
- Carrier **ON**
- Carrier **AUTO**
- Carrier **VSAT**
- Carrier **RTS**

C.1 Carrier Off

Modulator output is disabled.

C.2 Carrier On

Modulator output is turned off before reprogramming modulator functions that may alter the output spectrum through the front panel, and the user is required to enter “Yes” to re-enable output after the change. When using the terminal, the modulator is turned off while re-programming modulator functions that may alter the output spectrum, and the user is required to manually turn on the output after the reprogramming.

C.3 Carrier Auto

Modulator output is turned off before reprogramming modulator functions that may alter the output spectrum through the front panel, but the output is automatically turned on after the change. When using the terminal, the modulator is turned off while re-programming modulator functions that may alter the output spectrum, and but the output is automatically turned on after the change.

C.4 Carrier VSat

Modulator output is turned off before reprogramming modulator functions that may alter the output spectrum through the front panel, and the user is required to enter “Yes” to re-enable output after the change. When using the terminal, the modulator is turned off while re-programming modulator functions that may alter the output spectrum, and the user is required to manually turn on the output after the reprogramming (same as “Carrier On”). Additionally “VSat” mode disables the modulators output is the modems demodulator does not have signal lock. When signal lock returns to the demodulator, the modems modulator turns the carrier back on.

C.5 Carrier RTS

Modulator output is turned off before reprogramming modulator functions that may alter the output spectrum through the front panel, and the user is required to enter “Yes” to re-enable output after the change. When using the terminal, the modulator is turned off while reprogramming modulator functions that may alter the output spectrum, and the user is required to manually turn on the output after the reprogramming (same as “Carrier On”). Additionally “RTS” (Request To Send) mode enables the modulator’s output based on the RTS lead of the data interface. When RTS is enabled on the data interface, the modulator turns on the carrier, when the RTS is disabled the modulator turns off the carrier.

Web Browser Setup Guide

D

D.0 Introduction

The DMD20/50/2050 Web Browser allows for connection to Radyne products through the Ethernet port. The Web interface allows for complete control and monitoring of all equipment parameters and functions via a 10Base-T Ethernet connection. Utilization of the Web browser is dependent upon proper set-up of the TCP-IP menus. For TCP-IP setup guidelines refer to Appendix G in the manual. For additional information on the various WEB configurations and descriptions refer to the Remote Protocol Manual TM117.

To verify that the Radyne product is configured with the Web interface, use the products front panel <SYSTEM> control screen and verify that the WEB sub menu is visible. Contact the Radyne Customer Service Department if the WEB feature is not displayed on the front panel to identify the product is upgradeable.



NOTE

Utilization of the Web Browser requires proper setup of the TCP-IP menus. The unit is shipped from the factory with the Boot Mode configured as Default. Verify that the Boot Mode is set for Default. To access the unit via the Ethernet port, set the Boot Mode to IP TEST. This will set the IP parameter to an accessible address. If user wants to utilize an alternate address, user must set the Boot Mode to NON-VOL. Contact the IT manager for proper guidance to ensure setup is successful.

D.1 WEB Users Configuration

Radyne utilizes authentication and authorization together with a user database and a security Realm to design a secure site. This allows for the use of users and groups, where users are generally people that may access the system, and groups represent the position that users can hold.

The Modem's Front Panel has full administrative privileges, allowing the operator through the Front Panel Menu entries to perform monitor and control modem parameter settings, change other interface user names and passwords, and modify IP network settings.

The web user database consists of three users (USER 1, USER 2, USER 3), with initial default names of guest, oper, and admin. Access rights and authentication make it possible to restrict access to a specified set of web users. There are three levels of security privileges that can be assigned to any user. They are:

- **GUEST:** Users are able to navigate most of the site, and view modem parameter settings.
- **OPER:** Users can monitor and control modem parameter settings, and change their own authentication passwords.
- **ADMIN:** At this highest access right, users can monitor and control modem parameter settings, change any user's name and authentication password, and modify IP network settings. This setting has full access to the entire site.
- **NO GROUP:** Assigning this will block all access from the site.

Listed in the table below are the factory default user names and corresponding default passwords and settings:

USER ID	Access Group	Authentication Password	Web User
USER 1	GUEST	guest	guest
USER 2	OPER	oper	oper
USER 3	ADMIN	admin	admin



NOTE

All entries are case sensitive.

Using the Front Panel display and arrow keys, scroll thru the System menu until the WEB sub menu is displayed. Enter into the WEB menu and the following Sub menus will appear (order may vary).

1. **CONFIRMATION:** When ENABLED, a change in the Web Interface will prompt a secondary window to be displayed asking "Do you want to proceed". This will allow the user to either OK the change or cancel the change. When this feature is DISABLED, the command will be executed upon entering the command or parameter change. Commands pertaining to frequency and data rates cannot be overridden. The Confirmation command is located under PASSWORD SETUP/PREFERENCES.

2. **USER 1**: This will allow the operator to change the user name, assign the Access group, authorized password for "USER 1". Upon entering the following fields will be displayed:
 - a. **ACCESS GROUP**: This will allow the assignment of "No Group", "ADMIN", "OPER", or GUEST to USER 1.
 - b. **AUTH PASSWORD**: This will allow for the entry of the password for USER 1.
 - c. **USER RESET**: Using this command will allow the factory defaults (as listed in the table above) to be restored to USER 1. This can be used in the event that USER 1 is locked out due to password restriction.
3. **USER 2**: This has the same menu structure as USER 1.
4. **USER 3**: This has the same menu structure as USER 1.

Radyne's Web configuration allows for the support of 3 user profiles. These are configured through the 'PASSWORD/SETUP ACCESS' section in the Web Browser.

Change Web User Name

Any of the three available user names can be modified.

1. Go to <USERx> Submenu, press <ENTER> then move the cursor to any position and press <CLEAR> along with the 'Right Arrow' key to clear all text to the right of the cursor.
2. Use the arrow keys and the numerical keypad to set the desired user name. Press <ENTER> to accept the changes.

Change Authentication Password

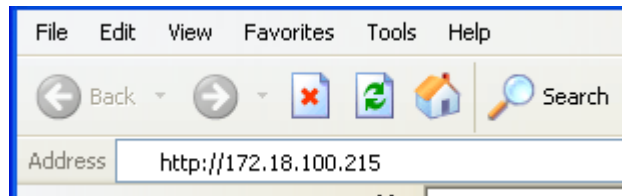
1. Go to <USERx> Submenu, press 'Down Arrow' to access its corresponding lower level sub-menus.
2. Go to <AUTH PASSWORD> Submenu, <ENTER> then move the cursor to any position and press <CLEAR> along with the 'Right Arrow' key to clear all text to the right of the cursor.
3. Use the arrow keys and the numerical keypad to set the desired password. Press <ENTER> to accept the changes.

Change Access Rights

1. Go to <USERx> Submenu, press 'Down Arrow' to access its corresponding lower level sub-menus.
2. Go to <ACCESS GROUP> Submenu, <ENTER> then move the cursor to any position and press <CLEAR> along with the 'Right Arrow' key to clear all text to the right of the cursor.

D.2 Modem Web Site

Once it is determined that the modem is reachable, start the internet browser and type the modem IP address in the address field then press the Enter key.

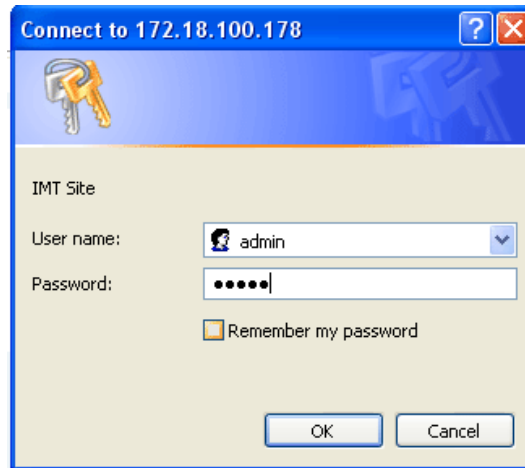


The equipment Introduction page will display general information, showing the equipment type, an overview of the equipment features and capabilities, and available hardware and software options. Complete technical specifications, and product options are also available in a PDF format at the bottom of the screen. Links to the Radyne Corporation website and Customer service is also accessible provided access to the internet is available.



WEB Browser Introduction Page

To navigate to other parts of the site, the modem needs to know who the user is to allow assignment of the proper level of transactions and resources. This authentication is done through the use of a username and password. The following screen will pop-up as soon as entry is requested into any other web pages (i.e. Password Set-up, IP Administration, and Monitor and Control). The factory default User Name and password defaults are listed above. Enter “admin” for the user name and “admin” for the password to gain access with full privileges to the other pages within the browser.



The screenshot shows a web browser window titled "Connect to 172.18.100.178". It features a login form with the following elements:

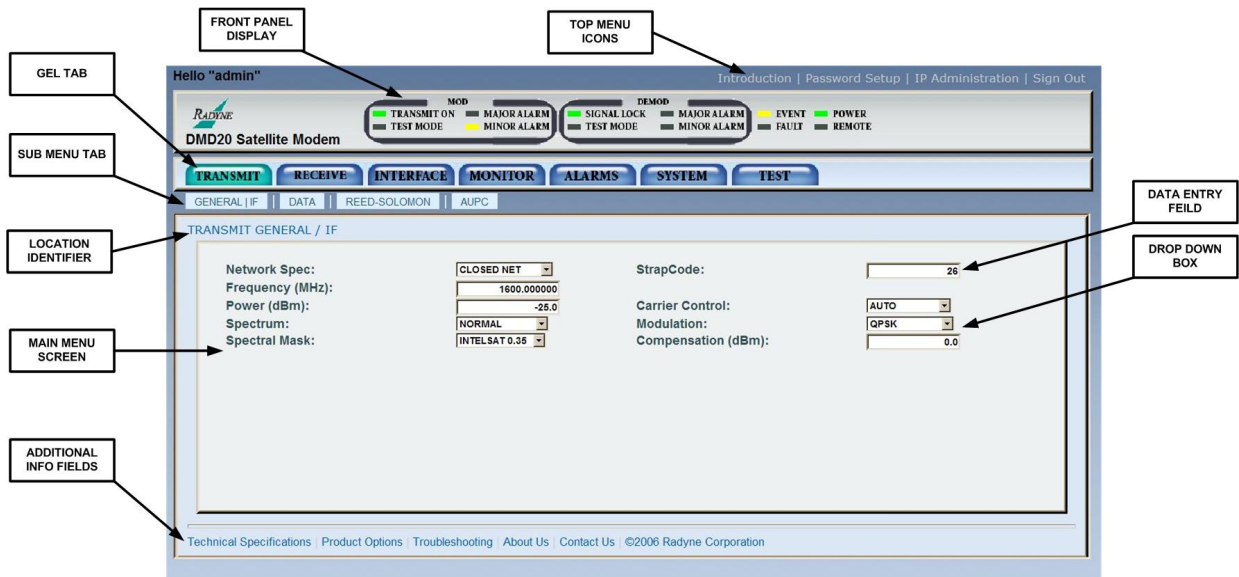
- IMT Site**: A label above the login fields.
- User name:**: A text box containing the value "admin".
- Password:**: A text box with masked characters (dots).
- Remember my password**: A checkbox that is currently unchecked.
- Buttons**: "OK" and "Cancel" buttons at the bottom right.

Login Window

Upon completion of a successful login, the user will be able to access the other screens within the Web Browser.

D.3 Web Page Appearance

This page displays the Monitor and Control section of the modem web interface. With a front panel look and feel, and an appearance that resembles the DMD2050 layout.



Monitor and Control Web Page

The top of the web page displays an alarm section reflecting the current front panel status of the modem. This block will be updated immediately whenever status is changed.

The navigation scheme consists of gel tabs that correspond to the front panel top-level menu selections. When the mouse is dragged over a tab, it gets highlighted with a rollover effect and a sub-level of selections is revealed. These correspond to the front panel sub-menus.

Below the menu gel tabs, the main menu section will reflect the current programmed control state. At the top of the page, breadcrumb location identifier reminds the user how the page is selected (Location Identifier).

There is a simple navigation scheme that consists of gel tabs centralized on the main page.





Efficient Drop & Insert



E.0 Introduction

The following paragraphs describe the menu structure and procedure for configuring a Radyne DMD20 / DMD2050 / DMD50 modem for Efficient Drop & Insert Mode.

E.1 Prerequisite

In order for a modem to be configured for efficient Drop & Insert, the modem must have a G.703 Interface card installed and Open Network Drop & Insert must be enabled. If you modem does not have the required hardware and/or feature set enabled, you will need to contact your Radyne salesperson to order the appropriate hardware and/or feature set upgrade. If your modem has the appropriate hardware, but the software revision is prior to AY, you will need to download the latest modem firmware from the Radyne FTP website.

The following menus illustrate how to determine whether or not your modem has the required hardware and feature set options.

SYSTEM		
HW/FW CONFIG		
FIRMWARE		
F05058-AY 6.1	or later required	
SYSTEM		
HW/FW CONFIG		
TERR INTFC BRD		
01-AS/4975	or equivalent required	
SYSTEM		
HW/FW CONFIG		
FEATURES		
UPGRADE LIST		
D&I		
INSTALLED	required	
ENH ASYNC		
INSTALLED	optional, required if desired	
AUPC		
INSTALLED	optional, required if desired	

E.2 Efficient Drop & Insert Mode

With Efficient Drop & Insert, the terrestrial interface selections, terrestrial framing modes, terrestrial to satellite mapping, ES to ES channel, satellite and terrestrial backward alarm functionality, and the In Station Prompt and Deferred Service alarm operation are identical to that of the Open Network standard. In addition, the selection and operation of Enhanced Async and AUPC are identical to their closed network IBS counterparts. For more information on these selections, refer to the appropriate section of the User's manual.

The following menu selections are utilized for controlling the additional functionality available with efficient Drop and Insert:

MODULATOR or DEMODULATOR

NETWORK SPEC

CLOSED NET

Efficient Drop & Insert is a Closed Network selection

SAT FRAMING

EFFICIENT D&I

The satellite frame type is Efficient Drop & Insert

DATA

DATA RATE (bps)

N x 64000

The data rate can be set to any N x 64 kbps rate based on the desired number of drop or insert slots. The following values of N are allowed based on the terrestrial interface and terrestrial framing types shown

T1	Any framing	Any N from 1 to 24
E1	Any PCM31	Any N from 1 to 31
E1	Any PCM30	Any N from 1 to 30 as TS 16 is automatically transmitted

SCRAMBLER CTRL

DISABLED

The Efficient Drop & Insert mode utilizes a frame synchronous energy dispersal technique that is always on, thus there is no need for any additional scrambling

INBAND RATE

150

300

600

1200

2400

4800

9600

19200

Available when enhanced async is enabled, this field allows the operator to set the Earth Station to Earth station in-band rate. In most cases, this should be set to the same rate or higher than the ES port baud rate. When this rate is set lower than the ES port baud rate, the user must insure that the actual transmission rate does not exceed the in-band rate, otherwise, characters will be dropped

E.2.1 Calculating the Required Satellite Bandwidth

In order to calculate the satellite bandwidth (i.e. the symbol rate), we must first calculate the Efficient D&I Rate (i.e. the data rate plus the overhead required for Efficient Drop & Insert). From there, the calculation of the required satellite bandwidth is identical to all other modes of operation and simply takes into account modulation type and forward error correction.

In this section, we will cover the calculation of the basic Efficient D&I Rate, as well as, the two cases that alter the basic rate.

E.2.2 Calculating the Basic Efficient D&I Rate

When E1 signaling is not required (all T1 and PCM31 cases) and Enhanced Async is not enabled (the Earth Station to Earth station link is the standard ES-ES), the Efficient D&I Rate for N timeslots is as follows:

$$\text{Efficient D\&I Rate} = \text{Data Rate} + (N * 250 \text{ bps})$$

In other words, the basic Efficient Drop & Insert Rate only requires 250 bps of overhead per slot, while at the same time providing all of the functionality found in the open network standard plus Automatic Uplink Power Control. By comparison, the open network standard requires 4267 bps per slot, so by utilizing Efficient Drop and Insert, Radyne customers can realize a bandwidth savings of over 4000 bps per slot.

E.2.3 Calculating the Efficient D&I Rate with E1 Signaling

When E1 signaling is enabled (PCM-30, PCM30C), an additional 2000 bps per slot are required to carry the E1 signaling. So the Efficient D&I Rate for N timeslots is:

$$\text{Efficient D\&I Rate increase} = N * 2000 \text{ bps}$$

With the open network standard requiring 4267 bps per slot, Efficient Drop and Insert provides a bandwidth savings of over 2000 bps per slot when E1 signaling is required.

E.2.4 Calculating the Efficient D&I Rate with Enhanced Asynchronous Overhead

The amount of overhead required to carry the Enhanced Async is driven by the in-band baud rate. The calculation is a two step process involving the in-band baud rate and the number of slots as follows:

$$X = \text{Truncation of } (\text{In-Band Baud Rate} / (N * 125))$$

$$\text{Efficient D\&I Rate increase} = X * N * 125 \text{ bps}$$

Because of the truncation, this increase in bandwidth is guaranteed to be less than the baud rate itself.

Summary and Examples:

The following examples further illustrate how to calculate the Efficient D&I rate which can be summarized for N timeslots as:

$$\text{Efficient D\&I Rate} = \text{Data Rate} + (N * 250 \text{ bps})$$

With E1 signaling add

$$N * 2000 \text{ bps}$$

With Enhanced Async add

$$(\text{Truncation}(\text{In-Band Baud} / (N * 125))) * (N * 125) \text{ bps}$$

Example 1a:

5 Drop Slots with T1-D4 framing, standard ES to ES overhead

For 5 Drop Slots, the Data Rate would be $5 * 64000$ or 320000 bps

The Efficient D&I Rate would be $320000 + (5 * 250) \text{ bps} = 321250 \text{ bps}$

The Open Network rate is over 20,000 bps higher at 341333 bps.

Example 1b:

Change to E1-PCM30 framing (E1 Signaling), standard ES to ES overhead

Add $5 * 2000 \text{ bps}$ to our previous calculation gives 331250 bps

Still saving over 10,000 bps compared to the open network standard.

Example 1c:

Change to Enhanced Async with In-Band Baud Rate of 1200

$$X = \text{Truncation of } (1200 / (5 * 125))$$

$$X = \text{Truncation of } (1.92)$$

$$X = 1$$

Add $1 * 5 * 125 \text{ bps}$ to our previous calculation gives 331875 bps

An increase of 625 bps to carry 1200 baud

Example 2a:

10 Drop Slots with T1-D4 framing, standard ES to ES overhead

For 10 Drop Slots, the Data Rate would be $10 * 64000$ or 640000 bps

The Efficient D&I Rate would be $640000 + (10 * 250) \text{ bps} = 642500 \text{ bps}$

The Open Network rate is over 40,000 bps higher at 682667 bps.

Example 2b:

Change to E1-PCM30 framing (E1 Signaling), standard ES to ES overhead

Add $10 * 2000$ bps to our previous calculation gives 662500 bps

Still saving over 20,000 bps compared to the open network standard.

Example 2c:

Change to Enhanced Async with In-Band Baud Rate of 1200

$X = \text{Truncation of } (1200 / (10 * 125))$

$X = \text{Truncation of } (0.96)$

$X = 0$

The rate stays at the previous value of 662500

With 10 slots, there is no increase required to carry 1200 baud Enhanced Async. It is passed transparently in the Efficient Drop & Insert overhead.

TCP/IP Ethernet Setup

F

F.0 Introduction

The modem supports SNMP, FTP protocols and the Web Browser. Utilization of the protocols is dependent upon proper set up of the TCP-IP menus. This document is to be used only as a guideline for setting up the TCP-IP menus. Contact the IT manager for proper guidance to ensure setup is successful. For additional information on the various WEB or SNMP configurations and descriptions refer to the Remote Protocol Manual (TM117).

F.1 TCP/IP Network Configuration

Using the Front Panel display and arrow keys, scroll thru the System menu until the TCP / IP sub menu is displayed. Each unit requires proper configuration with the correct network settings. Contact the IT manager for a valid IP address mask, Modem, server and router IP addresses.

Enter into the TCP / IP menu and the following Sub menus will appear, however the order may vary.

1. **Boot Mode:** This allows for the selection of the operating boot mode for the TCP / IP. Several selections are available and are described below. When configuring the modem for Web Browser, Boot Mode must be set to "NON-VOL". A brief description of the available selections are:
 - a. **Default:** If the Ethernet interface is not to be used, select this mode. No IP Address or mask changes will be allowed while in this mode of operation. The following parameters will be set and will not change until the boot mode is changed. The IP addresses are non accessible addresses.

• IP MASK	255.000.000.000
• MODEM IP ADDR	010.000.000.001
• SERVER IP ADDR	010.001.001.001
• ROUTER IP ADDR	010.000.001.001
 - b. **BOOTp:** When enabled, at boot time, the modem will use the Bootp Protocol to automatically get names, masks, and IP Addresses of the modem, router, and server from the Network Manager. This should be consistent with the tag expected by the users Bootp Server (see the next menu selection for setting the BOOTp TAG). If Bootp is not enabled, the modem will ignore the BOOTp Tag setting.
 - c. **NON-VOL:** This will allow for setting up all required IP Addresses and will store the information to the non-volatile memory. Upon power cycle, the modem will restore the saved settings into the correct fields.

- d. **IP TEST:** The IP Test selection is similar in behavior to the Default selection. When enabled, the following preset parameters will be programmed and will not change until the selection is changed. To edit these parameters, change the boot mode to NON-VOL.

▪ IP MASK	255.255.255.000
▪ MODEM IP ADDR	192.168.000.238
▪ SERVER IP ADDR	192.168.000.101
▪ ROUTER IP ADDR	192.168.000.102



NOTE

The DMD20/20LBST is shipped from the factory with the Boot Mode configured as Default. Verify that the Boot Mode is set for Default. To access the unit via the Ethernet port, set the selection to IP TEST. This will set the IP parameter to an accessible address. If user wants to utilize an alternate address, user must set the selection to NON-VOL. As an example, we will use a new modem IP address of 172.018.100.205 for the remaining of the TCP-IP setup procedure. Contact the IT manager for proper guidance to ensure setup is successful.

2. **BOOT SERVER TAG:** This allows for the selection of the operating boot tag when operating in the BOOTp Mode. The default setting of 206 is automatically selected when the boot mode is set to 'DEFAULT' (factory preset mode).
3. **MODEM HOST:** This displays the unit Host name that is operating, such as "DMD20". This is a read only display.
4. **IP ADDR MASK:** This will allow for the entry of the IP Address Mask. This will need to be entered based on the Network settings. Refer to your IP Administrator if you do not know this address for the correct address setting. Example IP Address Mask setting: 255.255.000.000.
5. **MODEM IP ADDR:** This will allow for the entry of the Modem's individual network IP Address. Each device on the network will have a unique address. Refer to the IT administrator for the correct address setting. Example Modem IP Address setting: 172.018.100.215.
6. **SERVER IP ADDR:** This allows for the setup of the Network Server IP Address. This section refers to the Host that will be used to optionally boot the DMD20 on power-up and is the SNMP Trap Server. This IP Address needs to be consistent with the Modem IP Address. Broadcast and loop back addresses will not be allowed. Example Server IP Address setting: 172.018.004.250.
7. **ROUTER IP ADDR:** This allows for the setup of the Network Router IP Address. If a router is present on the local network, and it is to be used, this address must be consistent with the IP Address Mask and the subnet of the modem. If no router is present, then the address should be set to a foreign address. Broadcast and loop back addresses will not be allowed. Router not used example: Router IP Address setting: 010.000.001.001.

8. **MODEM EADDR:** This displays the Modem (Unit) Ethernet Address. The Modem Ethernet Address is configured at the factory. It is a unique Radyne equipment identifier Address. Example: 0010650903EB
9. **ETHER RATE:** This displays the current Ethernet port data rate. If multiple rates are available, then a selection can be made to specify the Ethernet port data rate (10BaseT). Example Ethernet port Data Rate: 10 MBPS/HD

F.2 Network Configuration Summary

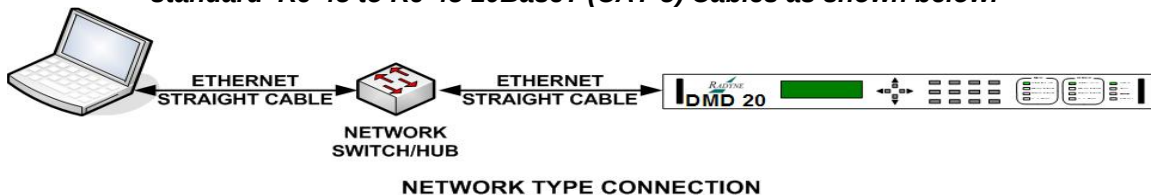
If the above steps were followed and the information was entered, then the following would be the TCP / IP configuration summary for a 'no router specified' setup:

1. Boot Mode = NON-VOL
2. Bootp Server Tag = 206
3. Modem Host= DMD20
4. IP Address Mask = 255.255.0.0
5. Modem IP Address = 172.18.100.215
6. Server IP Address = 172.18.4.250
7. Router IP Address = 010.000.001.001
8. Modem Ethernet Address= 0010650903EB
9. Ethernet Rate = 10 MBPS/HD

F.3 Ethernet Test

Connecting the Modem Ethernet Cable to a Network Link

1. *Connect the Network Switch (Hub) to the Modem Ethernet port (J9) using standard RJ-45 to RJ-45 10BaseT (CAT-5) Cables as shown below.*



DMD20 Ethernet Network Connection

Connecting the Modem Ethernet Cable Directly to a Computer (without a Network)

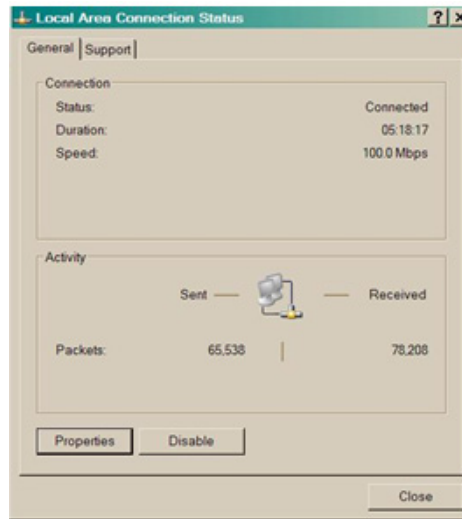
The user can directly connect to the equipment without connecting to a network. This will often occur at remote sites where a network is not available. To connect, the user will need an Ethernet Crossover (Null) cable. The pinout for this cable is as follows,

RJ45 Connector A	RJ45 Connector B
Pin #1	Pin #3
Pin #2	Pin #6
Pin #3	Pin #1
Pin #4	Pin #4
Pin #5	Pin #5
Pin #6	Pin #2
Pin #7	Pin #7
Pin #8	Pin #8



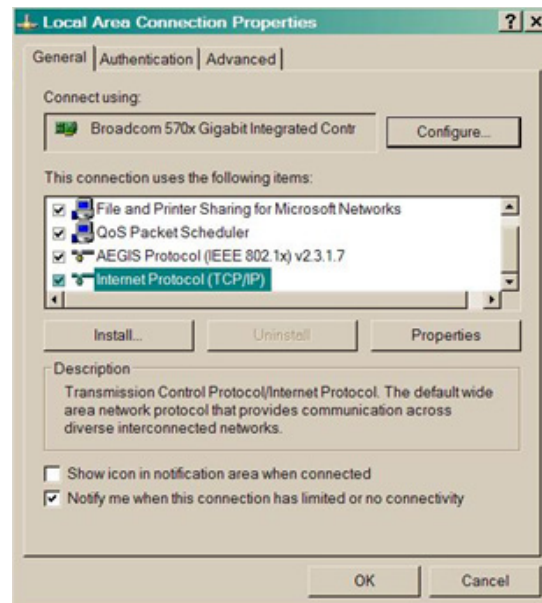
The Computer TCP/IP must be properly configured in order to obtain connectivity. The following set-up procedure can be used as a guide to aide in this setup. The following instructions apply only to Windows 2000 or XP Classic.

1. Click on the Start Button. Select Settings and click on the Control Panel Icon. Double click the Network Connections Icon.
2. Select the Local Area Connection icon for the applicable Ethernet adapter. Usually it is the first Local Area Connection listed. Double click the Local Area Connection. Click on the Properties icon.

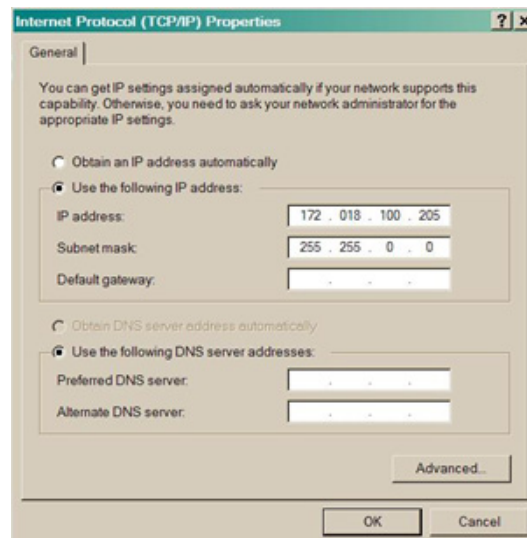


Local Area Connection Status Box

3. Make sure that the box next to the Internet Protocol (TCP/IP) is checked. Highlight Interconnect Protocol (TCP/IP) and click on the Properties button.

**Local Area Connection Properties Box**

4. Select “Use the following IP Address”. Enter in the IP Address that is offset by 5 or so numbers from the equipment address (the computer and the equipment that it is connecting to can not have identical addresses) and Subnet Mask (this is identical to the subnet mask programmed into the equipment) into the corresponding fields. Click the OK button to complete the PC Configuration. Note: some computers may require that the computer be restarted for the changes to take effect.

**Internet Protocol (TCP/IP) Properties Box**

5. To reconnect the computer to a network, select the “Obtain an IP address automatically” selection in the screen shown above.

F.4 Testing the Ethernet Connection using the Ping Program (Optional)

To verify that connectivity and settings are correct, use the Ping command to report if the Host (Equipment) is correctly responding. This is accomplished by opening the MSDOS Command Prompt and executing a Ping command as shown in the following example.

1. Open MSDOS Command Prompt. The Screen will display:

```
Microsoft Windows XP [Version 5.1.2600]  
(C) Copyright 1985-2001 Microsoft Corp.
```

2. At the Command Prompt Enter "ping 172.18.100.215" (Enter the IP Address of the equipment to be tested). The screen will display:

```
Microsoft Windows XP [Version 5.1.2600]  
(C) Copyright 1985-2001 Microsoft Corp.
```

```
C:\> ping 172.18.100.215
```

3. If the ping is successful the screen will display:

```
C:\>ping 172.18.100.215
```

```
Pinging 172.18.100.215 with 32 bytes of data:
```

```
Reply from 172.18.100.215: bytes=32 time=109ms TTL=64  
Reply from 172.18.100.215: bytes=32 time<1ms TTL=64  
Reply from 172.18.100.215: bytes=32 time=2ms TTL=64  
Reply from 172.18.100.215: bytes=32 time=123ms TTL=64
```

```
Ping statistics for 172.18.100.215:
```

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 0ms, Maximum = 123ms, Average = 58ms
```

4. If the ping is unsuccessful the screen will display:

```
C:\>ping 172.18.100.215
```

```
Pinging 172.18.100.215 with 32 bytes of data:
```

```
Request timed out.  
Request timed out.  
Request timed out.  
Request timed out.
```

```
Ping statistics for 172.18.100.215:
```

```
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Check the following items that may lead to the unsuccessful response:

- a. Verify that the correct cables are connected to the Ethernet port and that they are secured.
- b. The Link Light is illuminated.
- c. The IP Address that is used matches the Modem's IP Address.
- d. The Server and Modem are on the same subnet.



Glossary



A	
A	Ampere
AC	Alternating Current
ADC	Analog to Digital Converter
AGC	Automatic Gain Control
AIS	Alarm Indication System. A signal comprised of all binary 1s.
AMSL	Above Mean Sea Level
ANSI	American National Standards Institute
ASCII	American Standard Code for Information Interchange
ASIC	Application Specific Integrated Circuit
ATE	Automatic Test Equipment
B	
BER	Bit Error Rate
BERT	Bit Error Rate Test
Bit/BIT	Binary Digit or Built-In Test
BITE	Built-In Test Equipment
bps	Bits Per Second
BPSK	Binary Phase Shift Keying
BUC	Block Upconverter
Byte	8 Binary Digits

C	
C	Celsius
CATS	Computer Aided Test Software
CA/xxxx	Cable Assembly
CD-ROM	Compact Disk – Read Only Memory
CLK	Clock
cm	Centimeter
COM	Common
CPU	Central Processing Unit
CRC	Cyclic Redundancy Check. A system of error checking performed at the transmitting and receiving stations.
CW	Continuous Wave
C/N	Carrier to Noise Ratio
D	
DAC	Digital to Analog Converter
dB	Decibels
dBc	Decibels Referred to Carrier
dBm	Decibels Referred to 1.0 milliwatt
DC	Direct Current
Demod	Demodulator or Demodulated
DPLL	Digital Phase Locked Loop
DVB	Digital Video Broadcast
D&I	Drop and Insert
E	
E_b/N_0	Ratio of Energy per bit to Noise Power Density in a 1 Hz Bandwidth.
EEPROM	Electrically Erasable Programmable Read Only Memory
EIA	Electronic Industries Association
EMI	Electromagnetic Interference
ESC	Engineering Service Circuits
ES-ES	Earth Station to Earth Station Communication
ET	Earth Terminal

F	
F	Fahrenheit
FAS	Frame Acquisition Sync. A repeating series bits, which allow acquisition of a frame.
FCC	Federal Communications Commission
FEC	Forward Error Correction
FIFO	First In, First Out
FPGA	Field Programmable Gate Arrays
FW	Firmware
G	
g	Force of Gravity
GHz	Gigahertz
GND	Ground
H	
HSSI	High Speed Serial Interface
HW	Hardware
Hz	Hertz (Unit of Frequency)
I	
IBS	Intelsat Business Services
IDR	Intermediate Data Rate
I/O	Input/Output
IEEE	International Electrical and Electronic Engineers
IESS	INTELSAT Earth Station Standards
IF	Intermediate Frequency
INTELSAT	International Telecommunication Satellite Organization
ISO	International Standards Organization
I & Q	Analog In-Phase (I) and Quadrature Signals (Q)
J	
J	Joule

K	
Kbps	Kilobits per Second
Kbps	Kilobytes per Second
kg	Kilogram
kHz	Kilohertz
Ksps	Kilosymbols per Second
L	
LCD	Liquid Crystal Display
LED	Light Emitting Diode
LO	Local Oscillator
M	
mA	Milliampere
Mbps	Megabits per Second
MFAS	Multi-Frame Acquisition Sync. See FAS.
MHz	Megahertz
MIB	Management Information Base
Mod	Modulator or Modulated
ms or msec	Millisecond
M&C	Monitor and Control
N	
NC	Normally Closed
NO	Normally Open
ns	Nanoseconds
NVRAM	Non-Volatile Random Access Memory
N/C	No Connection or Not Connected
O	
OQPSK	Offset Quadrature Phase Shift Keying
P	
PC	Personal Computer
PD Buffer	Plesiochronous/ Doppler Buffer
PLL	Phase Locked Loop
ppb	Parts per Billion
ppm	Parts per Million
P/N	Part Number

Q	
QAM	Quadrature Amplitude Modulation
QPSK	Quadrature Phase Shift Keying
R	
RAM	Random Access Memory
RF	Radio Frequency
ROM	Read Only Memory
rms	Root Mean Square
RU	Rack Unit. 1 RU = 1.75"/4.45 cm
Rx	Receive (Receiver)
RxD	Receive Data
R-S	Reed-Solomon Coding. Reed-Solomon codes are block-based error correcting codes with a wide range of applications in digital communications and storage.
S	
SCC	Satellite Control Channel. A Radyne satellite format.
SEQ	Sequential
SYNC	Synchronize
T	
TBD	To Be Designed or To Be Determined
TM	Technical Manual
TPC	Turbo Product Codes
TRE	Trellis
TT	Terminal Timing
Tx	Transmit (Transmitter)
TxD	Transmit Data
U	
UART	Universal Asynchronous Receiver/Transmitter
UUT	Unit Under Test
V	
V	Volts
VAC	Volts, Alternating Current
VCO	Voltage Controlled Oscillator
VDC	Volts, Direct Current
VIT	Viterbi Decoding

W X Y Z	
W	Watt
Misc.	
μs	Microsecond
16QAM	16 Quadrature Amplitude Modulation
8PSK	8 Phase Shift Keying